

THE GRAIL COLLECTOR

Engineering Report — complete technical record

Converging roll-bond solar absorber with transparent insulation, PCM and vacuum insulation: design, CFD, reduced-order model, surrogate, optimisation, system model, annual and economic analysis, verification and validation

Item	Value
Project	B.Tech minor project — GRAIL Collector
Document type	Simple engineering report (working record). A formal engineering report will be prepared from it.
Revision	Rev 1.0, 25 September 2026 (dataset Rev 4.2; Simulink V21)
Authors / institution	[to be filled in by the authors]
Units	SI throughout; temperatures in kelvin (K) in all equations, °C only where stated
Status	All stages complete. V5 validation closed at method/component level. No experimental test of the GRAIL collector itself.

Contents

1. Summary of results

This report records everything done on the GRAIL collector in one place: the inputs, every method, every result, every figure and table, all verification checks and the validation against published experiments. It is written plainly so that each number can be traced to its source file.

Area	Key result
Collector	12-channel aluminium roll-bond absorber, 1.10×0.48 m (0.528 m^2 per module), channels converging along the flow ($G = 0.539935$, $\lambda = 1.0$), bridge 31.29 mm; glass + honeycomb TIM cover; PCM + VIP rear; optical factor 0.649
3-D CFD	Conjugate OpenFOAM model; developed Nusselt number 4.48 (GCI 2.1 %), H1-type condition; replaces the earlier 2.92
Reduced-order model (GRAIL-CHT)	Matches 3-D co-current solution within 0.01 % (η) and 0.6 % (plate std); Hottel-Whillier-Bliss check +0.22 %
Campaign	300 Latin-hypercube cases (150 alternating, 150 parallel), all converged, energy error < 0.5 %, none removed
Alternating vs parallel (equal inputs)	Alternating: efficiency -7.3% ($p = 2 \times 10^{-8}$), plate +13.9 K hotter; uniformity better only above 3.3 g/s (-13.4%), worse below 2.2 g/s ($+16.8 \%$)
Alternating vs parallel (same plate temperature, V7)	Efficiency equal ($\Delta\eta = -0.001$ points); R4 not reduced ($+0.04 \%$); alternating 8–11 % less uniform; needs ≈ 15 K cooler inlet
Mechanism	$\approx 40 \%$ of the heat each channel absorbs is returned to the plate in alternating flow (9.94 of 24.80 W)
ANN surrogate (Stage 3)	Test R^2 0.9994–0.9998 on 60 unseen cases; mean error 0.14 %
NSGA-II (Stage 5)	8 optima re-run in GRAIL-CHT agree within 0.13 efficiency points, 2.4 % std, 5.8 % Δp
Efficiency curves (Stage 6)	Parallel η_0 0.637, a_1 2.10 W/m ² K; alternating η_0 0.597, a_1 2.00 W/m ² K (a_1 up to 2.5 with measured VIP conductivity)
Annual, Berhampur (Stage 7)	POA 1,976 kWh/m ² /yr; yield at 40 °C inlet 1,030 (parallel) / 965 (alternating) kWh/m ² /yr
Economics	Best size 2 modules (1.06 m^2): solar fraction 86 % / 83 %; cost of heat Rs 4.34 / 4.49 per kWh vs Rs 4.95 for electric geyser; payback ≈ 7.4 / 7.6 years
System models	Python, GNU Octave, Scilab Xcos and MATLAB/Simulink agree; Simulink V20/V21 PASS in all 4 cases (solar heat within 0.05 %)
Validation (V5)	Method checked on tested roll-bond collectors (Del Col 2013): a_1 , a_2 within test uncertainty, η_0 within 8 % ($+6.0$ / $+7.8 \%$). TIM front loss matches Beikircher 2015. Partial pass, method level
Patent	First-pass search: combination appears new; inventive step weak–moderate; decision: publish, no filing

Statements that must be kept exactly as they are

1. Parallel flow is more efficient than alternating flow at equal inputs in every stage of the work.
2. At the same mean plate temperature the two arrangements have the same efficiency; alternating flow is less uniform at the design flow.
3. Within the studied range, converging the channels does not raise efficiency; the optimiser moves G towards 1. The converging channel can be argued only for plate uniformity.
4. All results are model results. The method was checked against published experiments; the GRAIL collector itself has not been tested.

2. Project definition

2.1 Objective

Design a flat-plate solar water-heating collector with a roll-bond aluminium absorber whose channels converge along the flow, add a low-loss cover and rear (glass + transparent insulation, phase-change material + vacuum insulation panel), and find out by simulation whether alternating (counter-current) flow in neighbouring channels is better than ordinary parallel flow. Then estimate its annual performance and cost for a household in Berhampur, Odisha.

2.2 Work chain

Step	What was done	Tool
1 CAD	Parametric 3-D model, 948 features, 131 parameters, verified gates	Autodesk Fusion
2 Geometry audit	Channel sections measured from the CAD (Gate 1)	Python (NURBS integration)
3 Mesh	Structured hexahedral channel meshes from the CAD surfaces	Python + OpenFOAM blockMesh
4 3-D CFD	Flow (simpleFoam) + conjugate heat transfer (chtMultiRegionSimpleFoam, frozenFlow)	OpenFOAM v1912
5 Reduced-order model	GRAIL-CHT: 2-D plate conduction + 1-D channel flow + glazing/radiation + rear stack	Python (own code)
6 Campaign	300 Latin-hypercube cases, Rev 4.2 dataset	GRAIL-CHT
7 Stage 3	ANN ensemble surrogate, GPR benchmark, SHAP	scikit-learn, shap
8 Stage 5	NSGA-II multi-objective optimisation + verification	pymoo + GRAIL-CHT
9 Stage 6	Virtual ISO 9806 test, efficiency curves, dynamic collector-tank model	GRAIL-CHT + Python
10 Stage 7	Annual simulation with PVGIS TMY; sizing; economics	Python
11 System models	Independent implementations	GNU Octave, Scilab Xcos, MATLAB/Simulink
12 V7	Alternating vs parallel at matched mean plate temperature	GRAIL-CHT
13 V5	Validation against published experimental work (16 papers screened)	HWB model, GRAIL-CHT
14 V1	First-pass patent/prior-art search	Google Patents, literature

2.3 Flags used in this report

Flag	Meaning
ENGINEERING_ASSUMPTION	A value chosen by the engineer without a datasheet or roadmap source. Must be stated as an assumption.
DESIGN_REVIEW_REQUIRED	A result or input that needs a design decision or better data before it is used in a design.
Deferred	Deliberately left out of scope (PCM latent heat, 3-D optics, full 3-D campaign, topology optimisation).

2.4 Rules followed

- Only the converging geometry is used. The old diverging geometry is never used.
- No data point was deleted or changed. Outliers were never removed.
- SI units; kelvin inside every T^4 term.

- No result was tuned to make alternating flow win.
- No experimental validation of the GRAIL collector is claimed.

3. Collector design and inputs

3.1 Description

One GRAIL module is a flat-plate collector 1.10 m long and 0.48 m wide. The absorber is two 1 mm aluminium sheets roll-bonded together, with 12 channels inflated between them. Each channel narrows from inlet to outlet. Above the absorber sit an air gap, a 10 mm honeycomb transparent-insulation layer, a second gap and the glass. Below it sit a 10 mm phase-change-material layer and a 20 mm vacuum insulation panel. In the alternating arrangement odd and even channels carry water in opposite directions; in the parallel arrangement all flow the same way.

3.2 Geometry

Quantity	Value	Source
Channel length L	1.100 m	Roadmap / CAD
Plate width W	0.480 m	Roadmap / CAD
Aperture per module	0.528 m ²	L × W
Number of channels	12	Roadmap
Channel pitch	40 mm	CAD
Upper / lower sheet thickness	1.0 / 1.0 mm	Roadmap
Inlet section area A _{in}	28.057 mm ²	Measured (Gate 1)
Outlet section area A _{out}	7.921 mm ²	Measured (Gate 1)
Inlet / outlet wetted perimeter	21.587 / 11.288 mm	Measured
Inlet / outlet hydraulic diameter	5.1989 / 2.8071 mm	Measured
Grading ratio G = Dh _{out} / Dh _{in}	0.539935	Measured
Grading exponent λ	1.0	Roadmap
Widest channel footprint	8.714 mm	Measured
Bridge (metal land) width	31.286 mm	pitch – footprint
Channel section	Lenticular: flat floor, root fillets, tangent dome	CAD

Table 3.1. Absorber geometry (as built, converging)

Axial grading law (hydraulic diameter along the flow, x from inlet):

$$Dh(x) = Dh_{in} \cdot [1 - (1 - G) \cdot (x / L)^{\lambda}], \quad \text{with } G = 0.539935 \text{ and } \lambda = 1.0$$

The ends do not depend on λ; λ shapes only the middle of the channel. The channel section is self-similar, so area scales with the square of the section scale.

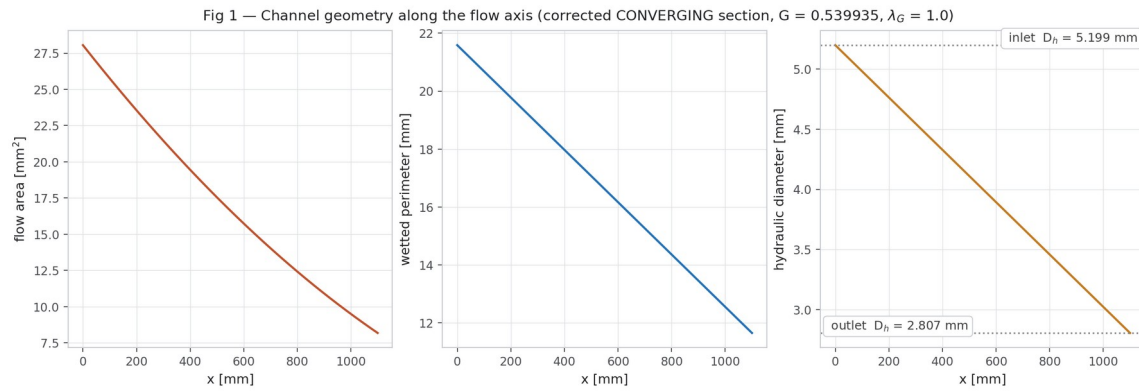


Figure 3.1. Absorber geometry (CAD)

Fig 2 — Channel section at three stations, measured on the CAD surface
the section SHRINKS along the flow

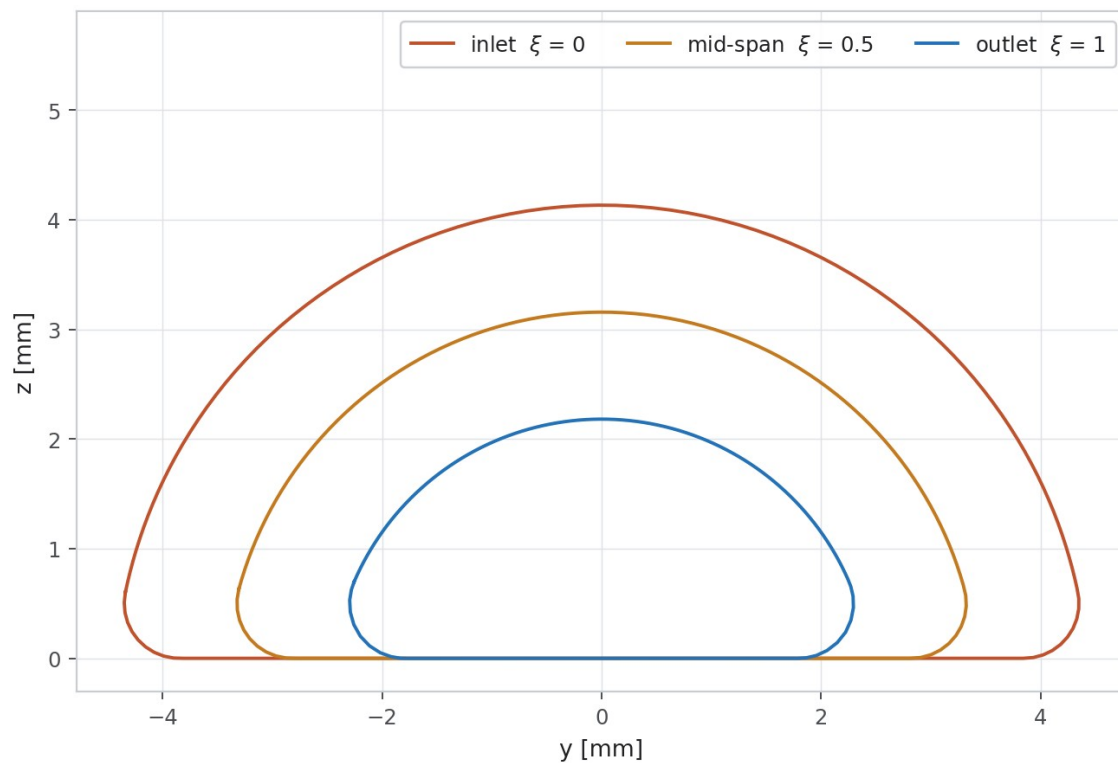


Figure 3.2. Channel cross-sections at inlet and outlet

3.3 Materials, optics and loss stack

Property	Value	Note
Absorber	AA1050 aluminium, $k = 229$ W/m-K	Roadmap
Absorber coating	$\alpha = 0.95$, $\varepsilon = 0.04$ (TiNOX-type selective)	Roadmap
Glazing	3.2 mm glass, $k = 1.05$ W/m-K, $\varepsilon = 0.88$; system $\tau = 0.833$	Roadmap
Transparent insulation (TIM)	10 mm honeycomb, $k = 0.075$ W/m-K, $\tau = 0.82$	Roadmap
Air gaps	Lower 7.45 mm, upper 4.30 mm, $k = 0.026$ W/m-K	CAD
Optical factor	$0.833 \times 0.82 \times 0.95 = 0.6489$	Absorbed flux = $0.6489 \times G_T$
PCM layer	10 mm, $k = 2.80$ W/m-K (sensible only in the models)	Roadmap; latent heat deferred

Property	Value	Note
Vacuum insulation panel (VIP)	20 mm, $k = 0.006 \text{ W/m}\cdot\text{K}$ (catalogue)	DESIGN_REVIEW_REQUIRED: measured in-collector 0.010–0.018
Rear outer film	$h_{\text{rear}} = 3.0 \text{ W/m}^2\text{K}$	ENGINEERING_ASSUMPTION
Water	$\rho \text{ 997}$, $c_p \text{ 4180 J/kg}\cdot\text{K}$, $\mu \text{ 8.55e-4 Pa}\cdot\text{s}$ (3-D); $k(T)$, $\mu(T)$ in GRAIL-CHT	—
Sky temperature	$T_{\text{sky}} = 0.0552 T_{\text{amb}}^{1.5}$	Swinbank
Wind coefficient	$h_{\text{wind}} = 5.7 + 3.8 v$	McAdams

Table 3.2. Materials and stack used in every model

3.4 CAD model identity

Property	Value
Document	Grail_Collector 2, version 3
Timeline features	948
Unhealthy features	0
Suppressed features	0
Occurrences	123
Bodies	145 (135 physical, 10 non-physical reference)
Non-solid bodies	0
Stray bodies at root	0
Sketches	285, every one with a closed profile
Joints	4, all healthy
External references	None
Design type / units	Parametric, millimetres
User parameters	131, no unresolved expression tokens
Materials in use	16, every body assigned

Table 3.3. CAD model verified state (Fusion, Grail_Collector 2 v3)

Source class	Count	What it means, and how far you may trust it
ROADMAP	73	The value is specified in the design roadmap, with the roadmap tag recorded in the comment. Quote freely, but cite the roadmap clause rather than the model.
MEASURED	9	Read back off the built geometry rather than typed in. These are results, not inputs, and they are the numbers to quote when describing what the model actually is.
DERIVED	3	Computed from other parameters by an expression. Correct by construction as long as its inputs are.
CHECKED	2	A hand calculation run against the model to test whether something invented is defensible. Weaker than MEASURED: it tests plausibility, not truth.
ENGINEERING_ASSUMPTION	23	No roadmap source. A physically reasonable value chosen so the model could be built. Never present one of these as a roadmap figure. 19 of them are actual dimensions.
Review and control	21	Flags, counters and records rather than dimensions, including the DESIGN_REVIEW register.

Table 3.4. Classes of the 131 CAD parameters (full register in Appendix A)

3.5 Mass and materials of the assembly

Group	Components	Bodies	Mass (kg)	Volume (mm ³)
00_FRAME	41	54	8.291	3,044,683
01_GLAZING	4	3	10.271	4,446,080
02_TIM	3	3	0.463	11,304,006
03_ABSORBER_ASSEMBLY	5	15	3.272	1,350,591
04_MANIFOLD	17	18	4.360	1,748,273
05_PCM_TRAY	10	9	6.095	5,939,571
06_INSULATION	6	5	2.209	11,794,743
07_BACKSHEET	2	1	1.921	710,007
08_SEALS	6	5	0.611	530,948
09_SENSING	9	8	0.030	21,712
10_MOUNTING	15	14	6.378	2,362,358
90_CFD_REFERENCE	3	3	7.491	3,046,916
90_CFD_STRIP_2CH_PERIODIC	1	3	1.767	225,105
90_CFD_STRIP_3CH_MASTER	1	4	2.650	337,626
TOTAL		145	55.810	

Table 3.5. Assembly groups

Material	Bodies	Mass (kg)
AA1050	19	17.385
Al6063T6	24	14.563
SolarGlass	2	9.491
RT55_EG	7	4.645
Steel	7	4.417
VIP_silica	1	2.090
Water300K	18	0.877
PMMA_thermo	1	0.780
EPDM	5	0.611
PC_honeycomb	1	0.456
Graphite	1	0.232
Aerogel	4	0.119
SS316	44	0.106
Electronics	8	0.030
Air	2	0.007
TiNOX	1	0.001

Table 3.6. Materials in the CAD assembly

4. Three-dimensional CFD

4.1 Solver and set-up

Item	Setting
Software	OpenFOAM v1912
Flow	simpleFoam, steady laminar, constant properties
Heat transfer	chtMultiRegionSimpleFoam with frozenFlow (flow held fixed, energy solved in fluid and solid)
Domain	2-channel periodic strip, 1100 × 80 mm, cut from the CAD plate; translational cyclic sides
Mesh	Structured hexahedra built from the CAD surfaces; axial NX 60/120/240, cross-section NU 48/64/96
Inlet	300 K, uniform velocity; 2.077072×10^{-4} kg/s per channel
Top boundary	$q'' - U_{\text{top}} (T - T_{\text{amb}})$ per plan area; $q'' = 519.1256 \text{ W/m}^2$, $U_{\text{top}} = 4.191176 \text{ W/m}^2\text{K}$ (benchmark ENGINEERING_ASSUMPTION)
Rear boundary	$h_{\text{rear}} (T - T_{\text{amb}})$, $3.0 \text{ W/m}^2\text{K}$; $T_{\text{amb}} = 298.15 \text{ K}$
Solid	AA1050, $k = 229 \text{ W/m-K}$
Convergence	All reported quantities change $\leq 2 \times 10^{-6}$ over the last 2000 iterations; energy balance 0.005 % (co-current), 0.035 % (alternating)

4.2 Problems found and fixed

- The buoyant p_rgh solver diverged on the thin O-grid cells. Repair: solve the flow with simpleFoam, then the conjugate energy equation with frozenFlow — the same steady solution, verified (zero pressure/momentum solves, flux unchanged to 1.5×10^{-8}).
- Solar input had been applied per curved area (+12.2 %); corrected to plan area.
- The meshed inlet carries 0.30 % less flow than nominal; the reduced-order model is run at the 3-D flow for comparison.
- The earlier Nusselt number 2.92 came from a fluid-only case with heat flux uniform around the perimeter (H2). The real aluminium wall is nearly isothermal around the perimeter (H1). Corrected closure: 4.48.

4.3 Grid study — axial and cross-section

grid	eta_co	eta_alt	std_co_K	std_alt_K	d_eta_pct	d_std_pct	d_spread_pct
NX60_NU64	0.51484	0.42212	5.7713	5.2543	-18.01	-8.96	-2.52
NX120_NU64	0.51597	0.41817	5.7907	5.5083	-18.95	-4.88	3.4
NX240_NU64	0.51594	0.4184	5.7842	5.4737	-18.91	-5.37	3.07
NX120_NU48	0.51605	0.41786	5.7921	5.5263	-19.03	-4.59	3.42
NX120_NU96	0.51586	0.41849	5.7888	5.481	-18.88	-5.32	3.2

Table 4.1. 3-D conjugate grid study (η , plate std, differences alternating vs co-current)

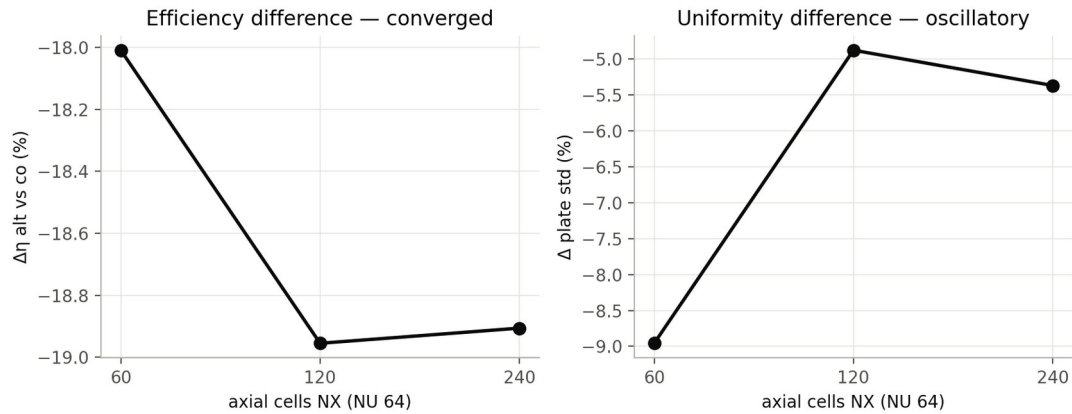


Figure 4.1. 3-D axial grid study

4.4 Nusselt closure

level	fluid_cells	Nu_developed_mean
NU48	80640	4.6455
NU64	138240	4.6022
NU96	322560	4.5557
extrapolated ($p=1.12$, GCI 2.1%)		4.48
H1 reference semicircular duct		4.088
campaign Rev1-3 (superseded)		2.9238

Table 4.2. Developed Nusselt number: grid levels, extrapolation, references

The developed Nu is flow-independent (4.60 at 1.0, 2.49 and 4.5 g/s) and axially converged (4.592 / 4.602 / 4.602). Reference: H1 semicircular duct 4.088 (Erdoğan & Imrak 2005; Shah & London table in Hesselgreaves 2001). The conjugate value lies above H1 because the section is not an exact semicircle and the wall has finite conduction.

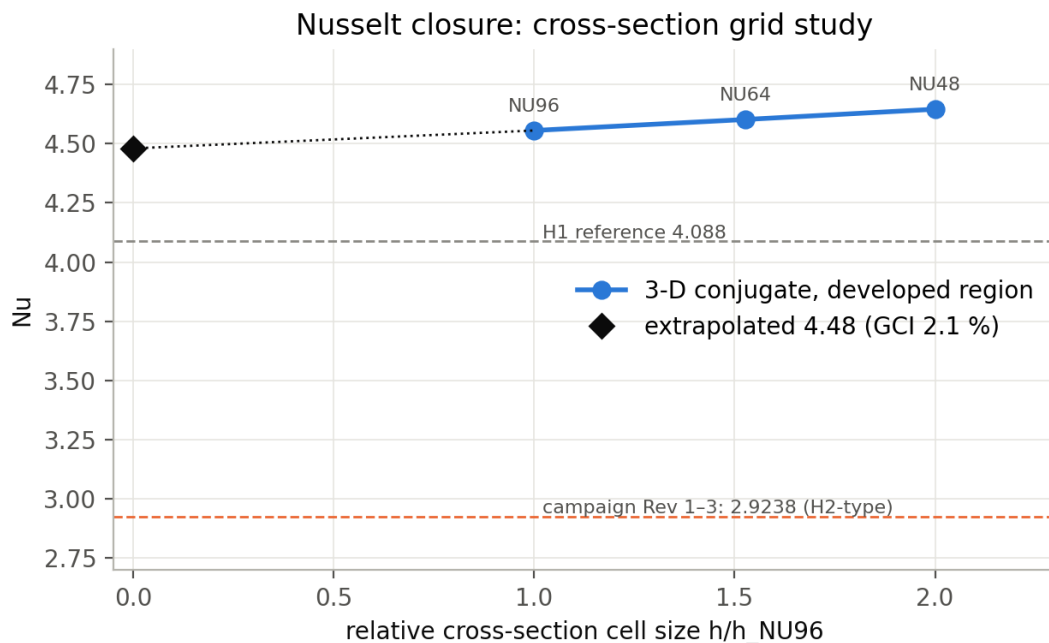


Figure 4.2. Cross-section grid study of Nu

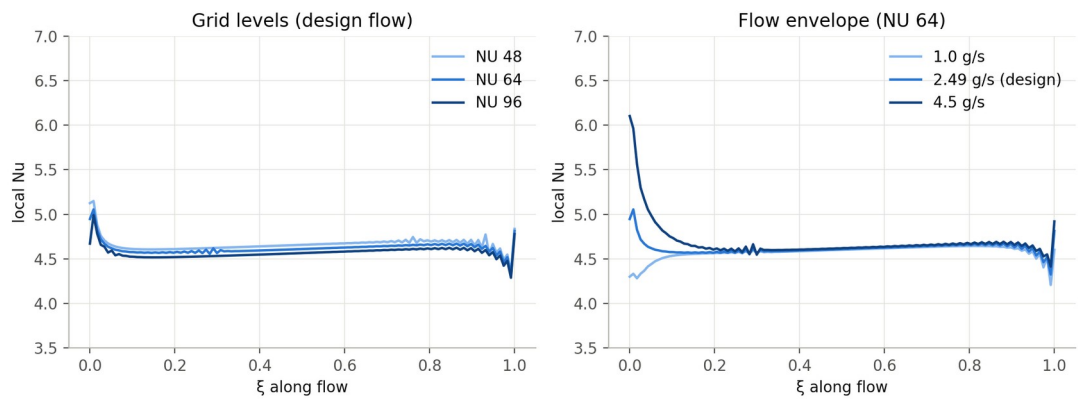


Figure 4.3. Local Nu along the channel

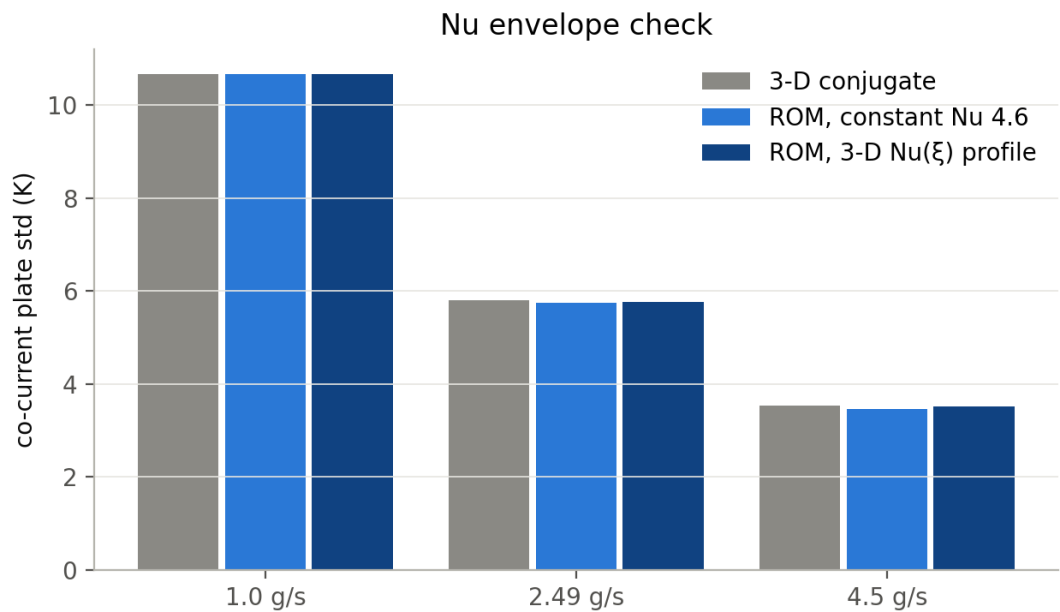


Figure 4.4. Nu envelope over the flow range

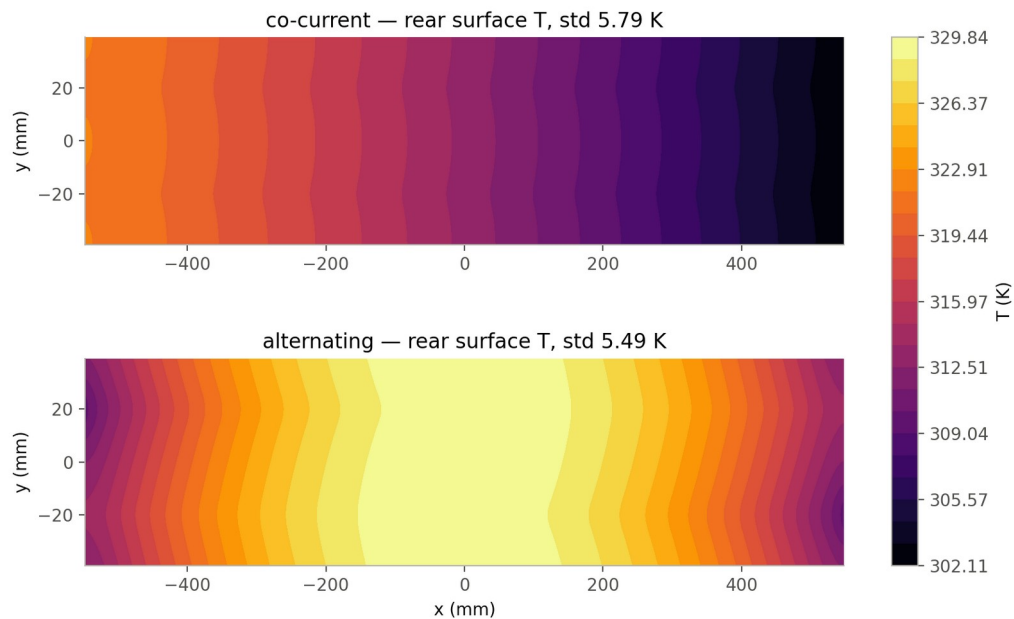


Figure 4.5. 3-D plate temperature maps, co-current and alternating

4.5 Mesh and computational domain

Meshes were generated directly from the CAD surfaces. The channel section is swept along the graded length so every cell follows the converging wall. These figures show geometry and mesh only and remain valid.

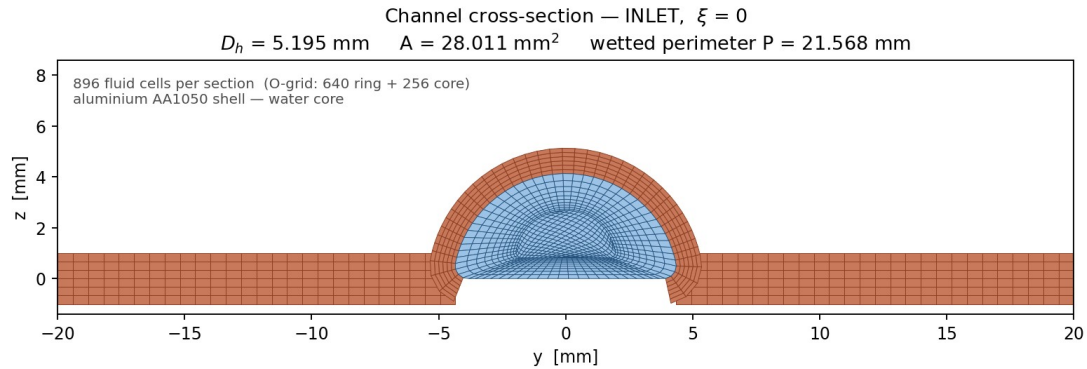


Figure 4.6. Channel section mesh at the inlet

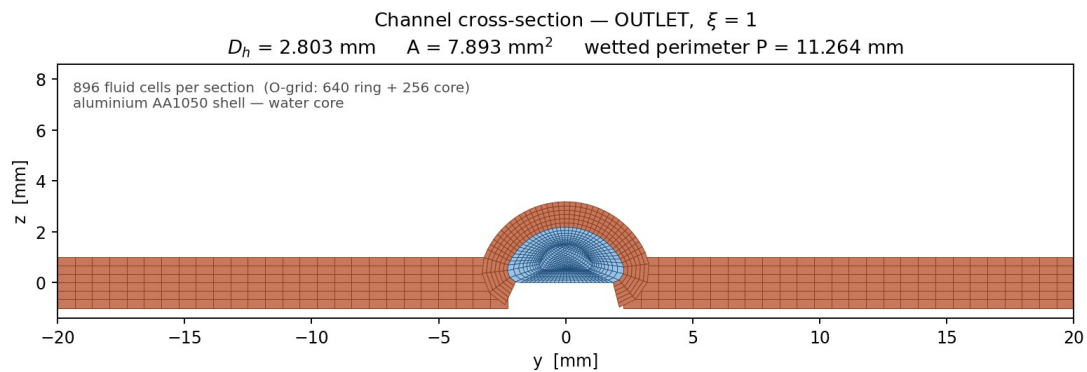


Figure 4.7. Channel section mesh at the outlet

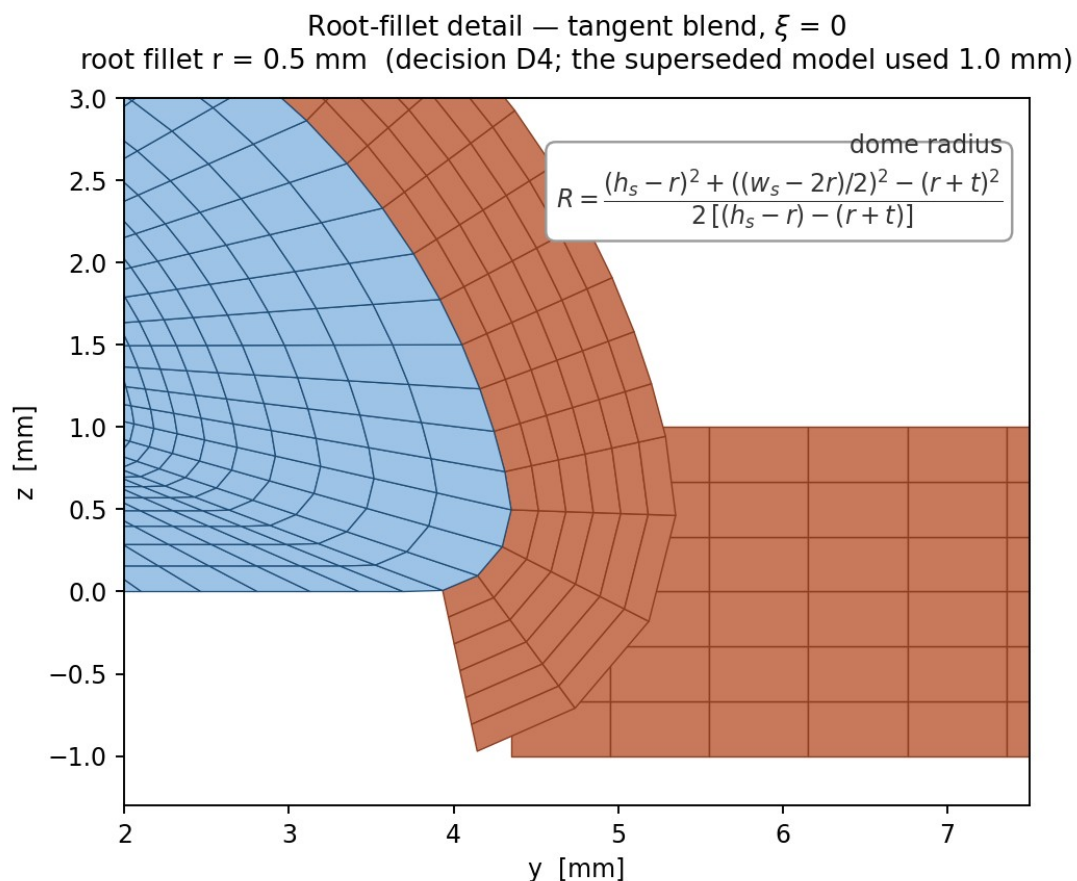
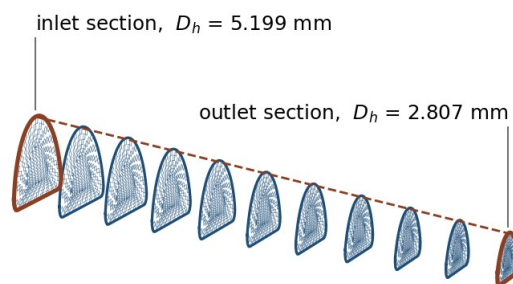


Figure 4.8. Root-fillet detail of the section mesh

Structured mesh swept along one channel (every 10th station shown) — the section SHRINKS



channel length 1100 mm (x); flow from inlet (left) to outlet (right); $G = 0.5399$

Figure 4.9. Swept structured mesh along the graded channel

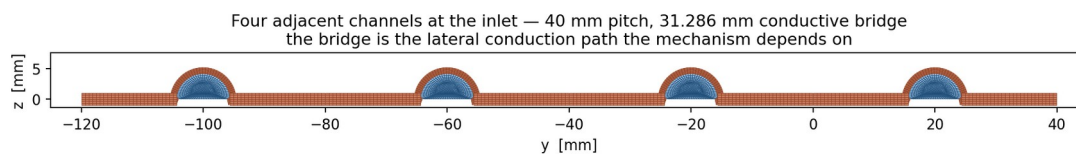


Figure 4.10. Multi-channel strip mesh

Fig M6 — Complete collector assembly, surface mesh

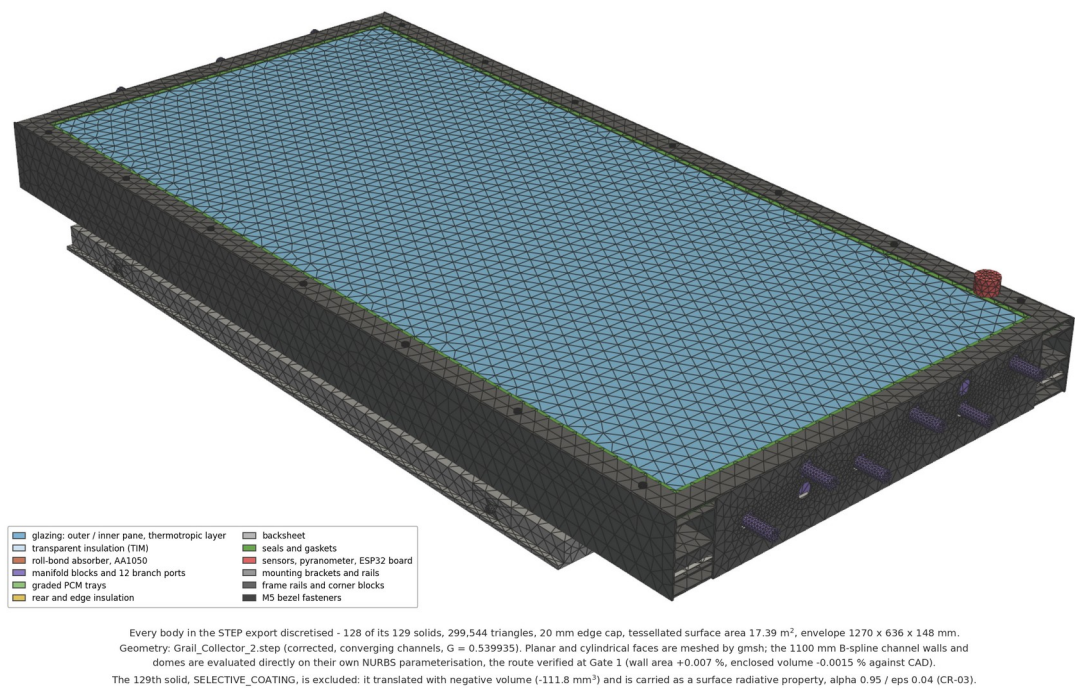


Figure 4.11. Assembly surface mesh

Fig M7 — The CFD mesh in position inside the assembly

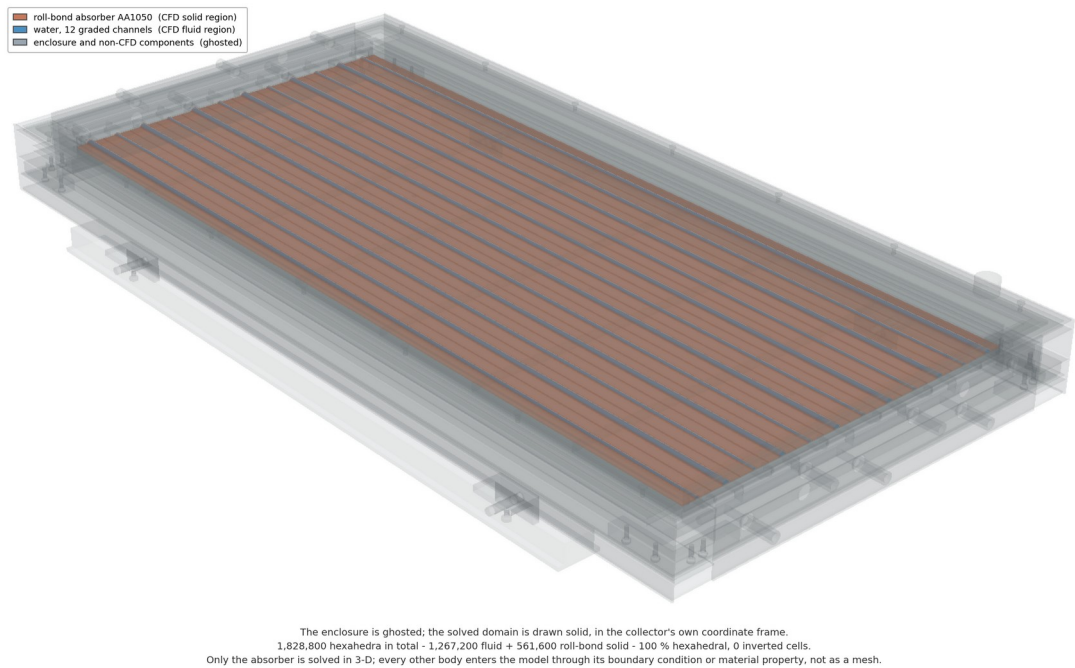


Figure 4.12. CFD domain located in the assembly

Fig M8 — Cut-away through the CFD mesh

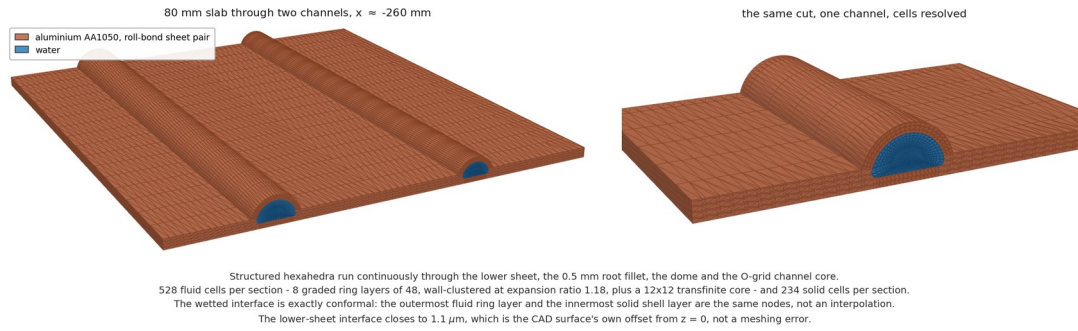


Figure 4.13. Cut-away of the collector stack

Fig M9 — CFD mesh, inlet end of the absorber

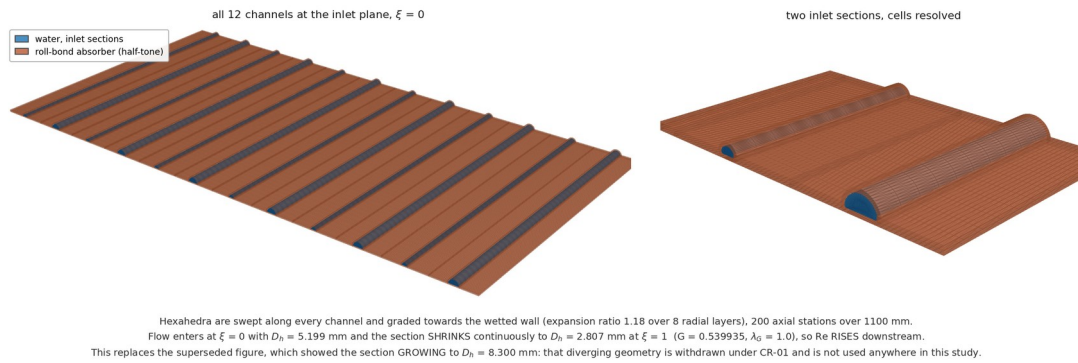


Figure 4.14. Inlet end of the absorber

Fig M10 — Full 12-channel absorber, complete CFD mesh

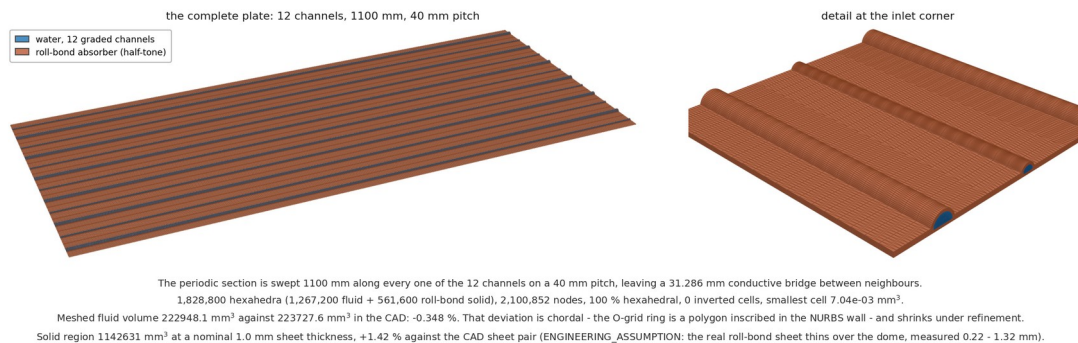
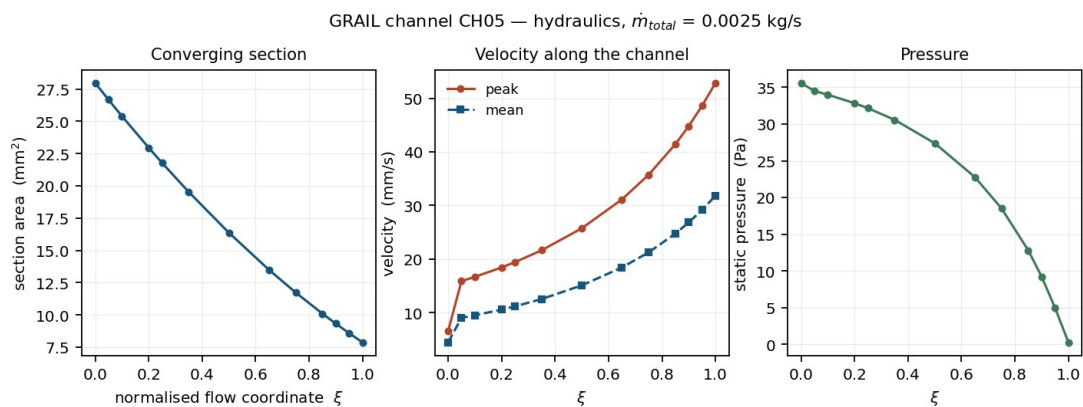
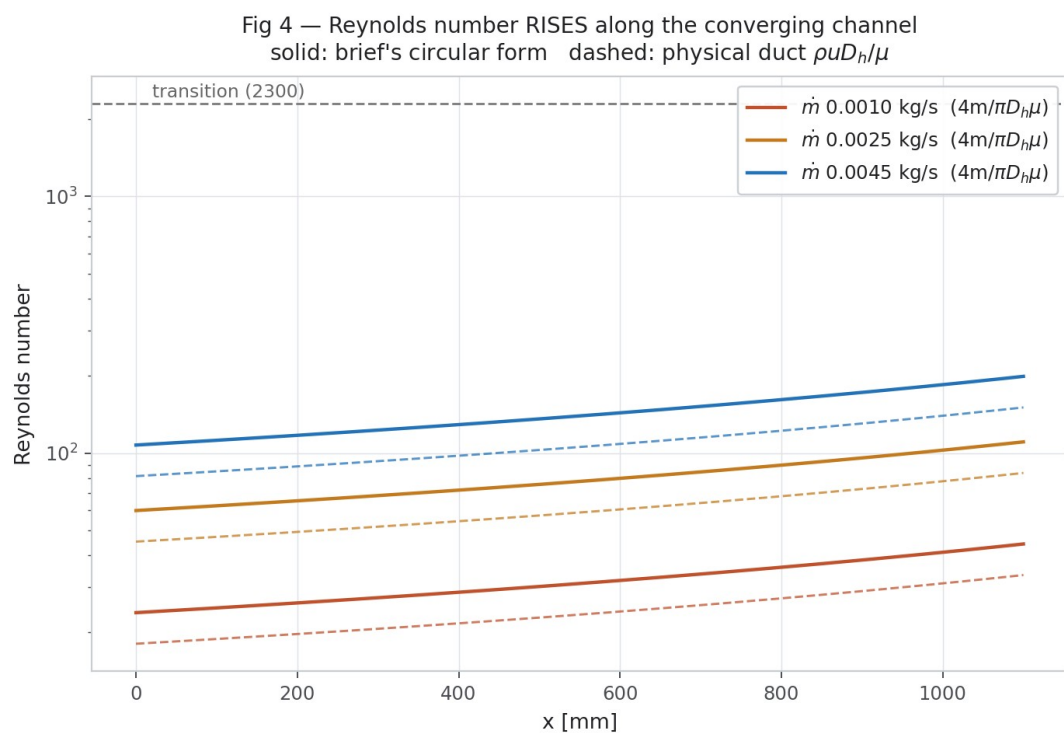
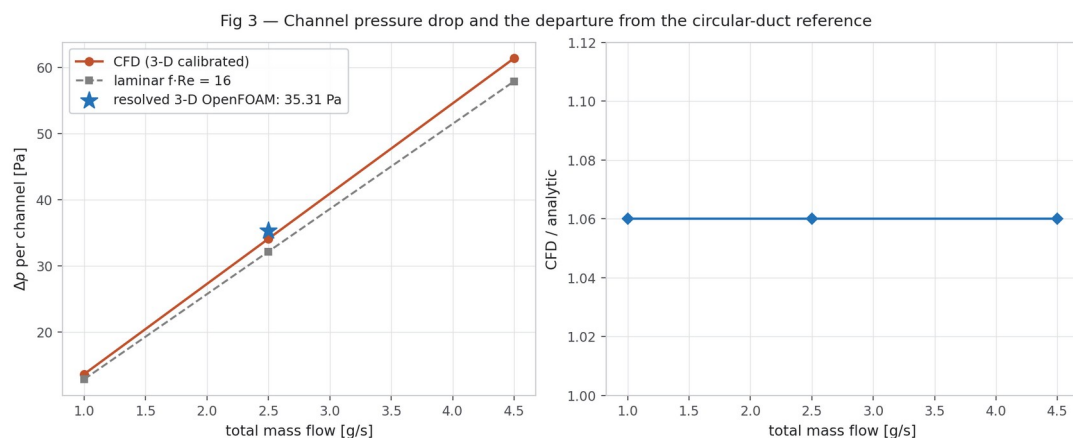


Figure 4.15. Full-plate mesh

4.6 Channel hydraulics (flow field)

The flow field does not depend on the thermal closure (constant properties), so the Rev 2 hydraulic results remain valid: laminar flow, pressure drop rising as the channel converges.



GRAIL CH05 — velocity magnitude, common scale, sections from the channel's own inlet

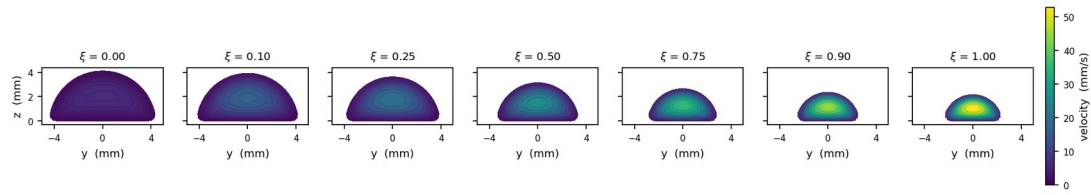
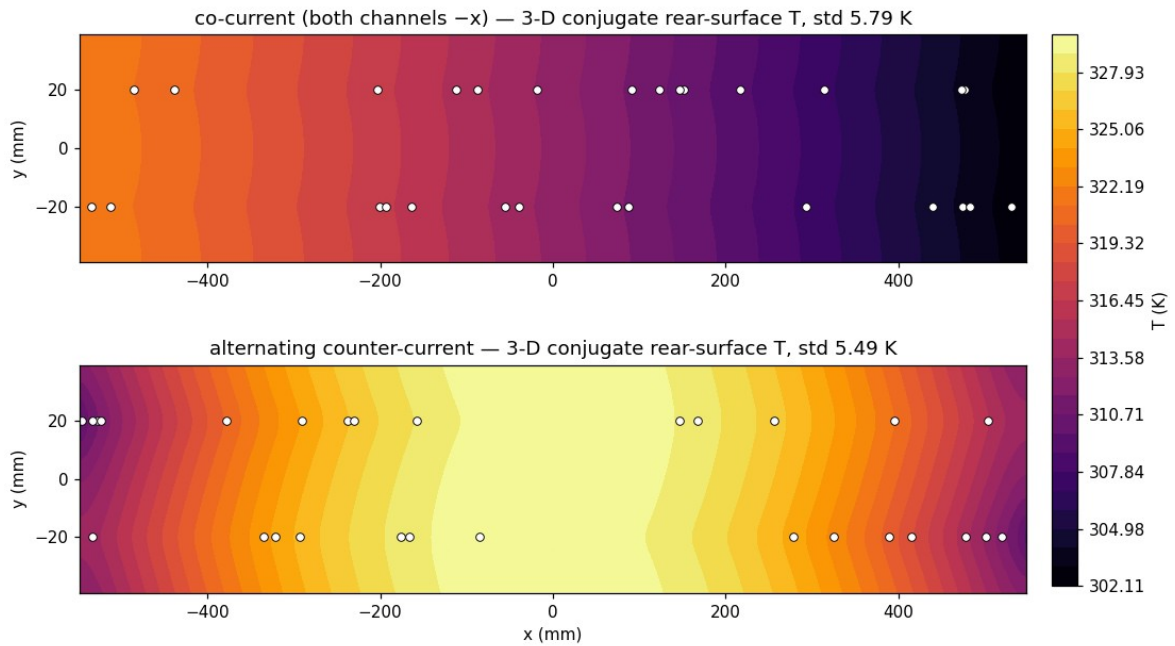


Figure 4.19. Velocity in cross-sections along the channel

4.7 Converged 3-D conjugate solution (video frames)

GRAIL Collector — converged 3-D conjugate CFD (OpenFOAM, NX 240, design flow). Tracers: bulk velocity, not particle paths.



tracer time 190 s

Figure 4.20. Converged rear-surface temperature, co-current vs alternating (NX 240, design flow)

5. Reduced-order conjugate model (GRAIL-CHT)

5.1 What it solves

- Plate: 2-D steady conduction in the aluminium sheets (upper + lower) on a 110×120 and 220×120 grid; the two grids are combined by Richardson extrapolation $2 f(220) - f(110)$.
- Channels: 1-D energy balance per channel, $\dot{m} c_p dT_f/dx = h P (T_{\text{wall}} - T_f)$, with $h = \text{Nu } k / D_h$, $\text{Nu} = 4.48$, $k(T)$ of water.
- Top loss: plate $\rightarrow$ glazing by conduction through the TIM and gaps (series resistance) plus radiation $\sigma(T_p^4 - T_g^4)/(1/\epsilon_p + 1/\epsilon_g - 1)$; glazing $\rightarrow$ ambient by wind convection and sky radiation $\epsilon_g \sigma(T_g^4 - T_{\text{sky}}^4)$. Glass temperature solved by Newton iteration at every cell. Temperatures in K.
- Rear loss: series resistance PCM + VIP + outer film, $U_{\text{rear}} = 1/(t_{\text{pcm}}/k_{\text{pcm}} + t_{\text{vip}}/k_{\text{vip}} + 1/h_{\text{rear}}) = 0.272 \text{ W/m}^2\text{K}$.
- Absorbed flux: $q_{\text{abs}} = 0.6489 G_T$ (uniform).
- Pressure drop: laminar integral along the graded channel with a calibration factor from the 3-D CFD.
- Convergence: outer-iteration residual tolerance 1×10^{-6} ; acceptance $|\text{energy error}| < 0.5 \%$ and $250 \text{ K} < T_{\text{plate}} < 500 \text{ K}$.

5.2 Checks of the model

Check	Result
GRAIL-CHT vs 3-D conjugate CFD, co-current	η within 0.01 %, plate std within 0.6 %
GRAIL-CHT vs 3-D, alternating	Overstates the uniformity benefit by ≈ 2.0 points and understates the efficiency penalty by ≈ 0.5 points (local Nu undefined where heat flow reverses) — carried as uncertainty
GRAIL-CHT with Nu 4.6 vs 3-D at NX 60	η 0.2 %, std 0.5 %, arrangement difference 0.1 points
Hottel-Whillier-Bliss (Duffie & Beckman) vs GRAIL-CHT	+0.22 % (co-current), U_L taken from the solver, not fitted
Axial convergence of GRAIL-CHT	Observed order $p = 1.01$; Richardson used
Energy balance	Every campaign case $< 0.5 \%$; measured range -0.00015 to $+0.00014 \%$

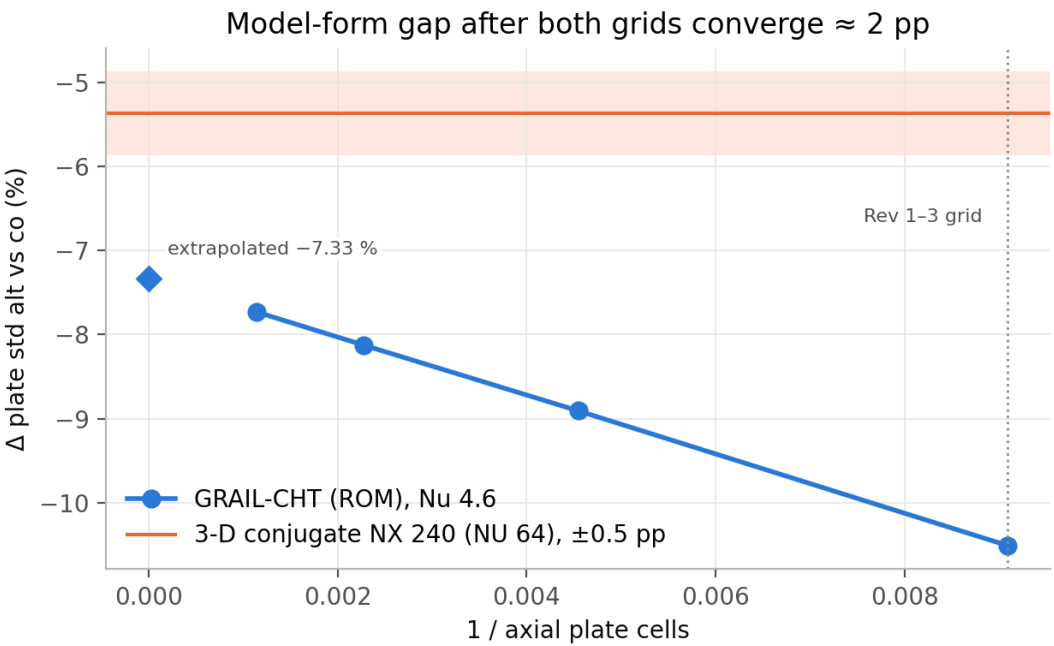


Figure 5.1. GRAIL-CHT axial convergence against the 3-D result

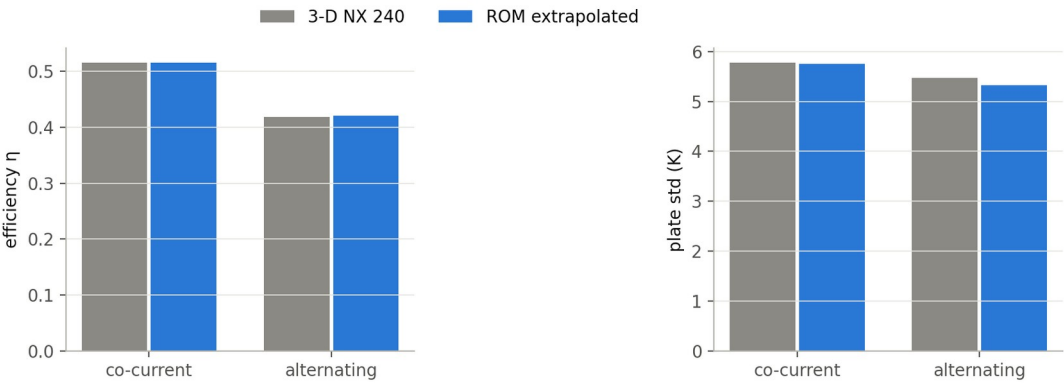


Figure 5.2. GRAIL-CHT vs 3-D CFD

6. Simulation campaign and dataset

6.1 Design of the campaign

Input	Range	Sampling
Irradiance G _T	400–1000 W/m ²	Latin hypercube
Inlet temperature T _{in}	288–333 K	Latin hypercube
Ambient temperature T _{amb}	283–313 K	Latin hypercube
Wind speed	0–5 m/s	Latin hypercube
Total mass flow	1.0–4.5 g/s (Re _{in} 22–151, laminar)	Latin hypercube
Bridge width	20–37 mm	Latin hypercube
Grading ratio G	0.35–1.0	Latin hypercube
Grading exponent λ	0.5–2.0	Latin hypercube
Arrangement	150 alternating + 150 parallel	Two independent draws (not paired)

Acceptance gates: converged to 10^{-6} , |energy error| < 0.5 %, $250\text{ K} < T_{\text{plate}} < 500\text{ K}$. Result: 300 / 300 accepted; no row deleted. The two groups are statistically identical in every input (Kolmogorov–Smirnov $p = 1.00$), so all comparisons are group-wise. ALT_n and PAR_n must never be paired.

6.2 Dataset summary (Rev 4.2, 300 rows, 59 columns)

Column	All: min / mean / median / max	Alternating: min / mean / median / max	Parallel: min / mean / median / max
Irradiance G _T (W/m ²)	400.1 / 700.0 / 701.0 / 999.5	400.1 / 700.0 / 700.0 / 998.9	403.4 / 700.1 / 701.1 / 999.5
Inlet T (K)	288.09 / 310.51 / 310.44 / 332.98	288.27 / 310.51 / 310.44 / 332.98	288.09 / 310.51 / 310.41 / 332.89
Ambient T (K)	283.04 / 298.00 / 298.02 / 312.87	283.04 / 298.00 / 298.01 / 312.83	283.04 / 298.00 / 298.02 / 312.87
Wind (m/s)	0.01 / 2.50 / 2.49 / 4.99	0.02 / 2.50 / 2.49 / 4.99	0.01 / 2.50 / 2.50 / 4.97
Total flow (kg/s)	0.00101 / 0.00275 / 0.00274 / 0.00449	0.00101 / 0.00275 / 0.00275 / 0.00449	0.00102 / 0.00275 / 0.00274 / 0.00448
Bridge (mm)	20.02 / 28.50 / 28.51 / 36.99	20.08 / 28.51 / 28.51 / 36.99	20.02 / 28.50 / 28.47 / 36.93
Grading ratio G	0.351 / 0.675 / 0.674 / 0.998	0.353 / 0.675 / 0.674 / 0.998	0.351 / 0.675 / 0.674 / 0.998
Exponent λ	0.503 / 1.250 / 1.247 / 1.999	0.503 / 1.250 / 1.249 / 1.999	0.505 / 1.250 / 1.246 / 1.991
Re at inlet	21.8 / 73.4 / 68.4 / 151.3	21.8 / 72.8 / 69.1 / 151.3	25.1 / 74.0 / 67.4 / 147.8
Channel Δp (Pa)	2.10 / 22.03 / 14.52 / 158.61	2.11 / 21.88 / 15.44 / 120.75	2.10 / 22.18 / 14.28 / 158.61
Outlet T (K)	298.17 / 330.76 / 331.87 / 373.41	298.34 / 329.53 / 329.82 / 370.28	298.17 / 331.99 / 333.82 / 373.41
Fluid ΔT (K)	6.18 / 20.25 / 17.28 / 58.27	6.18 / 19.02 / 16.41 / 46.85	6.82 / 21.48 / 17.82 / 58.27
Useful heat (W)	90.1 / 201.2 / 200.3 / 327.6	90.1 / 193.4 / 190.6 / 327.6	101.6 / 209.0 / 213.8 / 318.6
Efficiency η	0.3415 / 0.5414 / 0.5446 / 0.6732	0.3415 / 0.5209 / 0.5264 / 0.6732	0.4260 / 0.5620 / 0.5626 / 0.6653
Mean plate T (K)	295.48 / 330.32 / 330.96 / 394.41	299.48 / 337.27 / 337.02 / 394.41	295.48 / 323.38 / 324.54 / 356.50

Column	All: min / mean / median / max	Alternating: min / mean / median / max	Parallel: min / mean / median / max
Max plate T (K)	298.95 / 338.79 / 338.83 / 412.63	301.97 / 344.76 / 342.93 / 412.63	298.95 / 332.82 / 334.39 / 374.02
Plate std (K)	1.513 / 6.239 / 4.991 / 19.468	1.513 / 6.464 / 4.960 / 19.468	1.915 / 6.013 / 5.013 / 16.130
Plate spread (K)	6.33 / 22.53 / 17.82 / 74.19	6.38 / 25.29 / 19.61 / 74.19	6.33 / 19.77 / 16.57 / 52.84
Energy error (%)	-0.00015 / 0.00003 / -0.00001 / 0.00014	-0.00015 / 0.00007 / 0.00008 / 0.00014	-0.00004 / -0.00002 / -0.00002 / 0.00002
Outer iterations	93 / 401 / 274 / 1541	217 / 647 / 530 / 1541	93 / 156 / 135 / 345

Table 6.1. Statistics of the campaign inputs and outputs

case_id	arrangement	G_T_W_m2	T_in_K	T_amb_K	v_wind_ms	mdot_total_kg_s	bridge_m	g_ratio	lambda_G	T_out_K	eta	T_plate_mean_K	plate_std_K	dp_channel_Pa
ALT_000	alternating	833.0998	292.0654	289.5495	3.0065	0.0038	23.8296	0.9382	1.9269	308.5433	0.5874	311.4657	4.0634	14.3053
ALT_001	alternating	484.5321	326.7904	287.0671	3.3326	0.0028	20.2109	0.6723	0.5626	335.8182	0.4134	338.7247	2.8683	15.5155
ALT_002	alternating	419.9503	290.3044	297.2822	3.0624	0.0023	28.3827	0.8241	1.4522	303.7276	0.5787	308.8907	4.5555	11.973
ALT_003	alternating	882.5161	309.9996	296.5432	2.4391	0.0024	24.9194	0.6607	1.528	334.6626	0.5288	344.29	8.5077	11.263
ALT_004	alternating	475.1487	302.6358	311.7342	4.9431	0.0015	27.5294	0.7969	1.9054	324.2092	0.5295	336.7688	8.7875	5.4925
PAR_000	parallel	665.0261	289.923	289.9617	1.8451	0.0032	29.2248	0.5216	1.2415	305.7192	0.6082	299.6618	4.4378	44.9145
PAR_001	parallel	913.2767	295.4192	310.78	1.4079	0.0032	20.2386	0.4153	1.2974	318.9878	0.6455	309.8163	6.6274	59.4854
PAR_002	parallel	550.2224	294.4523	306.8768	0.7646	0.0015	34.4898	0.5823	0.8952	322.9835	0.6211	311.1776	7.9585	15.0681
PAR_003	parallel	990.3464	323.6548	283.0374	0.1349	0.0039	20.7219	0.8877	0.8737	340.66	0.5363	334.2487	4.7928	9.8807
PAR_004	parallel	750.5296	288.091	289.1408	4.707	0.003	27.5461	0.7223	0.5767	307.1957	0.6129	299.7744	5.3586	26.1354

Table 6.2. Sample rows (first five of each arrangement); full file: GRAIL_CFD_dataset_FINAL_rev4.2.csv

6.3 Alternating vs parallel at equal inputs

quantity	parallel	alternating	difference	mann_whitney_p
eta_mean	0.562	0.5209	-7.3%	2e-8
plate_std_median_K	5.013	4.96	-1.1%	0.76
plate_spread_median_K	16.57	19.61	+18.4%	0.001
T_plate_mean_diff_K			+13.9	2e-10
P90_diff_K			+12.6	2e-6

Table 6.3. Group comparison (150 vs 150, Mann-Whitney)

total_flow_g_s	n_alt	n_par	d_eta_mean_pct	d_std_median_pct	p_std
1.0-2.2	50	50	-12.7	16.8	0.033
2.2-3.3	50	50	-6.0	7.9	0.14
3.3-4.5	50	50	-3.5	-13.4	0.017

Table 6.4. Differences by flow tercile

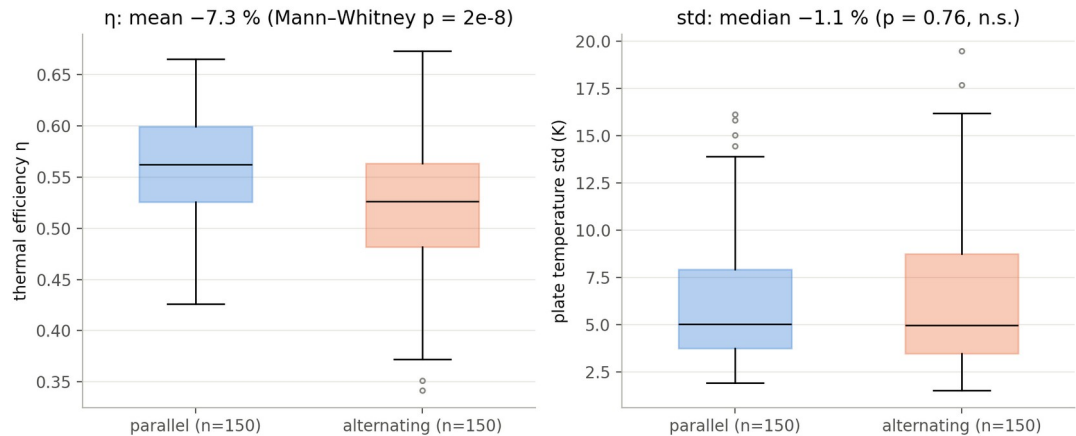


Figure 6.1. Arrangement groups

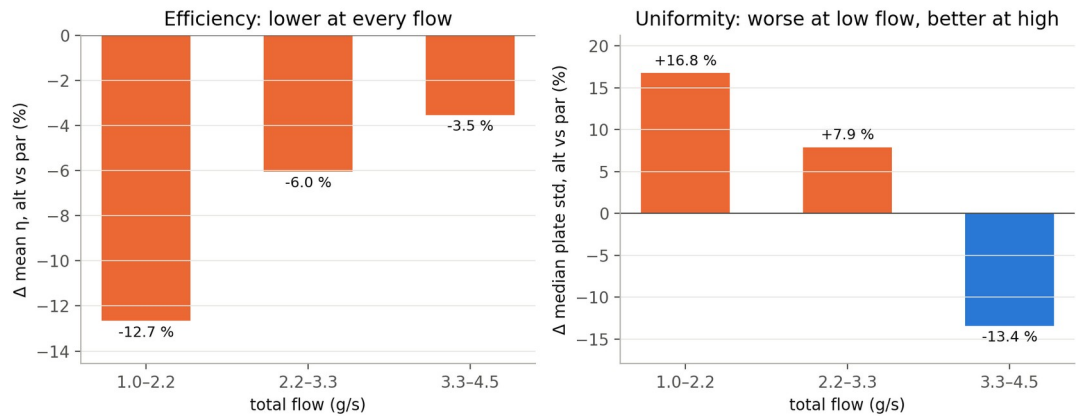


Figure 6.2. Arrangement effect by flow

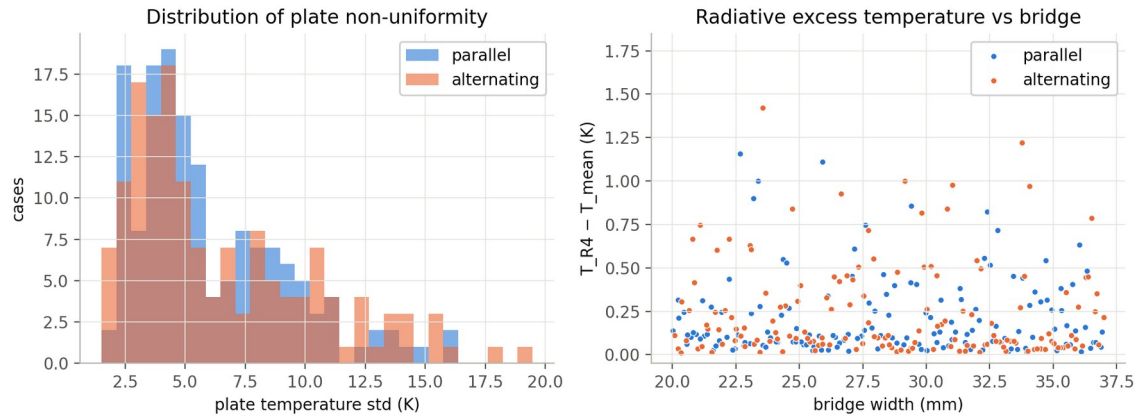


Figure 6.3. Plate-std distributions and radiative excess vs bridge width

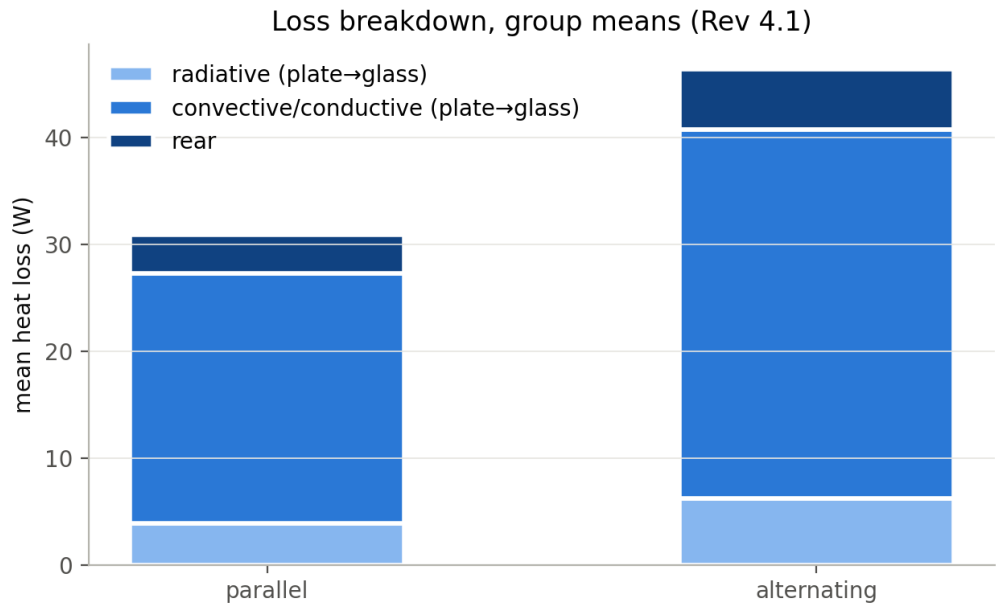


Figure 6.4. Loss breakdown (group means)

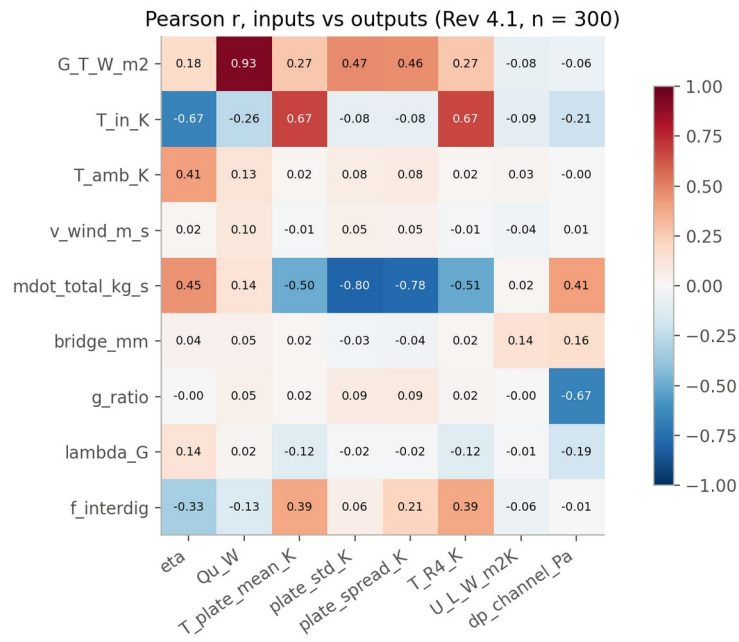


Figure 6.5. Correlation between inputs and outputs

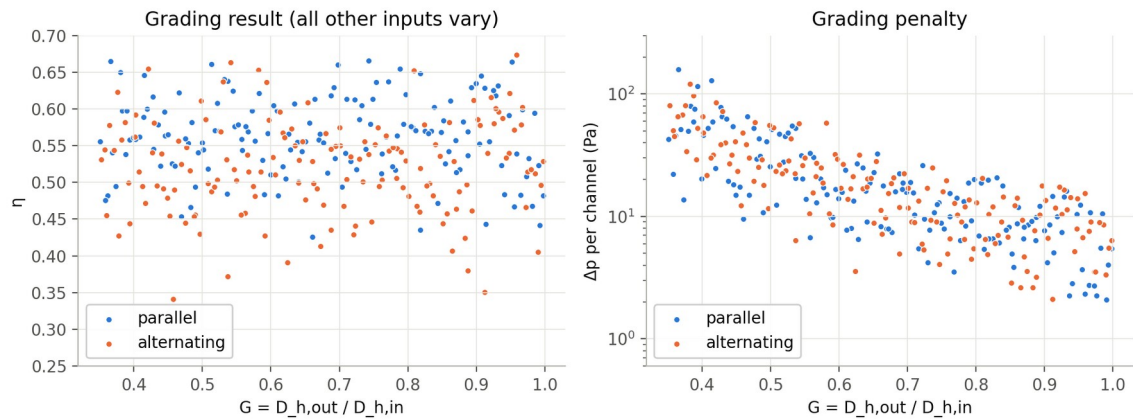


Figure 6.6. Efficiency and pressure drop against grading ratio

7. Mechanism, uncertainty and matched-temperature study (V7)

7.1 Why alternating flow loses efficiency

quantity	value
absorbed_per_channel_W	24.8
returned_per_channel_W	9.94
returned_fraction_pct	40.1
length_with_negative_qprime_pct	41
net_W	14.86
mdot_cp_dT_W	14.85

Table 7.1. Heat recirculation in alternating flow (3-D, design point)

Counter-flowing neighbours act as a counter-flow heat exchanger through the metal. About 40 % of the heat a channel absorbs is handed back to the plate over 41 % of its length. The fluid peaks mid-channel (328.6 K) and leaves cooler (317.1 K). The plate runs hotter and loses more heat to the front.

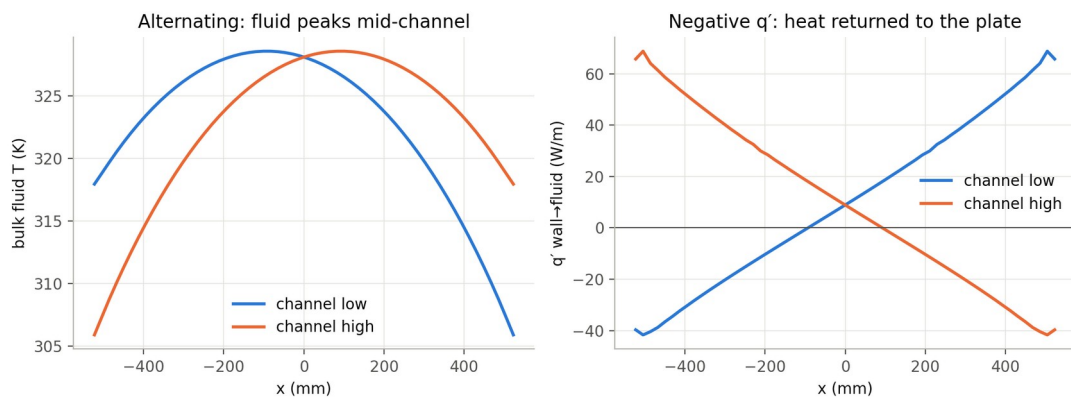


Figure 7.1. Heat recirculation along the channel, alternating

7.2 Uncertainty budget

source	d_eta	d_std
3D discretisation	±0.1 pp	±1.0 pp
ROM axial after Richardson	<0.1 pp	~0.4 pp
ROM model form (alternating)	0.5 pp	2.0 pp
Nu band 4.46-4.65	~0.1 pp	~1.1 pp
entry Nu at 4.5 g/s	<0.1%	<=2.1% of std
glazing/radiation model	not quantified	not quantified

Table 7.2. Uncertainty budget of the campaign results

7.3 V7 — matched mean plate temperature

Question: at the same mean plate temperature, does alternating flow reduce $R4 = \text{mean}(T^4)$ and raise efficiency?
Method: as-built geometry, 2.75 g/s, G 800 W/m², T_{amb} 303.15 K, wind 3 m/s. For each target the inlet temperature was found by a secant search until the plate mean matched within 0.001 K. All runs converged with energy error < 0.5 %.

Target plate mean (K)	Inlet needed alt / par (K)	η alt / par	$\Delta\eta$ (points)	$\Delta R4$ (%)	ΔT_{R4} (K)	Plate std alt / par (K)	Δ std (%)
320	291.3 / 306.7	0.59889 / 0.59890	-0.0012	0.036	0.029	6.68 / 6.17	+8.3
330	302.4 / 317.3	0.57245 / 0.57246	-0.0013	0.038	0.032	6.49 / 5.90	+9.9
340	313.5 / 328.0	0.54567 / 0.54568	-0.0014	0.037	0.032	6.26 / 5.63	+11.2

Table 7.3. V7 results

- Efficiency is the same (difference 0.001 points).
- R4 is not reduced; it rises 0.04 %.
- Alternating is 8–11 % less uniform at the design flow.
- Alternating needs an inlet ≈ 15 K cooler for the same plate temperature. That is why it is less efficient at equal inputs.

8. Stage 3 — ANN surrogate

- Inputs (9): G_T , T_{in} , T_{amb} , wind, total flow, bridge, grading ratio, grading exponent, arrangement flag.
- Targets: η , plate std, $\ln(\Delta p)$, mean plate temperature.
- Split: 240 training / 60 test, stratified by arrangement, seed 42. Test set never used in tuning.
- Tuning: 5-fold cross-validation over 6 architectures $\times$ 5 L2 penalties; tanh; L-BFGS; standardised data.
- Final: ensemble of 10 networks; baselines: linear regression and Gaussian-process regression.

Target	Architecture, α	CV R^2	ANN test R^2	ANN MAPE %	GPR test R^2	Linear test R^2
eta	[64, 32], 0.01	0.9994	0.9997	0.14	0.9999	0.8897
plate_std_K	[32, 32], 0.01	0.9993	0.9996	0.53	1.0000	0.8519
log_dp_Pa	[64, 32], 0.01	0.9992	0.9998	0.92	1.0000	0.9747
T_plate_mean_K	[64, 32], 0.01	0.9992	0.9994	0.06	1.0000	0.8813

Table 8.1. Surrogate accuracy on the 60 held-out cases

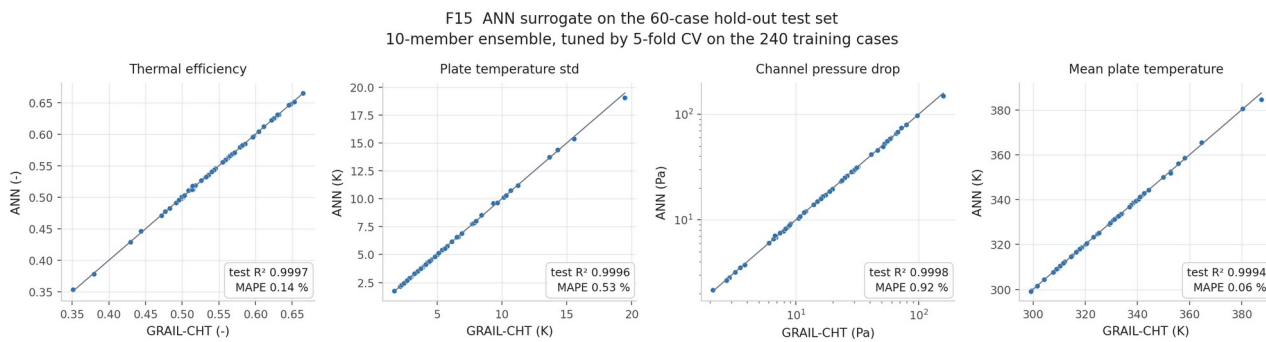


Figure 8.1. Parity plots, test set

SHAP: efficiency is driven by T_{in} , T_{amb} , flow and arrangement; the three geometric inputs are about two orders of magnitude weaker. Plate uniformity is driven by flow and irradiance; pressure drop by grading ratio and flow.

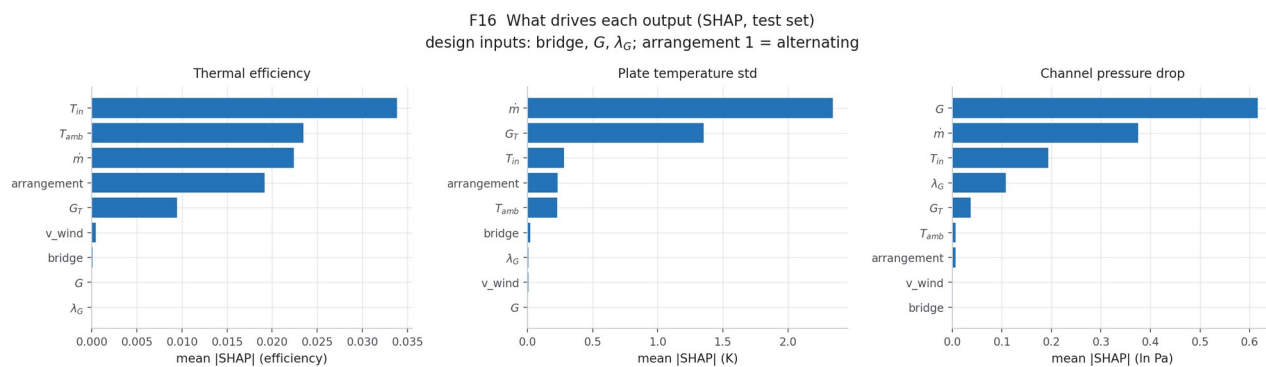


Figure 8.2. SHAP importance

9. Stage 5 — NSGA-II optimisation

- Variables: flow 1.0–4.5 g/s, bridge 20–37 mm, G 0.35–1.0, λ 0.5–2.0; each arrangement separately.
- Objectives: maximise η , minimise plate std, minimise pumping power.
- Reference point (ENGINEERING_ASSUMPTION, Berhampur clear-sky noon): G 800 W/m², T_{in} 313.15 K, T_{amb} 303.15 K, wind 3 m/s.
- Population 200, 250 generations, seed 7.

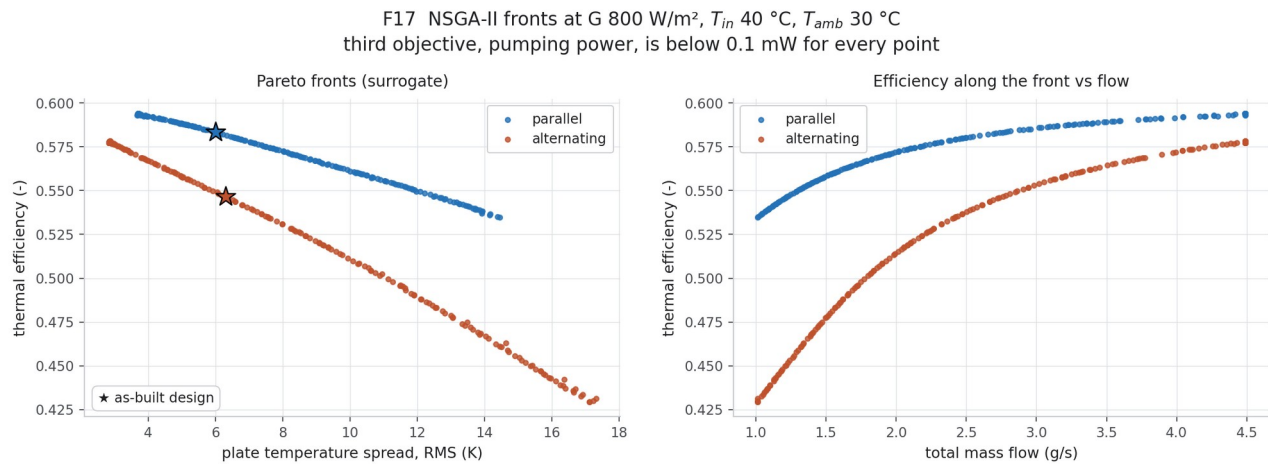


Figure 9.1. Pareto fronts with GRAIL-CHT verification points

Arrangement	Pick	Flow g/s	Bridge mm	G	λ	η ANN	η CHT	std ANN K	std CHT K	Δp err %
alternating	max_eta	4.49	31.9	0.998	0.50	0.5783	0.5784	2.87	2.83	5.8
alternating	min_std	4.49	37.0	0.708	1.35	0.5773	0.5782	2.83	2.87	0.0
alternating	knee	3.28	31.0	0.997	1.21	0.5599	0.5601	4.80	4.82	-0.5
parallel	max_eta	4.49	33.9	0.998	0.50	0.5940	0.5943	3.71	3.76	5.3
parallel	min_std	4.49	37.0	0.998	2.00	0.5931	0.5944	3.68	3.76	-0.3
parallel	knee	2.94	27.6	0.998	1.40	0.5852	0.5850	5.64	5.64	-0.7
alternating	as_built	2.75	31.3	0.540	1.00	0.5467	0.5466	6.29	6.27	-0.2
parallel	as_built	2.75	31.3	0.540	1.00	0.5834	0.5830	6.00	6.01	-0.0

Table 9.1. Selected optima and as-built design re-run in GRAIL-CHT

All points converged; the surrogate holds to 0.13 efficiency points. The optimum sits at the top flow bound with $G \rightarrow 1$ (straight channel). Channel pumping power is below 0.1 mW everywhere, so manifold and pump losses (not modelled) dominate in practice.

10. Stage 6 — Efficiency curves and system model

10.1 Virtual steady-state test (ISO 9806 form)

As-built geometry (bridge 31.29 mm, G 0.539935, λ 1.0) at the design flow 2.75 g/s. Fit form: $q = \eta_0 G - a_1 \Delta T - a_2 \Delta T^2$, $\Delta T = T_m - T_{amb}$ on the mean fluid temperature, on the 0.528 m² aperture and incident irradiance.

Arr.	G W/m ²	T _{in} K	T _{amb} K	Wind m/s	T _{out} K	ΔT_{fluid} K	Q _u W	η	Plate mean K	Plate std K	Δp Pa
alt	500	293.15	303.15	3.0	307.06	13.91	159.9	0.6056	311.23	4.20	37.4
alt	500	308.15	303.15	3.0	320.76	12.61	145.0	0.5491	324.70	3.89	27.9
alt	500	323.15	303.15	3.0	334.45	11.30	129.9	0.4921	338.08	3.54	21.7
alt	500	332.15	303.15	3.0	342.66	10.51	120.8	0.4575	346.08	3.32	19.1
alt	900	293.15	303.15	3.0	317.64	24.49	281.5	0.5923	325.18	7.48	33.4
alt	900	308.15	303.15	3.0	331.31	23.16	266.2	0.5603	338.68	7.21	25.3
alt	900	323.15	303.15	3.0	344.97	21.82	250.9	0.5279	352.09	6.89	20.0
alt	900	332.15	303.15	3.0	353.16	21.01	241.5	0.5083	360.09	6.67	17.7
alt	900	323.15	303.15	0.5	345.05	21.90	251.7	0.5297	352.19	6.92	20.0
alt	900	323.15	303.15	4.9	344.95	21.80	250.6	0.5273	352.05	6.88	20.0
par	500	293.15	303.15	3.0	307.93	14.78	169.9	0.6434	302.10	4.14	37.0
par	500	308.15	303.15	3.0	321.58	13.43	154.4	0.5847	316.25	3.77	27.7
par	500	323.15	303.15	3.0	335.21	12.06	138.6	0.5250	330.40	3.39	21.6
par	500	332.15	303.15	3.0	343.37	11.22	129.0	0.4886	338.89	3.15	19.0
alt	700	332.15	288.15	3.0	346.62	14.47	166.3	0.4500	351.36	4.58	18.5
par	900	293.15	303.15	3.0	319.20	26.05	299.5	0.6302	308.90	7.29	32.8
par	900	308.15	303.15	3.0	332.84	24.69	283.9	0.5974	323.02	6.92	25.0
par	900	323.15	303.15	3.0	346.46	23.31	268.0	0.5639	337.16	6.55	19.8
par	900	332.15	303.15	3.0	354.62	22.47	258.3	0.5436	345.63	6.31	17.5
par	900	323.15	303.15	0.5	346.48	23.33	268.1	0.5643	337.15	6.55	19.8
par	900	323.15	303.15	4.9	346.46	23.31	267.9	0.5638	337.16	6.54	19.8
par	700	332.15	288.15	3.0	347.61	15.46	177.7	0.4809	341.43	4.35	18.4

Table 10.1. Virtual test matrix (22 GRAIL-CHT runs)

Arrangement	η_0	a_1 W/m ² K	a_2 W/m ² K ²	Fit RMS (points)	Held-out errors (points)
alternating	0.5966	2.004	0.00038	0.15	0.24, -0.01, 0.15
parallel	0.6368	2.104	0.00000	0.23	0.15, 0.10, -0.04

Table 10.2. Efficiency correlations

Parallel's free fit gave $a_2 = -0.0006$ (unphysical); refitted with $a_2 = 0$. The fit covers $T_m - T_{amb}$ from -3 to 40 K. Wind changes η by < 0.3 points. With the measured in-collector VIP conductivity, a_1 would be 2.3–2.5 W/m²K (Section 14).

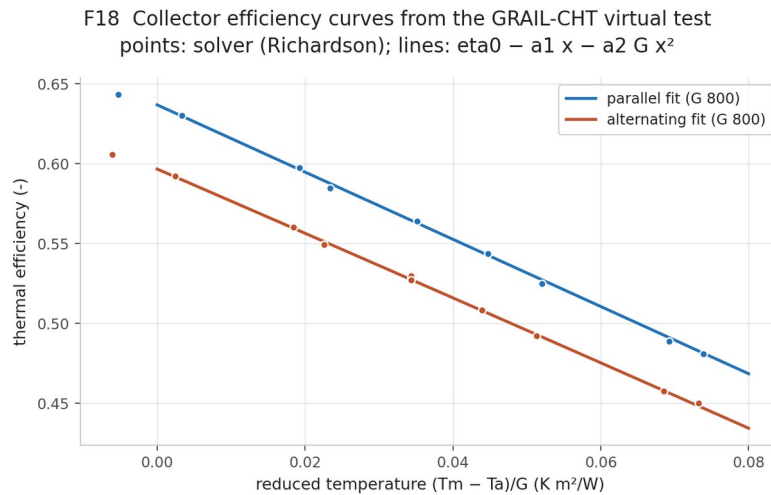


Figure 10.1. Efficiency curves

10.2 Dynamic collector-tank model

Collector: $C \frac{dT_c}{dt} = A [\eta_0 G_{\text{eff}} - a_1 (T_c - T_a) - a_2 (T_c - T_a)^2] - g_2 (T_c - T_t) P$
 Tank: $M c_p \frac{dT_t}{dt} = g_2 (T_c - T_t) P - UA (T_t - T_a) - m_t c_p (T_t - T_{\text{mains}})$
 Valve: $m_t = m_{\text{draw}} \cdot \min(1, (T_{\text{set}} - T_{\text{mains}})/(T_t - T_{\text{mains}}))$; electric top-up to T_{set}
 Pump P: on at $T_c - T_t \geq 7$ K, off at ≤ 2 K; off while $T_t > 358.15$ K

Parameter	Value
tilt_deg	19.3
azimuth_deg	180
transposition	Hay-Davies, albedo 0.20
iam	$K_b = 1 - b_0 (1/\cos(\theta) - 1)$, $b_0 = 0.10$, $K_b = 0$ beyond 80 deg; diffuse $K_d = K_b(60 \text{ deg}) = 0.90$
n_modules	4
aperture_m2	2.112
mdot_per_module_kg_s	0.00275
C_eff_J_m2K	33500
C_eff_basis	Al sheets 4.87 + channel water 7.46 + PCM sensible only 16.0 + 0.5x(glass 6.72 + TIM 0.36 + VIP 3.2) kJ/m2K; PCM latent heat NOT modelled (deferred)
tank_kg	100
tank_UA_W_K	1.5
T_set_K	318.15
daily_draw_kg	100
draw_profile	40% 06-08h, 20% 12-13h, 40% 18-20h IST
mains	trailing 24-h mean ambient
pump_on_dT_K	7
pump_off_dT_K	2
tank_max_K	358.15
pump_power_W	15

Parameter	Value
geyser_efficiency	0.95

Table 10.3. System-model inputs (base case record; recommended size is 2 modules)

Other fixed values: $C = 33.5 \text{ kJ/m}^2\text{K} \times A$; $g2 = 2 \dot{m} \text{ cp}$ with $\dot{m} = 0.00275 \text{ kg/s}$ per module.

11. Stage 7 — Annual performance and economics, Berhampur

11.1 Weather

- PVGIS TMY, 19.313° N, 84.810° E, 2005–2023, 8,760 hours.
- Time alignment tested: $GHI = DNI \cos z + DHI$ closes to 0.17 W/m² at the stated UTC stamp (14.3 W/m² with a half-hour shift).
- Plane: 19.3° tilt, south, Hay-Davies, albedo 0.2 → POA 1976 kWh/m²/yr (after incidence-angle modifier 1855); GHI 1865.
- Cross-check at 35° tilt: 1,935 vs 1,965 kWh/m²/yr in the PVGIS PV report (–1.5 %).

11.2 Collector-only yield

Month	POA kWh/m ²	Parallel @40 °C	Parallel @60 °C	Alternating @40 °C	Alternating @60 °C
Jan	185.7	96.2	84.6	90.1	79.0
Feb	185.1	97.8	86.5	91.7	80.8
Mar	205.6	109.3	96.7	102.5	90.4
Apr	186.9	99.1	86.5	92.9	80.8
May	177.8	95.1	81.6	89.2	76.2
Jun	149.8	78.4	65.8	73.5	61.5
Jul	124.6	61.5	48.9	57.6	45.5
Aug	134.4	67.8	55.0	63.5	51.2
Sep	136.7	70.3	57.9	65.8	54.0
Oct	161.6	84.5	72.3	79.2	67.4
Nov	170.8	89.5	77.6	83.8	72.4
Dec	157.1	80.4	68.9	75.3	64.3

Table 11.1. Monthly collector yield per m² at fixed inlet temperature (kWh/m²)

Arrangement	Yield @40 °C inlet (kWh/m ² /yr)	Yield @60 °C	Annual efficiency on POA @40 °C	Operating hours @40 °C
parallel	1030	882	52.1 %	3858
alternating	965	823	48.8 %	3853

Table 11.2. Annual collector-only yield

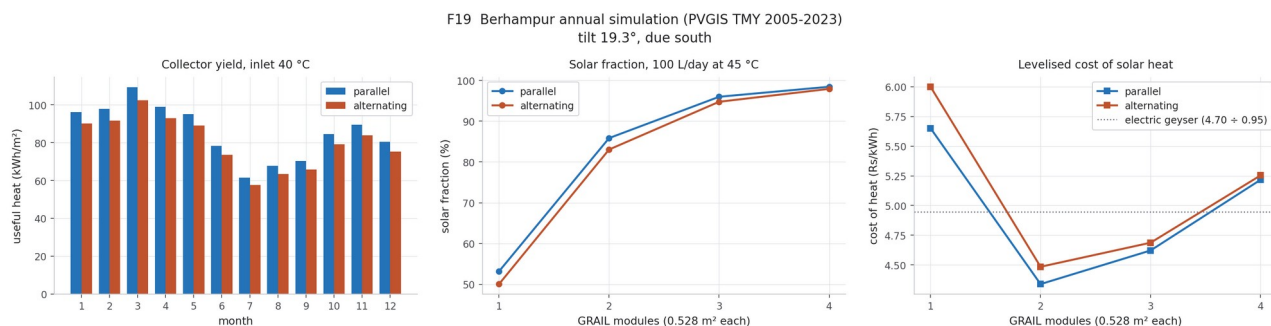


Figure 11.1. Monthly yield, sizing and cost of heat

11.3 Household system sizing (100 L/day at 45 °C)

Arr.	Mo d.	Area m ²	SF	Solar kWh/yr	Tank loss kWh	Max Tc K	Capex Rs	Payback yr	NPV Rs	Cost of heat Rs/kWh
alt	1	0.53	50.0 %	385	111	329	16500	10.8	-3464	6.00
alt	2	1.06	83.0 %	638	242	349	21000	7.6	2526	4.49
alt	3	1.58	94.7 %	728	402	389	25500	8.0	1620	4.69
alt	4	2.11	97.9 %	753	509	427	30000	9.2	-1976	5.25
par	1	0.53	53.2 %	409	118	331	16500	10.1	-2461	5.65
par	2	1.06	85.8 %	660	266	353	21000	7.4	3435	4.34
par	3	1.58	96.0 %	738	436	405	25500	7.9	2053	4.62
par	4	2.11	98.4 %	757	533	442	30000	9.1	-1748	5.22

Table 11.3. Sizing sweep (15 years, 8 % discount, 1 % O&M, 15 W pump)

Economic input	Value
tariff_Rs_kWh	4.7
tariff_basis	TPSODL LT domestic 51-200 kWh/month slab, 470 paise/kWh (FY 2025-26, retained FY 2026-27); electricity duty excluded (conservative)
tariff_high_Rs_kWh	5.7
capex_Rs	30000
capex_range_Rs	22000 – 45000
capex_basis	ENGINEERING_ASSUMPTION: market 100 LPD flat-plate system Rs 20,000-30,000 + installation Rs 3,000-10,000; GRAIL prototype cost unknown
om_frac_per_yr	0.01
life_yr	15
discount	0.08

Table 11.4. Economic inputs

- Two modules give the lowest cost of heat: SF 86 % (parallel) / 83 % (alternating), Rs 4.34 / 4.49 per kWh vs Rs 4.95 for a geyser (Rs 4.70/kWh at 95 %).
- Four modules are oversized: the tank sits at its limit 533–726 h/yr, the collector stagnates at 427–442 K. DESIGN_REVIEW_REQUIRED: overheat protection for arrays of 3 or more modules.
- Capital cost is an ENGINEERING_ASSUMPTION (Rs 12,000 + Rs 4,500 per module); the GRAIL prototype cost is unknown.

12. System models: Python, Octave, Xcos and Simulink

The same collector–tank system was built four times, independently, to check the equations and the code. All use the same hourly input file (G_eff, T_amb, T_mains, draw for 8,760 h).

Implementation		What it is				Agreement			
Python (Stage 7)		Reference model, 60 s step, draw as pulse at the start of each hour				Baseline			
GNU Octave 8.4		Same equations as .m files; own solar-position routine				SF within 0.000004; solar heat within 0.003 kWh/yr; peak Tc within 0.17 K			
Scilab Xcos 2024		24-block diagram, LSodar, continuous draw				Solar heat within 0.04 % of Octave (same continuous-draw equations)			
MATLAB/Simulink (V20, V21)		Built by script with add_block/add_line, ode45, data embedded				All 4 cases PASS vs Xcos: solar heat within 0.05 %, max Tc within 0.94 K			
Arrangement	Modules	Q_coll kWh	Q_loss kWh	Q_solar kWh	Q_load kWh	Pump h	SF	Tt end K	Max Tc K
alternating	2	875.1	247.5	626.0	768.9	2835	81.42 %	311.68	347.96
parallel	2	921.9	271.7	648.4	768.9	2840	84.34 %	313.17	352.03
alternating	4	1263.2	509.2	749.7	768.9	2153	97.51 %	334.68	429.85
parallel	4	1293.0	534.2	754.1	768.9	2056	98.08 %	338.79	443.82

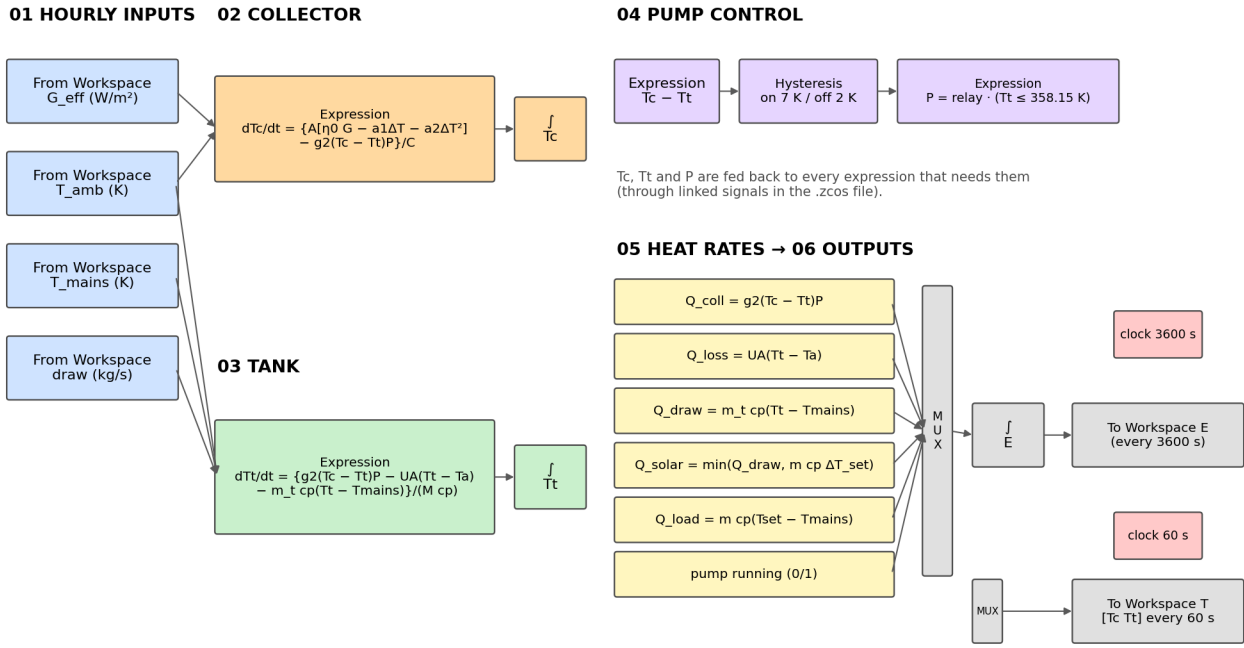
Table 12.1. Xcos annual results (continuous draw)

Arrangement	Modules	Solar kWh	SF %	Max Tc K	Tank end K	Closure kWh	vs Xcos solar %	vs Xcos max Tc K	Verdict
alternating	2	626.28	81.45	347.963	311.435	4.9e-12	+0.0408	+0.0009	PASS
parallel	2	648.55	84.35	352.021	312.899	−3.1e-11	+0.0205	−0.0089	PASS
alternating	4	749.79	97.52	430.787	334.159	−6.0e-11	+0.0108	+0.9375	PASS
parallel	4	754.13	98.08	444.211	338.204	−1.5e-10	+0.0068	+0.3892	PASS

Table 12.2. Simulink V20 run in MATLAB Online (V21: same numerics, OVERALL PASS)

The continuous-draw models (Xcos, Simulink) give a solar fraction 1.5–1.6 points lower at 2 modules than the Stage 7 pulse-draw baseline. The ranking and the 2-module sizing do not change; the Stage 7 values remain the reported results. The larger max-Tc differences at 4 modules happen at stagnation (> 420 K) and depend on step placement; annual energy agrees within 0.011 %.

Scilab Xcos model GRAIL_system.zcos — block structure (redrawn for print; same 24 blocks and equations)



Solver LSodar, tolerance 1e-6, max step 60 s; one year runs in 7-10 s. Inputs held constant within each hour.

Figure 12.1. Scilab Xcos model — block structure (redrawn for print from GRAIL_system.zcos)

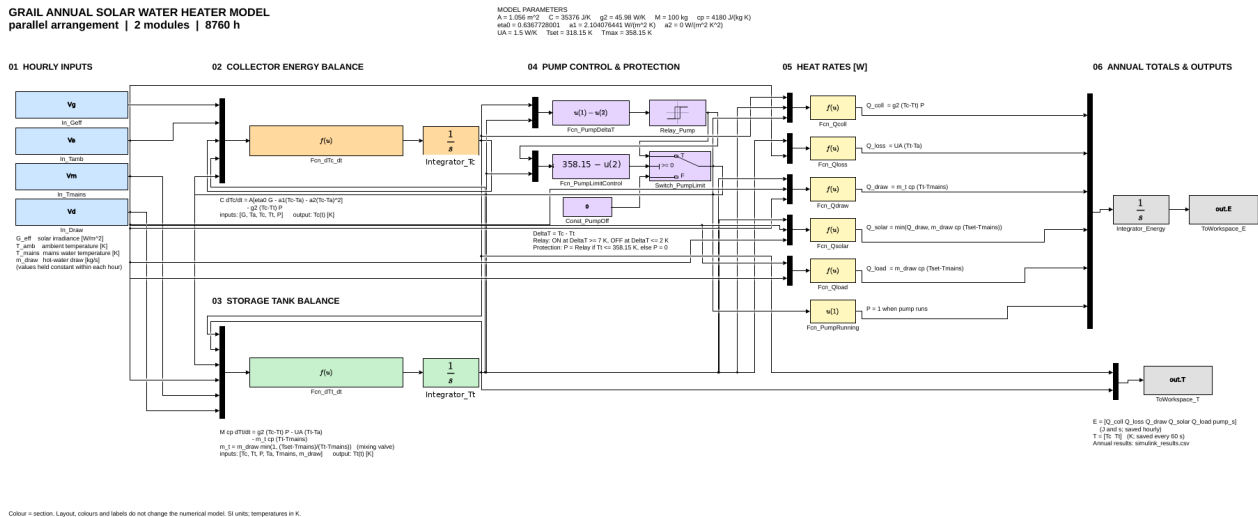


Figure 12.2. MATLAB/Simulink V21 block diagram (parallel, 2 modules); two duplicate labels removed for print — V22 script fixes them at source

13. Verification record (model checks)

Verification asks: are the equations solved correctly? Validation (Section 14) asks: do they describe reality?

#	Check	Result	Status
1	3-D grid convergence of Nu	GCI 2.1 %, p = 1.12	PASS
2	3-D energy balance from raw fields	0.005 % / 0.035 %	PASS
3	3-D iteration convergence	$\leq 2 \times 10^{-6}$	PASS
4	frozenFlow honoured	0 pressure/momentum solves; flux unchanged 1.5×10^{-8}	PASS
5	GRAIL-CHT vs 3-D (co-current)	η 0.01 %, std 0.6 %	PASS
6	GRAIL-CHT vs HWB textbook model	+0.22 %	PASS
7	Campaign energy balance, 300 cases	all < 0.5 % (max 0.00015 %)	PASS
8	Campaign convergence	300 / 300	PASS
9	Independent dataset audit (Rev 4.1)	3 derived-column groups corrected; thermal results unchanged	PASS after fix
10	ANN hold-out	$R^2 \geq 0.9994$	PASS
11	NSGA-II optima re-run in solver	≤ 0.13 points η	PASS
12	Efficiency-curve held-out points	≤ 0.24 points	PASS
13	Tank first law, annual	$\approx 10^{-10}$ kWh	PASS
14	Weather time alignment	0.17 W/m ² closure	PASS
15	Python vs Octave vs Xcos vs Simulink	≤ 0.05 % solar heat	PASS
16	V7 matched-temperature searches	matched within 0.001 K; energy < 0.5 %	PASS

14. Validation against published experimental work (V5)

V5 verdict

COMPLETE — PARTIAL PASS at method and component level (closed 25 Sep 2026).

The modelling method was checked against measurements on similar collectors. The GRAIL collector itself has not been built or tested; it is not experimentally validated.

14.1 Screening rules

- (a) measured data given in tables;
- (b) data pass an independent energy check;
- (c) the inputs needed to rerun the case are stated.

14.2 All papers screened (16)

#	Paper	Measured?	Tables	Energy check	Inputs	Use
1	Gunjo et al. 2017, Renew. Energy	CFD vs test	–	Fails (CFD exceeds solar input)	–	Rejected
2	Bharathiraja et al. 2024, J. Energy Storage 96	Yes, outdoor	Yes	Pass (64.8 / 73.0 % vs 64.7 / 71.7 %)	Partly	Comparison
3	Vahidinia & Khorasanizadeh 2024, Energy 304	No	–	Pass (quoted points)	No	Pointer
4	Zareie et al. 2024, Energy 286	Yes, simulator	Plot	Not checkable	Partly	Rejected
5	Parthiban et al. 2025, Appl. Therm. Eng. 258	No	System	n/a	Yes	Comparison (TIM τ 0.89)
6	Ding et al. 2025, Int. J. Therm. Sci. 208	Heat sink	Partly	n/a	Yes	Physics support
7	Li et al. 2024, Appl. Therm. Eng. 251	No	n/a	n/a	n/a	Method support
8	Alawi et al. 2024, Eng. Appl. AI 133	No	n/a	n/a	n/a	ANN comparison
9	Del Col et al. 2013, Energy 58	Yes, EN 12975	Yes, U95	Pass	Mostly	REFERENCE — method check
10	Mansour 2013, Energy 60	Yes	Plots	Model table passes	Model	Model benchmark
11	Zheng et al. 2024, Appl. Energy 355	Yes	Plot	~11 % uncertainty	Yes	Comparison
12	Sakib et al. 2026, ECM:X 31	No	n/a	n/a	n/a	Method comparison
13	Beikircher et al. 2015, SOLMAT 141	Yes	Yes	Pass (η_0 0.89 < τ_α 0.91)	Partly	REFERENCE — component check
14	Kessentini et al. (CTTC/UPC)	Yes	Fitted line	Plausible	Partly	Comparison (η = 0.732 – 7.19 x)
15	Robles et al. 2014, Sol. Energy 110	Yes	Plots	Not checkable	No	Background
16	Oyinlola & Shire 2018, BSERT	Lab rig	Plots	n/a	Rig	Background

14.3 Method check — Del Col et al. 2013 (tested roll-bond collectors)

The same Hottel-Whillier-Bliss fin model with Klein's top-loss correlation was applied without any tuning. Stated inputs were used as given (aperture 1.81 m², 28 channels, Ø 3.7 mm, 1.5 mm Al, α/ϵ , 0.02 kg/s·m², G 800–1000 W/m², T_{amb} 8–12 °C, wind 1.4 m/s). The 11 unstated inputs (glass τ and ϵ , width, Al k, insulation, edge loss, Nu, bond footprint, wind coefficient, tilt, ambient) were sampled over typical ranges in 2,000 Monte Carlo runs (ENGINEERING_ASSUMPTION).

Collector	Coefficient	Model nominal	Model 95 % band	Measured $\pm$ U95	Inside U95?
Black roll-bond	η_0	0.862	0.848–0.876	0.7995 $\pm$ 0.0108	No (+7.8 %)
	a1 W/m ² K	5.37	5.05–5.77	4.69 $\pm$ 0.79	Yes
	a2 W/m ² K ²	0.0185	0.0174–0.0198	0.0288 $\pm$ 0.0121	Yes
Semi-selective roll-bond	η_0	0.785	0.773–0.798	0.7409 $\pm$ 0.0110	No (+6.0 %)
	a1 W/m ² K	4.26	3.99–4.60	4.55 $\pm$ 0.69	Yes
	a2 W/m ² K ²	0.0118	0.0113–0.0125	0.0131 $\pm$ 0.0100	Yes

Table 14.1. Model vs measured efficiency coefficients

Tm* (m ² K/W)	0	0.01	0.02	0.03	0.04	0.06	0.08
Black: deviation %	+7.9	+7.6	+7.7	+8.2	+9.2	+14.3	+31.1
Semi-selective: deviation %	+6.0	+6.8	+7.9	+9.1	+10.8	+15.7	+25.8

Table 14.2. Efficiency-curve deviation, model vs measured (G = 1000 W/m²); absolute deviation $\leq$ 0.076 points

Result: the heat-loss side (a1, a2) is reproduced within the test uncertainty. η_0 is over-predicted by 6–8 %, from optical losses (frame shading, soiling, absorptance) that the ideal model leaves out. Relative deviation grows at high Tm* only because the measured efficiency becomes small.

14.4 Component check — Beikircher et al. 2015 (TIM and vacuum insulation)

Plate – ambient (K)	GRAIL front U (W/m ² K)	Conduction part	Radiation part
20	2.057	1.768	0.289
40	1.961	1.660	0.301
60	1.949	1.623	0.326
80	1.959	1.603	0.356

Table 14.3. GRAIL front heat loss (wind 3 m/s) vs measured 1.95–2.2 W/m²K for honeycomb covers → PASS

Rear case	k (W/m·K)	Rear U, GRAIL stack (W/m ² K)
GRAIL input (catalogue VIP)	0.006	0.272
Measured in collector, silica VSI	0.010–0.018	0.43–0.69
Measured, perlite VSI	0.020–0.034	0.75–1.07

Table 14.4. Rear loss: GRAIL is optimistic → a1 2.1 becomes 2.3–2.5 W/m²K (DESIGN_REVIEW_REQUIRED)

14.5 Comparison of GRAIL with tested collectors

Collector	η_0	a1 W/m ² K	a2 W/m ² K ²	Source
GRAIL parallel	0.637	2.10 (2.3–2.5)	0	Model

Collector	η_0	$a_1 \text{ W/m}^2\text{K}$	$a_2 \text{ W/m}^2\text{K}^2$	Source
GRAIL alternating	0.597	2.00	0.0004	Model
Roll-bond black	0.800 ± 0.011	4.69 ± 0.79	0.029	Del Col 2013
Roll-bond semi-selective	0.741 ± 0.011	4.55 ± 0.69	0.013	Del Col 2013
VSI prototype, AR glass	0.89	3.28	0.014	Beikircher 2015
Honeycomb/slat TIM collector	0.732	7.19	–	Kessentini (poor coating ϵ 0.5)

Table 14.5. GRAIL trades optical efficiency for about half the heat loss; curves cross at $T_m^* \approx 0.036\text{--}0.050 \text{ m}^2\text{K/W}$

14.6 V5 verdict against the < 8 % criterion

Item	Result
Heat-loss coefficients a_1 , a_2	PASS — inside measured 95 % uncertainty
Optical efficiency η_0	PASS (< 8 %), outside test uncertainty
Efficiency curve $T_m^* \leq 0.02$	PASS (< 8 %)
Efficiency curve $T_m^* > 0.03$	Relative deviation > 8 % (≤ 0.076 points absolute)
TIM front loss	PASS
VIP rear loss	Model optimistic; carried as uncertainty

Allowed wording: "The modelling method was checked against published experimental data. It reproduces the measured heat-loss coefficients of tested roll-bond collectors within test uncertainty and their optical efficiency within 8 %. No prototype of the present collector was tested." Not allowed: "experimentally validated".

15. Patent first pass (V1)

An engineering first-pass search (Google Patents, Justia, OSTI and literature; not a professional search, not legal advice) found no document combining a roll-bond absorber with channels converging along the flow by a grading law, alone or with the TIM/PCM/VIP stack.

Document	Status	Relation to GRAIL
EP2313705A1 / WO2010064078A1 roll-bond absorber	Withdrawn / ceased	Uniform parallel channels
EP1916486A2 solar collector system	Withdrawn	Different sizes between channels, not a taper
CN103557603A inflated flat-plate collector	Rejected	Fixed section
DE10344084A1 adapted flow resistance	Expired	No taper along the flow
EP2522927A2 aerogel + honeycomb TIM collector	Granted	Freedom-to-operate note if an aerogel layer is used
Li 2024, Ding 2025 (heat sinks)	Literature	Tapered / variable-width channels in heat sinks

Assessment: novelty possible; inventive step weak-moderate (alternating flow shows no benefit at matched plate temperature; the optimiser moves G towards 1). Decision recorded: publish, no patent filing.

16. Assumptions, open design items and limitations

Tag	Item
ENGINEERING_ASSUMPTION	Whole manifold geometry (header bore, length, risers, ports)
ENGINEERING_ASSUMPTION	3-D benchmark loss model (U_{top} 4.19, h_{rear} 3.0); constant water properties in 3-D
ENGINEERING_ASSUMPTION	Tilt 19.3° S; IAM b0 0.10; C_{eff} 33.5 kJ/m ² K; tank 100 kg, UA 1.5 W/K; 100 L/day at 45 °C; draw profile; mains = 24 h mean ambient; pump 15 W; geyser 95 %; capex model; 15 yr, 8 %, 1 % O&M
ENGINEERING_ASSUMPTION	NSGA-II reference point; design flow 2.75 g/s
ENGINEERING_ASSUMPTION	Unstated inputs of the Del Col check (Monte Carlo ranges)
DESIGN_REVIEW_REQUIRED	VIP conductivity: use a measured in-collector value (0.010–0.018 W/m·K)
DESIGN_REVIEW_REQUIRED	Stagnation above 420 K for ≥ 3 modules: overheat protection needed
DESIGN_REVIEW_REQUIRED	Converging-channel benefit: not supported for efficiency in the range studied
Open (C7, C8)	Manifold dimensions and frame wall thickness
Open (V6)	Supplier datasheets for TIM, PCM, VIP
Deferred	PCM latent heat; 3-D radiation/optics; full 3-D campaign; topology optimisation (Stage 4)

16.1 Limitations

- Simulation only; no measurement of the GRAIL collector.
- Optical efficiency may be optimistic by up to $\approx 7\%$ (Del Col check); a_1 may be up to 2.5 W/m²K (Beikircher check).
- Single collector node and fully mixed tank in the annual model; capital cost assumed.
- Alternating-flow uniformity overstated by ≈ 2 points in GRAIL-CHT (Nu undefined where heat flow reverses).

17. Conclusions

- A converging roll-bond absorber with a glass + honeycomb TIM cover and PCM + VIP rear was modelled from CAD through 3-D CFD to annual economics, with every step verified.
- The corrected conjugate Nusselt number is 4.48 (GCI 2.1 %); GRAIL-CHT with this closure reproduces the 3-D solution.
- Alternating flow lowers efficiency by 7.3 % at equal inputs; at the same plate temperature efficiency is equal and uniformity is worse at design flow. The cause is heat returned to the plate ($\approx 40\%$).
- The ANN surrogate ($R^2 \geq 0.9994$) and NSGA-II optimisation are verified against the solver; the optimum favours high flow and straight channels for efficiency.
- The collector has η_0 0.637 and a_1 2.1 (up to 2.5) $\text{W/m}^2\text{K}$ — about half the heat loss of tested roll-bond collectors — and outperforms them above $T_m^* \approx 0.04\text{--}0.05 \text{ m}^2\text{K/W}$.
- In Berhampur a 2-module system covers 83–86 % of a 100 L/day load at Rs 4.34–4.49 per kWh, cheaper than an electric geyser, payback ≈ 7.5 years.
- Four independent system models agree. The method was checked against published experiments (partial pass). A prototype test is the next step.

18. Files and folders

Folder	Contents
00_README.md	Package guide
01_Roadmap	Roadmap Rev I, completion status Rev 4.2
02_Reports	CFD report Rev 4.1, CAD report, correction report, audit addendum, paper draft, this report
03_CAD	Fusion (.f3d), STEP, strips, gate JSON, geometry scripts
04_Figures	F01–F14 (current), legacy figures
05_Videos	3-D conjugate CFD video
06_Datasets	Rev 4.2 dataset, dictionary, workbook with Stages 3–7
07_Results_JSON	Benchmark, Nu profiles, ROM runs, uncertainty
08_Tables	T1–T6
09_Records_MD	Provenance records, audit outputs
10_Code	Model, CFD, post-processing, report scripts
11_Validation_V5	Screening of 16 papers, V5 final record
12_Stages_3-5-6-7	ANN, NSGA-II, efficiency curves, annual/economics
13_Octave_system_model	Octave model, compare_with_python.m
14_Xcos_block_diagram_model	GRAIL_system.zcos
15_Run_in_VSCode	One launcher for all re-runnable stages
16_JSON_Python_Mirror	65 JSON results as Python
17_Simulink_MATLAB_model	GRAIL_simulink_complete_FINAL_v21.m, run records
18_V7_matched_mean_T	V7 record and results
19_V5_DelCol_method_check	Del Col method check
20_Patent_prior_art	V1 first pass
21_V5_Beikircher_component_check	Beikircher component check

Appendix A. CAD parameter register

Parameter	Value	Expression	Source note
ap_L	1100.00 mm	1100 mm	ROADMAP A1 plate length
ap_W	480.00 mm	480 mm	ROADMAP A2 plate width
ap_t	2.00 mm	2.0 mm	ROADMAP A3 plate thickness = 2 x 1.0mm roll-bond
n_ch	12	12	ROADMAP A4 number of channels
ch_pitch	40.00 mm	ap_W / n_ch	ROADMAP A5 DERIVED channel pitch
ch_w_in	8.7009537 mm	ch_h_in * 8.0 / 3.8	DERIVED from ch_h_in holding ROADMAP A10 aspect ratio 2.105 (the 6.1 table ratio 8.0/3.8).
ch_h_in	4.132953 mm	4.132953 mm	RE-SOLVED 2026-09-14 after r_fillet moved to 0.5 mm. Chosen so the built inlet section gives Dh_in = 5.2000 mm exactly per ROADMAP A6; measured 5.199326. ch_w_in is driven from this holding the A10 aspect ratio 8.0/3.8 = 2.105; measured 2.1053. Previous value 4.02401 mm belonged to the r_fillet 1.0 mm construction.
ch_AR	2.10526	ch_w_in / ch_h_in	ROADMAP A10 DERIVED aspect ratio, target 2.1
Dh_in	5.20 mm	5.2 mm	ROADMAP A6 nominal inlet hydraulic diameter
G_ratio_A7_baseline	1	1.00	ROADMAP A7 stated baseline = NO grading
G_ratio_S61_fusion	0.54	0.54	ROADMAP 6.1 Fusion table value and A7 definition: the Dh RATIO Dh_out/Dh_in. NOW MATCHED BY THE GEOMETRY - measured 0.540014. This is the parameter that drives Dh_out.
G_ratio	0.527853	0.527853	AS-BUILT SECTION SCALE at the outlet, solved 2026-09-14 so the measured Dh ratio equals the ROADMAP A7/6.1 value 0.54 (achieved 0.540014). The channel CONVERGES. NOTE: this parameter is a SECTION SCALE, not the roadmap Dh ratio - the two differ because r_fillet is fixed and does not scale with the section. The roadmap Dh ratio lives in G_ratio_S61_fusion. Sweep range per A7 is 0.35-1.50 on the Dh ratio, but this construction floors the SECTION SCALE at 0.483915; below that the dome inverts silently.
Dh_out	2.808 mm	Dh_in * G_ratio_S61_fusion	DRIVEN per ROADMAP 6.1: Dh_out = Dh_in * G_ratio. Uses G_ratio_S61_fusion (0.54), the roadmap Dh ratio. NOTE the parameter named G_ratio in this file holds the SECTION SCALE (0.527853), which is not the same quantity because r_fillet is fixed and does not scale. As-built measured value is in Dh_out_geom.

Parameter	Value	Expression	Source note
lam_G	1	1.0	ROADMAP A8 grading profile exponent. Baseline 1.0, range 0.5-3.0, SWEPT. SET TO THE A8 BASELINE 1.0 on 2026-09-14. PREVIOUSLY 2.0, justified only by the REFERENCE_IMAGE label Exponential. That image has since been shown to contradict A7, 6.1 and 8.2 on the grading DIRECTION, so it is not a valid source for the EXPONENT either, and the value reverts to the roadmap baseline. The law is $Dh(x) = Dh_{in} * [1 - (1 - G_{ratio}) * (x/ap_L)^{lam_G}]$, so lam_G 1 is linear, below 1 concave (taper early), above 1 convex (taper late). ENDPOINTS ARE INDEPENDENT OF lam_G - at $x = ap_L$ the scale equals G_ratio whatever the exponent - so Dh_in, Dh_out, the Dh ratio and the Reynolds band are unchanged by this. Only the interior profile and the fluid volume change. F10 should sweep it 0.5-3.0.
P_ratio	1	1.00	ROADMAP A9 pitch grading ratio, 1.0 = uniform pitch
r_fillet	0.50 mm	0.5 mm	ROADMAP A11 channel root fillet. Baseline 1.0 mm, range 0.5-2.0, Role FIXED. REDUCED to 0.5 mm by DECISION D4 on 2026-09-14, because at 1.0 mm the tangent-blend section degenerates at $h_s = 3.0$ mm and floors the section scale at 0.745525, which makes the ROADMAP 6.1 grading G_ratio 0.54 geometrically unreachable. 0.5 mm is the lower end of the A11 range, so this stays inside the roadmap. CONSEQUENCE TO REPORT: A11 Role reads FIXED, so Rev H must record that the fillet was moved within its stated range, and 6.3 notes the fillet carries the lateral conduction path, so F10 should sweep it alongside w_bridge.
f_interdig	1	1	ROADMAP B1 1 = alternating counter-current ACTIVE
n_alt	1	1	ROADMAP B2 channels per flow-direction group
w_bridge_B3_baseline	34.80 mm	34.8 mm	ROADMAP B3 stated baseline bridge width
w_bridge	31.2990463 mm	$ch_pitch - ch_w_in * \max(1; G_ratio)$	DERIVED minimum lateral conduction bridge = $ch_pitch - ch_w_in * G_ratio$. 26.195 mm at the as-built grading. ROADMAP B3 states 34.8 mm - see DESIGN_REVIEW_W_BRIDGE. Filing 1 claims a bridge width SELECTED to reduce the peak-to-mean plate temperature difference, so this is a claimed dimension and must be swept (Rev H F10, 20-60 mm), not assumed.
alpha_abs	0.95	0.95	ROADMAP C1 solar absorptance TiNOX
eps_abs	0.04	0.04	ROADMAP C2 emittance
coat_t	0.0003 mm	0.0003 mm	ROADMAP 5.1 sputtered TiNOX ~0.3 um
tau_glz	0.91	0.91	ROADMAP C5 glazing transmittance
glz_t	3.20 mm	3.2 mm	ROADMAP 5.1 low-iron tempered glass
glz_L	1160.00 mm	1160 mm	ROADMAP 5.2 BOM glazing length
glz_W	560.00 mm	560 mm	ROADMAP 5.2 BOM glazing width
t2_t	1.00 mm	1.0 mm	ROADMAP 5.1 thermotropic PMMA layer
T_sw	75	75	ROADMAP 5.1 thermotropic switching temperature in degC

Parameter	Value	Expression	Source note
tim_t	10.00 mm	10 mm	ROADMAP D1 transparent insulation thickness
tim_cell	3.50 mm	3.5 mm	ROADMAP D2 honeycomb cell size, cutaway block only
gap_t	15.00 mm	15 mm	ROADMAP D5 sealed dry air gap
vip_t	20.00 mm	20 mm	ROADMAP D6 vacuum insulation panel thickness
vip_W	500.00 mm	500 mm	ROADMAP 5.2 BOM VIP width
edge_t	15.00 mm	15 mm	ROADMAP D8 aerogel edge insulation
graph_t	0.40 mm	0.4 mm	ROADMAP 5.1 compressed graphite interface
pcm_t_S51	11.60 mm	11.6 mm	ROADMAP 5.1 layer stack PCM thickness
pcm_t_S61	10.00 mm	10 mm	ROADMAP 6.1 Fusion table PCM thickness, agrees with BOM 4.65 kg
tray_t	0.80 mm	0.8 mm	ROADMAP 5.2 BOM ribbed aluminium PCM tray
back_t	1.00 mm	1.0 mm	ROADMAP 5.1 recycled aluminium backsheet
frame_L	1200.00 mm	1200 mm	ROADMAP 5.0 gross length
frame_W	600.00 mm	600 mm	ROADMAP 5.0 gross width
frame_d	92.00 mm	92 mm	ROADMAP 5.1 frame overall depth
tilt	33.0 deg	33 deg	ROADMAP E6 nominal tilt
tilt_min	10.0 deg	10 deg	ROADMAP E6 lower adjustable limit
tilt_max	45.0 deg	45 deg	ROADMAP E6 upper adjustable limit
DESIGN_REVIEW_VIP_MASS	0	0	CONFIRMED AT SOURCE, ROADMAP CORRECTION REQUIRED. 5.2 lists the VIP as fumed silica 1120 x 500 x 20 mm at 1.05 kg, which implies 93.75 kg/m ³ , while 5.3 gives VIP fumed silica 190 kg/m ³ - the roadmap contradicts itself. Fumed-silica VIP cores are 160-200 kg/m ³ , so 5.3 is correct and the 5.2 mass is wrong; it should read 2.13 kg for the BOM dimensions. The CAD uses 190 kg/m ³ on a 1100 x 500 x 20 panel (trimmed to ap_L to clear the manifolds) = 2.090 kg. No CAD change required.
DESIGN_REVIEW_TIM_DENSITY	0	0	CLOSED 2026-09 as a FALSE ALARM. ROADMAP 5.3 lists PC honeycomb homogenised density 85 kg/m ³ , not 851 (transcription error in the earlier note). Model uses 85 -> TIM_PANEL 0.456 kg vs BOM 5.2 value 0.42 kg. No conflict remains.
frame_bot_z	-58.80 mm	frame_top_z - frame_d	DERIVED frame bottom, holds ROADMAP 92mm envelope
Dh_in_geom	5.199685 mm	5.199685 mm	MEASURED on the built solid at the inlet face after the 2026-09 grading correction. Target 5.2000 (ROADMAP A6). Error -0.0130 percent.
Dh_out_geom	2.809592 mm	2.809592 mm	MEASURED on the built solid at the outlet face after the 2026-09 grading correction. Target 2.8080 = Dh_in * 0.54 (ROADMAP 6.1). Error -0.0104 percent. The channel now CONVERGES, per A7, 6.1 and 8.2.
CFD_mdot_per_channel	0.0002083	2.083E-04	kg/s per channel at the ROADMAP E4 design point mdot 0.0025 kg/s over n_ch 12. Apply as Mass Flow on the 3 cold port faces per end.

Parameter	Value	Expression	Source note
CFD_Re_min	23.9	23.9	MEASURED-geometry Reynolds at mdot 0.001 kg/s on the CONVERGING channel, inlet Dh 5.1993 mm. ROADMAP 8.2 states the band 30-210. Correction C1 is now APPLIED in the geometry, so 8.2 and the model agree.
CFD_Re_max	198.8	198.8	MEASURED-geometry Reynolds at mdot 0.0045 kg/s, outlet Dh 2.8077 mm. Full span across the mass-flow range is 23.9 to 198.9, against ROADMAP 8.2 stating 30-210. Deeply laminar; the Laminar Flow interface stands.
CFD_swept_ratio	211	211	channel length / Dh_in. High aspect ratio - swept mesh along the channel is mandatory per ROADMAP 8.5.
CFD_suppress_count	57	57	Components to suppress before STEP export: 30 FASTENER_, 5 SEAL_, 14 MOUNT_, 8 SENSOR_ = 57, PLUS SELECTIVE_COATING = 58 bodies total. All four group counts were re-verified by component scan in the 2026-09 audit and match exactly. ROADMAP 8.3 handles the coating as a Thin Layer, not a meshed solid. The 12 PORT_ components are NOT suppressed - they carry the CFD inlet and outlet faces.
DESIGN_REVIEW_G_RATIO	0	0	CLOSED 2026-09-14 BY FIX. The grading now CONVERGES per ROADMAP A7, 6.1 and 8.2. Section scale solved to 0.527853 giving measured Dh_in 5.199326 and Dh_out 2.807707 mm, a Dh ratio of 0.540014 against the roadmap 0.54. Reynolds now spans 23.9 to 198.9 across mdot 0.001-0.0045, matching the 8.2 band 30-210. The REFERENCE_IMAGE diverging value 1.596 is superseded and was never a roadmap source. Applied by walking G_ratio down in ~20 small steps so the outlet sketch tangent constraints tracked continuously; a single jump fails.
G_ratio_sweep_min	0.35	0.35	ROADMAP A7 lower bound, verified at source.
G_ratio_sweep_max	1.5	1.50	CORRECTED AT SOURCE. ROADMAP A7 gives G_ratio range 0.35-1.50, baseline 1.00. The previous 1.60 exceeded the roadmap range and is withdrawn. NOTE: the as-built G_ratio 1.629546 and the REFERENCE_IMAGE Dh ratio 1.596 are BOTH outside A7 entirely.
DESIGN_REVIEW_W_BRIDGE	0	0	CLOSED AT SOURCE. ROADMAP B3 lists w_bridge as SWEPT with baseline 34.8 mm and range 20-60 mm. The derived value 26.195 mm is INSIDE that range, so there was never a conflict - only a difference from the baseline. The earlier pitch-minus-Dh hypothesis is withdrawn as unnecessary. Rev H F10 still sweeps 20-60 per B3. NOTE: if the grading is corrected to converging, the widest section moves to the INLET and the true minimum bridge becomes 40 - 8.4716 = 31.53 mm, which is much nearer the B3 baseline 34.8 - further corroboration that converging is the intended direction.
glz_n_panes	2	2	DECISION D2: the second pane is intentional DOUBLE GLAZING, not a modelling error. Must be added to ROADMAP 5.1 layer stack, 5.2 BOM and the U_L derivation.

Parameter	Value	Expression	Source note
gap_air_total	11.75 mm	11.75 mm	MEASURED total AIR depth = gap_t - glz_inner_t. The ROADMAP D5 envelope gap_t = 15 mm is honoured exactly (z 1.0 to 16.0), but 3.2 mm of it is glass, so the cavity is TWO gaps of 7.45 and 4.30 mm, not one of 15 mm. The U_L derivation must use two cavities.
DESIGN_REVIEW_GLZ_DOUBLE	0	0	CLOSED by decision D2 as a GEOMETRY question: the inner pane is intentional double glazing, not a stray body, and ROADMAP 5.1 / 5.2 / 6.1 now list it (correction C9). The 5.1 gap_t 15 mm envelope is preserved exactly; see gap_air_lower 7.45, glz_inner_t 3.2, gap_air_upper 4.30. NOTE: the OPTICAL consequence was missed at the time and is now tracked separately as DESIGN_REVIEW_GLZ_TRADE, because it decides whether the pane should stay at all.
MATERIALS_ASSIGNED	135	135	Hardware bodies carrying a ROADMAP 5.3 material (plus 3 CFD reference bodies = 138 total). 15 custom materials were created at the 5.3 densities. Previously every body was Fusion default Steel, so all mass output produced before this pass is void.
DRY_MASS_AS_MODELLED	43.456	43.456	MEASURED HARDWARE dry mass in kg at ROADMAP 5.3 densities, excluding the 90_CFD_REFERENCE duplicates, water and air. Re-measured 2026-09-14 after the converging grading correction and the r_fillet change.
DESIGN_REVIEW_COATING_T	0	0	CLOSED 2026-09 BY FIX. SELECTIVE_COATING_TiNOX was modelled at 0.05 mm, 167x the specified thickness, carrying 0.1365 kg of mass that does not exist. The OffsetFaces1 distance that drives the shell is now bound to the parameter coat_t (0.0003 mm per ROADMAP 5.1, sputtered TiNOX). Rebuilt result: volume 182.69 mm ³ , mass 0.000822 kg, measured uniform thickness 0.000301 mm against the 0.000300 target. Because the offset is now parametric, the thickness can never drift from coat_t again. 0 unhealthy features after the change.
DESIGN_REVIEW_BOM_TOTAL	0	0	CONFIRMED AT SOURCE, ROADMAP CORRECTION REQUIRED. The 5.2 table TOTAL row reads 17.82 kg, but its own line items sum to 22.42 kg (5.20+0.78+0.42+2.85+0.35+0.21+4.65+1.30+1.05+0.12+2.90+1.95+0.44+0.20; the selective coating is listed with no mass). The stated total is low by 4.60 kg. This is an arithmetic error in the roadmap, not a CAD issue. Correct the TOTAL row to 22.42 kg in Rev H.

Parameter	Value	Expression	Source note
DESIGN_REVIEW_FILLET_FLOOR	0	0	CLOSED 2026-09-14 BY DECISION D4. The outlet sketches build the dome radius with denominator $2*((h_s - r_{\text{fillet}}) - (r_{\text{fillet}} + ap_{\text{sheet_t}}))$, which passes through zero at $h_s = 2*r_{\text{fillet}} + ap_{\text{sheet_t}}$ and inverts the dome below it. At r_{fillet} 1.0 mm that floored the section scale at 0.745525 and made the roadmap G_ratio 0.54 UNREACHABLE. r_{fillet} was reduced to 0.5 mm, which ROADMAP A11 permits (range 0.5-2.0), lowering the threshold to h_s 2.0 mm and the floor to scale 0.483915. 0.527853 now sits safely above it. WARNING retained: never set the section scale below 0.4840 in this construction - at 0.62 with the old fillet the sketch solved with $R = -0.8703$ mm and Fusion reported only a warning, so an inverted dome can pass unnoticed.
DECISION_D4_FILLET	1	1	DECISION D4, taken 2026-09-14. r_{fillet} reduced from 1.0 to 0.5 mm, inside the ROADMAP A11 range 0.5-2.0, so that the 6.1 grading G_ratio 0.54 becomes geometrically achievable. It was the only option making A7, 6.1 and 8.2 simultaneously true. REJECTED ALTERNATIVES: hold r_{fillet} 1.0 and accept a Dh ratio near 0.78, which fails 6.1 and reaches only Re 135 against the stated 210; or hold the A7 baseline 1.00 with no grading, which abandons the graded absorber that 5.1 names. REV H MUST RECORD: A11 Role reads FIXED, so the move within its range is a documented deviation; and 6.3 notes the fillet carries the lateral conduction path, so F10 should sweep r_{fillet} alongside w_{bridge} .
DESIGN_REVIEW_PCM_T	0	0	ROADMAP CORRECTION PENDING, no CAD change. ROADMAP 5.1 layer stack gives PCM 11.6 mm; 6.1 Fusion table gives 10 mm. The CAD uses 10 mm (pcm_t_S61) because it reproduces the 5.2 BOM mass of 4.65 kg, which 11.6 mm would not. Both competing values are kept as pcm_t_S51 and pcm_t_S61 . ACTION: reconcile 5.1 against 6.1 and the BOM in Rev H. Previously this conflict was noted only inside the pcm_t comment and had no flag, so it was invisible to the item count.
DESIGN_REVIEW_VIP_L	0	0	ROADMAP CORRECTION PENDING, no CAD change. ROADMAP 5.2 BOM specifies the VIP as 1120 mm long, but at that length it overlaps the manifold blocks at $x = +/-550$. The CAD trims it to $ap_L = 1100$ mm. ACTION: restate the VIP length in 5.2, or state the clearance that makes 1120 mm fit. Previously noted only inside the vip_L comment with no flag.

Parameter	Value	Expression	Source note
tau_glz_sys	0.833	0.833	ROADMAP C9 (added in Rev H, correction C11): glazing SYSTEM transmittance for ALL panes, DERIVED. First-order estimate for two identical low-iron panes of tau 0.91 and reflectance 0.08: $\tau^2/(1-\rho^2) = 0.8281/0.9936 = 0.833$. C5 tau_glz 0.91 is a SINGLE-PANE value and its range 0.88-0.92 cannot represent the built double-glazed stack. ROADMAP 8.4 now builds the absorber boundary condition from tau_glz_sys, not tau_glz. TREAT AS AN ESTIMATE: it ignores angular dependence, spectral mismatch and the AR/TiO2 coatings on the outer pane. Compute properly in the optical model before quoting any efficiency.
DESIGN_REVIEW_GLZ_T RADE	1	1	OPEN - NEW, raised by correction C11. The second glazing pane (decision D2, kept because the built collector has it) costs about 8.5 percent of absorbed flux: at the design point q falls from $800 \times 0.91 \times 0.82 \times 0.95 = 567.1 \text{ W/m}^2$ to $800 \times 0.833 \times 0.82 \times 0.95 = 519.1 \text{ W/m}^2$, and the ROADMAP Section 1 optical penalty rises from 18 to about 25 percent. In exchange it splits the 15 mm cavity into 7.45 and 4.30 mm, which lowers U_L. WHETHER THAT TRADE IS FAVOURABLE AT THE 40 K DESIGN DIFFERENCE HAS NOT BEEN COMPUTED. ACTION: run single-pane against double-pane at matched U_L method and matched mass flow before freezing the design. If the double-glazed stack loses, delete GLASS_INNER_PANE and revert roadmap correction C9. This is a design decision, not a modelling one.

Table A1. Roadmap-sourced parameters

Parameter	Value	Expression	Source note
CFD_inlet_area	63.617 mm ²	63.617 mm ²	MEASURED area of each port face at x=±635. 12 faces total, all identical. = $\pi \times \text{port_id}^2/4$ with port_id 9 mm.
glz_inner_t	3.20 mm	3.2 mm	MEASURED inner pane thickness, GLASS_INNER_PANE z 8.500 to 11.700. Same 3.2 mm low-iron tempered glass as the outer pane. Mass 4.294 kg, currently ABSENT from the 5.2 BOM.
gap_air_lower	7.45 mm	7.45 mm	MEASURED lower air cavity, absorber top to inner pane, z 1.050 to 8.500.
gap_air_upper	4.30 mm	4.30 mm	MEASURED upper air cavity, inner pane to TIM, z 11.700 to 16.000.
FLUID_MASS_AS_MODE LLED	0.445	0.445	MEASURED hardware water plus sealed air in kg: 12 channel voids, 4 manifold voids, 2 air cavities. Excludes the 90_CFD_REFERENCE fluid duplicates.
MASS_assembly_kg	43.901	43.901	MEASURED v17. Mass of the REAL assembly, groups 00_FRAME through 10_MOUNTING, fluid included: $8.291 + 10.271 + 0.463 + 3.272 + 4.360 + 6.095 + 2.209 + 1.921 + 0.611 + 0.030 + 6.378 \text{ kg}$. Use this figure, never the number Fusion reports for the whole document.

Parameter	Value	Expression	Source note
MASS_phantom_cfd_kg	11.908	11.908	WARNING, MEASURED v17. Mass carried by NON-PHYSICAL reference bodies that duplicate real geometry: 90_CFD_REFERENCE 7.491 kg, 90_CFD_STRIP_3CH_MASTER 2.650 kg, 90_CFD_STRIP_2CH_PERIODIC 1.767 kg. A whole-document mass query returns 55.810 kg, which is 11.908 kg too high. This does not affect any dimension or any CFD result, but it WILL corrupt a BOM export or a mass figure quoted into the thesis. Subtract this, or exclude every 90_group.
CFD_suppress_occurrences	58	58	MEASURED v17. Occurrences to suppress before a CFD export: 30 FASTENER, 14 MOUNT, 5 SEAL, 8 SENSOR, 1 SELECTIVE_COATING. NOTE these 58 occurrences carry 72 BODIES, not 58 - fastener components hold more than one body each. Count occurrences, not bodies, when checking the set. The 12 PORT_ components are deliberately NOT in this set; they are part of the flow path. SEALED_AIR_GAP bodies are also excluded - they are glazing cavity air, not seals.
CFD_suppress_bodies	72	72	MEASURED v17. Body count behind CFD_suppress_occurrences. Recorded because 58 and 72 have been confused before.

Table A2. Measured parameters

Parameter	Value	Expression	Source note
ap_sheet_t	1.00 mm	ap_t / 2	DERIVED single aluminium sheet thickness
pcm_t_max	12.8571429 mm	pcm_t * 2 * pcm_grade_ratio / (1 + pcm_grade_ratio)	DERIVED thickest PCM zone
pcm_t_min	7.1428571 mm	pcm_t * 2 / (1 + pcm_grade_ratio)	DERIVED thinnest PCM zone

Table A3. Derived parameters

Parameter	Value	Expression	Source note
CFD_hdr_dP_fraction	4.8	4.8	CHECKED in v16. Header pressure drop as a percentage of channel pressure drop at the E4 design point (mdot 0.0025 kg/s, 12 channels, water rho 984 kg/m3, mu 4.9e-4 Pa.s). Laminar Hagen-Poiseuille estimate: channel 19.43 Pa over 1100 mm at mean Dh 3.9 mm, header 0.93 Pa over 480 mm at 9 mm bore. 4.8 per cent, below the 10 per cent rule of thumb, so expected channel-to-channel maldistribution is a few per cent. NOTE the header bore area 63.617 mm2 is only 0.378 times the 168.331 mm2 total inlet area of the six channels it feeds, which fails the common area rule of thumb; it passes anyway because the channels are long and narrow enough to dominate the resistance. Section 8.1 excludes manifolds from the CFD domain, so this is never checked by simulation and rests on this hand calculation alone.
CFD_hdr_Re	361	361	CHECKED in v16. Reynolds number in the header bore at the design point, 6 channels per header. Laminar, consistent with the channel range 23.9 to 198.8.

Table A4. Checked parameters

Parameter	Value	Expression	Source note
pcm_zones	7	7	ENGINEERING_ASSUMPTION number of discrete axial PCM grading zones
pcm_grade_ratio	1.8	1.8	ENGINEERING_ASSUMPTION thickest zone / thinnest zone, mean held at pcm_t
frame_wall	2.50 mm	2.5 mm	ENGINEERING_ASSUMPTION 6063-T6 extrusion wall thickness
hdr_len	45.00 mm	45 mm	ENGINEERING_ASSUMPTION manifold block length outboard of absorber end
hdr_bore	9.00 mm	9 mm	ENGINEERING_ASSUMPTION non-branching header bore diameter, no Murray-law branching
port_od	15.00 mm	15 mm	ENGINEERING_ASSUMPTION external port boss outside diameter
port_id	9.00 mm	9 mm	ENGINEERING_ASSUMPTION external port bore
hdr_slot_d	10.00 mm	10 mm	ENGINEERING_ASSUMPTION depth the absorber plate inserts into the manifold slot
hdr_x	17.50 mm	17.5 mm	ENGINEERING_ASSUMPTION header bore axis distance inboard from absorber end face
hdr_z_cold	-11.00 mm	-11 mm	ENGINEERING_ASSUMPTION cold header bore axis height, below channel plane
hdr_z_hot	14.00 mm	14 mm	ENGINEERING_ASSUMPTION hot header bore axis height, above channel plane
riser_d	9.00 mm	9 mm	ENGINEERING_ASSUMPTION vertical riser bore connecting header to channel transition
port_len	40.00 mm	40 mm	ENGINEERING_ASSUMPTION external port protrusion beyond manifold block
port_y	120.00 mm	120 mm	ENGINEERING_ASSUMPTION external port transverse position
blk_z_bot	-18.00 mm	-18 mm	ENGINEERING_ASSUMPTION manifold block bottom
blk_z_top	21.00 mm	21 mm	ENGINEERING_ASSUMPTION manifold block top
frame_top_z	33.20 mm	ap_sheet_t + gap_t + tim_t + t2_t + glz_t + 3 mm	ENGINEERING_ASSUMPTION frame top plane, 3mm lip over glazing
frame_prof_y	55.00 mm	55 mm	ENGINEERING_ASSUMPTION Y-side frame profile width, gives 490mm clear aperture
frame_prof_x	40.00 mm	40 mm	ENGINEERING_ASSUMPTION X-side frame profile width, gives 1120mm clear aperture

Parameter	Value	Expression	Source note
DESIGN_REVIEW_FRAM E_MASS	0	0	CONFIRMED AT SOURCE, ROADMAP CORRECTION REQUIRED. 5.2 gives 3.6 m of recycled Al 6063-T6 at 2.90 kg = 0.806 kg/m = 298 mm ² of section at 2700 kg/m ³ . For the 92 mm deep snap-fit glazed profile of 5.1 that implies a wall of about 0.9 mm, below the practical minimum for structural 6063-T6 extrusion (1.2-1.5 mm). The BOM mass is therefore not achievable for the stated geometry. The CAD uses frame_wall 2.5 mm (an ENGINEERING_ASSUMPTION) giving 2.30 kg/m; a realistic 1.5 mm snap-fit wall would give roughly 4.9 kg for 3.6 m. ACTION: set a manufacturable wall in 5.1 and restate the 5.2 frame mass; then frame_wall can be driven from it.
DESIGN_REVIEW_MANIF OLD_MASS	0	0	CONFIRMED AT SOURCE, ROADMAP CORRECTION REQUIRED. 5.2 gives two CNC aluminium manifold blocks at 0.35 kg total = 0.175 kg each = 64.8 cm ³ . The roadmap specifies NO manifold dimensions anywhere, so every manifold dimension in the CAD is an ENGINEERING_ASSUMPTION (hdr_len, hdr_bore, blk_z_bot, blk_z_top, hdr_x). A block that spans the 480 mm absorber width and carries two headers plus six ports cannot be 64.8 cm ³ ; the modelled block is 739 cm ³ = 2.07 kg. ACTION: either state manifold dimensions in the roadmap or restate the 5.2 mass. No CAD change until the roadmap decides.
DESIGN_REVIEW_EDGE_ INS	0	0	CLOSED-ACCEPTED as ENGINEERING_ASSUMPTION. ROADMAP D8 edge_t 15 mm cannot fit between the 480 mm absorber and the frame inner face. Modelled as four 15 mm aerogel strips around the VIP/tray perimeter in the rear cavity; mass 0.119 kg vs BOM 0.12 kg confirms the interpretation.
ENG_ASSUMPTION_TiN OX_RHO	4500	4500	ENGINEERING_ASSUMPTION, NOT FROM THE ROADMAP. ROADMAP 5.3 lists the TiNOx coating with NO density - the row reads "boundary condition", alpha 0.95 / eps 0.04 - and 5.2 lists the selective coating with NO mass. The CAD needs some density to own a solid body, so 4500 kg/m ³ was assumed (typical for TiNOx on aluminium). Now immaterial: after the coating was rebuilt at coat_t 0.0003 mm its mass is 0.000822 kg, consistent with the roadmap carrying it as massless. Never quote this density as a roadmap value.

Table A5. Engineering assumptions

Parameter	Value	Expression	Source note
G_ratio_img	1.596	1.596	SUPERSEDED 2026-09-14. REFERENCE_IMAGE Dh_out 8.3 / Dh_in 5.2 = diverging. Retained only as provenance. It is outside the A7 range 0.35-1.50 and contradicts 6.1 and 8.2; it was never a roadmap source and must not be used.
vip_L	1100.00 mm	ap_L	ACTIVE 1100 mm. NOTE: BOM 5.2 specifies 1120 mm, which overlaps the manifold blocks at x=±550. Trimmed to the absorber length. See DESIGN_REVIEW_VIP_L for the 1120 vs 1100 conflict; it is a roadmap text issue, not a CAD one.

Parameter	Value	Expression	Source note
pcm_t	10.00 mm	pcm_t_S61	ACTIVE mean PCM thickness. NOTE: 11.6 vs 10 See DESIGN_REVIEW_PCM_T for the 11.6 vs 10 conflict; it is a roadmap text issue, not a CAD one.
DESIGN_REVIEW_DH	0	0	CLOSED 2026-09-14 BY REBUILD. Re-measured on the CONVERGING loft: Dh_in 5.199326 mm against the A6 target 5.2000 (-0.013 percent) and Dh_out 2.807707 mm against the 6.1 target 2.8080 (-0.010 percent). The old +0.54 percent outlet residual belonged to the superseded diverging loft and no longer exists. ch_h_in was re-solved to 4.132953 mm to hold A6 exactly after the fillet change; ch_w_in follows at 8.700954 mm, aspect ratio 2.1053 against the A10 target 2.1.
DESIGN_REVIEW_TIM_SUPPORT	0	0	CLOSED. A 2 mm ledge was added to the two long frame rails at z 14-16 and TIM_PANEL widened to +/- 244 mm so it bears 2 mm each side. End rails cannot carry a ledge because the manifold blocks occupy that space.
CFD_defeature_tol	0.50 mm	0.5 mm	COMSOL Remove Short Edges tolerance, retained as good practice only. The 0.1358 mm lands were ELIMINATED AT SOURCE by raising the manifold transition-pocket y-margin from 0.4 to 2.0 mm (land now 1.736 mm inlet / 4.402 mm outlet). Both CFD domains now measure ZERO edges below 0.5 mm.
SURFACE_OFFSET_SUPPRESSED	1	1	Timeline item 145 SurfaceOffset1 had 12 broken references and produced 13 orphaned surface bodies. SUPPRESSED, not deleted, so nothing is lost and it can be repaired later. After suppression: 0 unhealthy features, 0 stray root bodies, and all solid volumes bit-identical (LOWER 528000.0, UPPER 640217.3, FLUID 746959.9, CFD_ABS 2652326.9 mm3).
CFD_REFERENCE_MASS_EXCLUDED	7.919	7.919	Mass in kg of the 3 bodies in 90_CFD_REFERENCE, EXCLUDED from DRY_MASS_AS_MODELLED and FLUID_MASS_AS_MODELLED because they are CFD export duplicates, not hardware. CFD_ABSORBER_SOLID_MASTER is the defeatured union used for the COMSOL solid domain; CFD_FLUID_CIRCUIT_ODD and _EVEN sum to 746959.8 mm3, which equals the 12 channel voids plus 4 manifold voids to within 0.3 mm3, confirming they are duplicates and not extra volume. Counting them was what inflated an earlier mass check to 50.878 kg.

Parameter	Value	Expression	Source note
MOUNT_BRACKET_FIX	3	3	2026-09 audit fix. Three of the four MOUNT_BRACKET occurrences carried spurious transform overrides: XMIN_YMIN was translated +700 mm in X, and both YMAX brackets carried a 180 degree rotation about X plus (0 or 700, 552, -149.6) mm. Effect: all four brackets sat at X 320..380 (the two XMIN ones fully coincident with the XMAX ones, 29131.68 mm ³ of solid-solid overlap) and the YMAX pair sat 30 mm inboard and upside down, clashing with the YMAX tilt rail and missing its lock-bolt bore. The four bracket SKETCHES were always correct and were not touched. Fixed with three named FIX_POS_ move features that apply the exact inverse of each occurrence override. Verified: all four brackets now mirror-symmetric at X +/- (320..380), Y +/- (272..310), Z -90.8..-58.8, and a full 138-body interference test returns ZERO real hardware interferences.
ROADMAP_CORRECTION_REQUIRED	6	6	Roadmap TEXT corrections, all confirmed at source and all OUTSIDE the CAD: (1) 5.2 TOTAL reads 17.82 kg but the line items sum to 22.42 kg; (2) 5.2 VIP mass 1.05 kg contradicts the 5.3 density 190 kg/m ³ , should be 2.13 kg; (3) 5.2 frame mass 2.90 kg implies a 0.9 mm wall, below manufacturable minimum; (4) 5.2 manifold mass 0.35 kg is unachievable and no manifold dimensions are stated anywhere; (5) 5.1 PCM 11.6 mm contradicts 6.1 PCM 10 mm; (6) 5.2 VIP length 1120 mm overlaps the manifolds. Each carries its evidence in its own DESIGN_REVIEW_ parameter.
ROADMAP_VERIFIED_AT_SOURCE	1	1	Sections 4.1 (Group A), 4.2 (Group B), 5.1, 5.2, 5.3, 6.1 and 8.2 were read directly from GRAIL_Collector_Roadmap.docx on 2026-09-14 rather than from working notes. Findings recorded in the individual DESIGN_REVIEW_ parameters. Two earlier working-note claims were WITHDRAWN on that evidence: the w_bridge pitch-minus-Dh hypothesis (B3 is simply SWEPT 20-60 and 26.195 is in range) and the idea that 8.2 needed correcting (8.2 is right; the as-built grading direction is what is wrong).
CFD_MASTER_DELTA	452.934	452.934	Volume difference in mm ³ between CFD_ABSORBER_SOLID_MASTER (2608699.950) and the four hardware solids it unions - absorber lower and upper sheets plus both manifold blocks (2608247.016), measured 2026-09-14 after the grading rebuild. The master carries slightly MORE material because its manifold transition pockets were deliberately shrunk (y-margin raised from 0.4 to 2.0 mm) to eliminate sub-0.5 mm lands before meshing. Expected and documented, not an error. The CFD fluid circuits match the 12 channel voids plus 4 manifold voids to 0.000 mm ³ exactly.
CFD_strip_n_ch	3	3	Section 8.1: alternating counter-current is a multi-channel effect; strip carries 3 channels (odd-even-odd) so the middle channel has a counter-flowing neighbour on both sides.

Parameter	Value	Expression	Source note
CFD_strip_y_min	-80.00 mm	-80 mm	Strip cut plane, low y. Lands on the bridge midline between CH04 and CH05 - 11.098 mm clear of the nearest channel feature.
CFD_strip_y_max	40.00 mm	40 mm	Strip cut plane, high y. Bridge midline between CH07 and CH08.
CFD_strip_width	120.00 mm	CFD_strip_y_max - CFD_strip_y_min	Strip width = 3 x ch_pitch = 120 mm.
CFD_strip_x_half	550.00 mm	550 mm	Strip x half-length. Solid trimmed to the channel ends so fluid inlet/outlet faces are coplanar with the solid end faces. Manifolds excluded - Section 8.1 names no-manifold-maldistribution as a limitation.
DESIGN_REVIEW_STRIP_BC	1	1	OPEN. Section 8.1 specifies a three-channel strip WITH SYMMETRY. A plain symmetry (adiabatic) plane at a bridge midline is not exact under alternating flow: the neighbour across the plane runs the opposite direction, so the field is glide-symmetric, not mirror-symmetric. Baseline kept as written in 8.1. Read results from the MIDDLE channel only and treat the two outer channels as buffers; or switch to a 2-channel domain with translational periodicity in y (period 2 x ch_pitch = 80 mm), which is exact and cheaper. Decide before the production sweep.
DESIGN_REVIEW_EVEN_CH_ASYM	0	0	CLOSED in v16. Tested by sampling 169 points on each fluid body lateral wall and querying pointContainment against the strip solid: CH05, CH06 and CH07 all returned 169 ON, 0 INSIDE, 0 OUTSIDE. The interface is conformal on every channel, with no gap and no interpenetration. All three fluid bodies also carry identical face areas (7.920, 28.055, 18073.283 mm ²), so the even channel is an exact mirror. The 66.927 mm ³ union deficit and 29.868 mm ³ sample-box difference were boolean simplification artifacts, not geometry.
CFD_export_exclude_90	1	1	v17. Any export of the FULL model for CFD must also exclude the three 90_ reference groups (90_CFD_REFERENCE, 90_CFD_STRIP_3CH_MASTER, 90_CFD_STRIP_2CH_PERIODIC). They duplicate real geometry and would appear as interpenetrating solids in the mesher. They are hidden in the browser for this reason, but hidden is not excluded.

Parameter	Value	Expression	Source note
DESIGN_REVIEW_REQUI RED	1	1	CAD-OPEN ITEMS (2): GLZ_TRADE and STRIP_BC. All GEOMETRIC conflicts are resolved: the channel converges per A7 and 6.1, the fillet floor is cleared by D4, the conjugate interfaces are conformal on every channel (169 of 169 boundary points, verified v16), and the timeline carries 0 unhealthy and 0 suppressed features. The two items that remain are MODELLING DECISIONS, not geometry, and neither can be closed by moving a face. GLZ_TRADE needs the single-pane against double-pane comparison at matched U_L method. STRIP_BC needs a choice between the three-channel symmetry domain and the two-channel periodic domain; both are built. This comment was stale between v16 and v17 - it still read ONE open item after STRIP_BC was raised. Update it whenever a DESIGN_REVIEW parameter is opened or closed.

Table A6. Review flags, counters and control parameters

Appendix B. CFD export domains and CAD verification gates

Component	90_CFD_STRIP_3CH_MASTER
Extent in x	–550 to +550 mm, the full channel length
Extent in y	–80 to +40 mm, width 120 mm = 3 × channel pitch
Channels	CH05, CH06, CH07 — odd, even, odd
Cut planes	Both on bridge midlines, 11.098 mm clear of the nearest channel feature
Solid volume	281,693.638 mm ³
Fluid bodies	3 × 18,643.963 mm ³
Boundary condition	Symmetry, as the roadmap specifies — see the caveat below

Table B1. Three-channel symmetry strip

Component	90_CFD_STRIP_2CH_PERIODIC
Extent in y	–80 to 0 mm, width 80 mm = 2 × channel pitch
Channels	CH05, CH06 — one of each direction
Solid volume	187,817.190 mm ³ , about two thirds of the three-channel domain
Boundary condition	Translational periodicity, period 80 mm
Periodic faces	Exactly 2200.000000 mm ² each, identical rectangles in the bridge material

Table B2. Two-channel periodic strip

Setting	Value
Occurrences to suppress	58 — 30 fastener, 14 mount, 5 seal, 8 sensor, 1 selective coating
Bodies behind those 58	72 — fastener components hold more than one body each
Deliberately NOT suppressed	The 12 PORT_ components; they are part of the flow path
Also excluded from a full export	All three 90_ groups — they duplicate real geometry and would appear as interpenetrating solids
Minimum edge length	0.50000 mm, no edges below the meshing floor, no sliver faces
Reynolds range	23.9 to 198.8 — laminar throughout, no turbulence model required

Table B3. Export settings

Gate	Result	Detail
Timeline healthy	PASS	948 features, 0 unhealthy
No suppressed features	PASS	0
All bodies solid	PASS	145
No stray bodies at root	PASS	0
No duplicate body names	PASS	0 duplicates
No degenerate bodies	PASS	none under 1 mm ³
All bodies within envelope	PASS	0 outliers
All sketches closed	PASS	285 sketches
Joints healthy	PASS	4 joints
No external references	PASS	0
Every body has a material	PASS	0 default or unassigned
Parameter expressions resolve	PASS	0 unresolved tokens

Gate	Result	Detail
Channel converges	PASS	Dh ratio 0.540339
Dome section not inverted	PASS	4.133 mm against a 2.0 mm threshold
Conjugate interfaces conformal	PASS	169 of 169 points on boundary, every channel
Review summary consistent	PASS	exactly 2 open, matching the register

Table B4. CAD verification gates

Appendix C. Dataset data dictionary (Rev 4.2)

column	unit	category	definition_and_source
case_id	-	METADATA	Case identifier. ALT_n = alternating, PAR_n = parallel. ALT_n and PAR_n are independent LHS samples and are NOT matched pairs
geometry_version	-	METADATA	Geometry build used (converging CAD, Gate A verified)
solver	-	METADATA	GRAIL-CHT reduced-order conjugate model (not a 3-D CFD solver)
arrangement	-	MODEL INPUT (sampled)	Flow arrangement: alternating (counter-current) or parallel
f_interdig	-	MODEL INPUT (sampled)	1 = alternating, 0 = parallel
G_T_W_m2	W/m ²	MODEL INPUT (sampled)	Incident solar irradiance on the aperture
T_in_K	K	MODEL INPUT (sampled)	Water inlet temperature
T_amb_K	K	MODEL INPUT (sampled)	Ambient air temperature
v_wind_m_s	m/s	MODEL INPUT (sampled)	Wind speed
mdot_total_kg_s	kg/s	MODEL INPUT (sampled)	Total water mass flow
mdot_channel_kg_s	kg/s	DERIVED (computed from other columns)	mdot_total / 12
bridge_mm	mm	MODEL INPUT (sampled)	Bridge (metal land) width between channels
g_ratio	-	MODEL INPUT (sampled)	Grading ratio $G = Dh_{out} / Dh_{in}$
lambda_G	-	MODEL INPUT (sampled)	Grading exponent of the hydraulic-diameter profile
Dh_in_mm	mm	DERIVED (computed from other columns)	Inlet hydraulic diameter (same for all cases)
Dh_out_mm	mm	DERIVED (computed from other columns)	$Dh_{in} \times g_ratio$
A_in_mm2	mm ²	DERIVED (computed from other columns)	Inlet cross-section area (same for all cases)
A_out_mm2	mm ²	DERIVED (computed from other columns)	$A_{in} \times (Dh_{out}/Dh_{in})^2$
Re_in	-	DERIVED (computed from other columns)	$mdot_channel \cdot Dh_{in} / (A_{in} \cdot \mu)$ — non-circular definition (Rev 4.1)
Re_out	-	DERIVED (computed from other columns)	$mdot_channel \cdot Dh_{out} / (A_{out} \cdot \mu)$

column	unit	category	definition_and_source
mu_Pa_s	Pa·s	DERIVED (computed from other columns)	Water viscosity, Vogel correlation at $\frac{1}{2}(T_{in}+T_{out})$; evaluated with the nx220 T_out (≤ 0.17 % difference)
dp_channel_Pa	Pa	MODEL OUTPUT (solved)	Channel pressure drop: calibrated laminar integral (calibration factor from 3-D CFD); a model result, not a per-case 3-D CFD result
W_pump_W	W	DERIVED (computed from other columns)	$dp_channel \times \dot{m}_{total} / 997$
q_abs_W_m2	W/m ²	DERIVED (computed from other columns)	Absorbed flux = $G_T \times 0.833 \times 0.82 \times 0.95$ (optical factor 0.6489)
T_sky_K	K	DERIVED (computed from other columns)	$0.0552 \times T_{amb}^{1.5}$
h_wind_W_m2K	W/m ² K	DERIVED (computed from other columns)	$5.7 + 3.8 \times v_{wind}$
T_out_K	K	MODEL OUTPUT (solved)	Mixed-mean outlet temperature
dT_fluid_K	K	DERIVED (computed from other columns)	$T_{out} - T_{in}$
Qu_W	W	DERIVED (computed from other columns)	Useful heat = $\dot{m}_{total} \times 4180 \times dT_{fluid}$
eta	-	DERIVED (computed from other columns)	Thermal efficiency = $Qu / (G_T \times 0.528 \text{ m}^2)$, on INCIDENT irradiance
T_plate_mean_K	K	MODEL OUTPUT (solved)	Area-mean plate temperature
T_plate_max_K	K	MODEL OUTPUT (solved)	Maximum plate temperature
T_plate_min_K	K	MODEL OUTPUT (solved)	Minimum plate temperature
plate_spread_K	K	DERIVED (computed from other columns)	$T_{plate_max} - T_{plate_min}$
plate_std_K	K	MODEL OUTPUT (solved)	Standard deviation of plate temperature (uniformity metric)
P90_K	K	MODEL OUTPUT (solved)	90th-percentile plate temperature
P95_K	K	MODEL OUTPUT (solved)	95th-percentile plate temperature
P99_K	K	MODEL OUTPUT (solved)	99th-percentile plate temperature
R4_K4	K ⁴	MODEL OUTPUT (solved)	Plate mean of T^4 (radiative functional)

column	unit	category	definition_and_source
Tmean_pow4_K4	K ⁴	DERIVED (computed from other columns)	(T_plate_mean) ⁴ — NOTE: fourth power of the mean, not mean of T ⁴
T_R4_K	K	DERIVED (computed from other columns)	R4 ^{^(1/4)} , effective radiative plate temperature
Q_solar_W	W	DERIVED (computed from other columns)	ABSORBED solar power = q_abs × 0.528 (not incident)
Q_rad_W	W	MODEL OUTPUT (solved)	Net radiative heat flow plate→glazing (signed; negative = heat gain)
Q_conv_W	W	MODEL OUTPUT (solved)	Net convective/conductive heat flow plate→glazing (signed)
Q_rear_W	W	MODEL OUTPUT (solved)	Net rear heat flow plate→ambient (signed; negative when plate below ambient)
U_L_W_m2K	W/m ² K	DERIVED (computed from other columns)	(Q_solar – Qu)/(0.528 (T_plate_mean – T_amb)); undefined/unreliable when T_plate_mean – T_amb < 5 K
T_glass_mean_K	K	MODEL OUTPUT (solved)	Mean glazing temperature
lateral_bridge_W	W	MODEL OUTPUT (solved)	Heat conducted laterally through the bridges (≈0 in parallel flow by symmetry)
energy_error_pct	%	SOLVER RECORD	Solver energy-balance diagnostic (nx220 run)
outer_iters	-	SOLVER RECORD	Outer iterations (max of the two plate grids)
residual	-	SOLVER RECORD	Final outer-iteration residual (nx220 run)
converged	-	SOLVER RECORD	Convergence flag (tolerance 1e-6)
Nu_closure_input	-	FIXED MODEL INPUT (closure / constant)	**MODEL INPUT, NOT A PER-CASE CFD OUTPUT.** Fixed Nusselt closure applied to every case; derived once from grid-studied 3-D conjugate OpenFOAM CFD (4.48, GCI 2.1 %). Constant across all 300 cases by design (developed Nu was flow-independent in 3-D: 4.60 at 1.0, 2.49 and 4.5 g/s)
Nu_closure_basis	-	METADATA	Provenance of Nu_closure_input: conjugate 3-D benchmark, H1-type thermal condition
dataset_rev	-	METADATA	Integer revision field; file is Rev 4.2 (see filename and change log)
k_model	-	METADATA	Water thermal conductivity model k(T)
outer_cap	-	METADATA	Outer-iteration cap (3000)
plate_nx	-	METADATA	Plate-grid treatment: Richardson 2·f(220) – f(110)
plate_ny	-	METADATA	Plate grid cells across width (120)

Table C1. All 59 columns

Appendix D. CAD body inventory

Group	Body	Material	Volume	Mass
FRAME	FASTENER_M5_BEZEL_01	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_01_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_02	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_02_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_03	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_03_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_04	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_04_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_05	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_05_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_06	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_06_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_07	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_07_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_08	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_08_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_09	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_09_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_10	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_10_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_11	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_11_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_12	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_12_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_13	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_13_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_BEZEL_14	SS316	141.372	0.0011
FRAME	FASTENER_M5_BEZEL_14_HEAD	SS316	190.852	0.0015
FRAME	FASTENER_M5_CORNER_XMAX_Y MAX_1	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MAX_2	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MAX_3	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MAX_4	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MIN_1	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MIN_2	SS316	548.993	0.0043

Group	Body	Material	Volume	Mass
FRAME	FASTENER_M5_CORNER_XMAX_Y MIN_3	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMAX_Y MIN_4	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MAX_1	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MAX_2	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MAX_3	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MAX_4	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MIN_1	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MIN_2	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MIN_3	SS316	548.993	0.0043
FRAME	FASTENER_M5_CORNER_XMIN_Y MIN_4	SS316	548.993	0.0043
FRAME	FRAME_BOTTOM	Al6063T6	286,275	0.7729
FRAME	FRAME_CORNER_BLOCK_XMAX_Y MAX	Al6063T6	46,821.903	0.1264
FRAME	FRAME_CORNER_BLOCK_XMAX_Y MIN	Al6063T6	46,821.903	0.1264
FRAME	FRAME_CORNER_BLOCK_XMIN_Y MAX	Al6063T6	46,821.903	0.1264
FRAME	FRAME_CORNER_BLOCK_XMIN_Y MIN	Al6063T6	46,821.903	0.1264
FRAME	FRAME_SIDE_BOTTOM	Al6063T6	996,611.429	2.6909
FRAME	FRAME_SIDE_LEFT	Al6063T6	180,102.184	0.4863
FRAME	FRAME_SIDE_RIGHT	Al6063T6	180,102.184	0.4863
FRAME	FRAME_SIDE_TOP	Al6063T6	996,611.429	2.6909
FRAME	FRAME_TOP_BEZEL	Al6063T6	204,258	0.5515
GLAZING	GLASS_INNER_PANE	SolarGlass	1,717,760	4.2944
GLAZING	GLASS_OUTER_PANE	SolarGlass	2,078,720	5.1968
GLAZING	THERMOTROPIC_LAYER_T2	PMMA_thermo	649,600	0.7795
TIM	SEALED_AIR_GAP_LOWER	Air	3,665,606.365	0.0044
TIM	SEALED_AIR_GAP_UPPER	Air	2,270,400	0.0027
TIM	TIM_PANEL	PC_honeycomb	5,368,000	0.4563
ABSORBER_ASSEMBLY	ABSORBER_LOWER_SHEET	AA1050	528,000	1.4282
ABSORBER_ASSEMBLY	ABSORBER_UPPER_SHEET	AA1050	598,692.05	1.6195
ABSORBER_ASSEMBLY	FLUID_CH01	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH02	Water300K	18,643.963	0.0186

Group	Body	Material	Volume	Mass
ABSORBER_ASSEMBLY	FLUID_CH03	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH04	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH05	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH06	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH07	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH08	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH09	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH10	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH11	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	FLUID_CH12	Water300K	18,643.963	0.0186
ABSORBER_ASSEMBLY	SELECTIVE_COATING_TiNOX	TiNOX	171.05	0.0008
MANIFOLD	MANIFOLD_BLOCK_LEFT	AA1050	739,041.35	1.9991
MANIFOLD	MANIFOLD_BLOCK_RIGHT	AA1050	739,041.35	1.9991
MANIFOLD	MANIFOLD_FLUID_LEFT_COLD	Water300K	50,842.13	0.0507
MANIFOLD	MANIFOLD_FLUID_LEFT_HOT	Water300K	57,109.461	0.0569
MANIFOLD	MANIFOLD_FLUID_RIGHT_COLD	Water300K	50,842.13	0.0507
MANIFOLD	MANIFOLD_FLUID_RIGHT_HOT	Water300K	57,109.461	0.0569
MANIFOLD	PORT_COLD_INLET_LEFT_N200	AA1050	4,523.893	0.0122
MANIFOLD	PORT_COLD_INLET_LEFT_N40	AA1050	4,523.893	0.0122
MANIFOLD	PORT_COLD_INLET_LEFT_P120	AA1050	4,523.893	0.0122
MANIFOLD	PORT_COLD_INLET_RIGHT_N200	AA1050	4,523.893	0.0122
MANIFOLD	PORT_COLD_INLET_RIGHT_N40	AA1050	4,523.893	0.0122
MANIFOLD	PORT_COLD_INLET_RIGHT_P120	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_LEFT_N120	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_LEFT_P200	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_LEFT_P40	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_RIGHT_N120	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_RIGHT_P200	AA1050	4,523.893	0.0122
MANIFOLD	PORT_HOT_OUTLET_RIGHT_P40	AA1050	4,523.893	0.0122
PCM_TRAY	GRAPHITE_INTERFACE	Graphite	211,152.876	0.2323
PCM_TRAY	PCM_CONTAINMENT_TRAY	AA1050	450,509.019	1.2186
PCM_TRAY	PCM_ZONE_01	RT55_EG	538,734.277	0.4741
PCM_TRAY	PCM_ZONE_02	RT55_EG	610,305.045	0.5371
PCM_TRAY	PCM_ZONE_03	RT55_EG	682,100.546	0.6002
PCM_TRAY	PCM_ZONE_04	RT55_EG	753,978.514	0.6635
PCM_TRAY	PCM_ZONE_05	RT55_EG	825,774.016	0.7267
PCM_TRAY	PCM_ZONE_06	RT55_EG	897,651.984	0.7899
PCM_TRAY	PCM_ZONE_07	RT55_EG	969,365.018	0.8530
INSULATION	EDGE_INSULATION_XMAX	Aerogel	124,178.572	0.0186

Group	Body	Material	Volume	Mass
INSULATION	EDGE_INSULATION_XMIN	Aerogel	124,178.572	0.0186
INSULATION	EDGE_INSULATION_YMAX	Aerogel	273,192.858	0.0410
INSULATION	EDGE_INSULATION_YMIN	Aerogel	273,192.858	0.0410
INSULATION	VIP_PANEL	VIP_silica	11,000,000	2.0900
BACKSHEET	BACKSHEET_PANEL	AA1050	710,007.124	1.9206
SEALS	SEAL_BACKSHEET_PERIMETER	EPDM	202,982.412	0.2334
SEALS	SEAL_GLAZING_LOWER	EPDM	204,410.221	0.2351
SEALS	SEAL_GLAZING_UPPER	EPDM	102,205.111	0.1175
SEALS	SEAL_MANIFOLD_LEFT	EPDM	10,675.2	0.0123
SEALS	SEAL_MANIFOLD_RIGHT	EPDM	10,675.2	0.0123
SENSING	SENSOR_DS18B20_1	Electronics	49.087	0.0001
SENSING	SENSOR_DS18B20_2	Electronics	49.087	0.0001
SENSING	SENSOR_DS18B20_3	Electronics	49.087	0.0001
SENSING	SENSOR_DS18B20_4	Electronics	49.087	0.0001
SENSING	SENSOR_DS18B20_5	Electronics	49.087	0.0001
SENSING	SENSOR_DS18B20_6	Electronics	49.087	0.0001
SENSING	SENSOR_ESP32_BOARD	Electronics	7,280	0.0102
SENSING	SENSOR_PYRANOMETER_Si	Electronics	14,137.167	0.0198
MOUNTING	MOUNT_BRACKET_XMAX_YMAX	Al6063T6	29,131.681	0.0787
MOUNTING	MOUNT_BRACKET_XMAX_YMIN	Al6063T6	29,131.681	0.0787
MOUNTING	MOUNT_BRACKET_XMIN_YMAX	Al6063T6	29,131.681	0.0787
MOUNTING	MOUNT_BRACKET_XMIN_YMIN	Al6063T6	29,131.681	0.0787
MOUNTING	MOUNT_LOCK_BOLT_XMAX_YMAX	Al6063T6	578.053	0.0016
MOUNTING	MOUNT_LOCK_BOLT_XMAX_YMIN	Al6063T6	578.053	0.0016
MOUNTING	MOUNT_LOCK_BOLT_XMIN_YMAX	Al6063T6	578.053	0.0016
MOUNTING	MOUNT_LOCK_BOLT_XMIN_YMIN	Al6063T6	578.053	0.0016
MOUNTING	MOUNT_PIVOT_PIN_XMAX_YMAX	Al6063T6	3,534.292	0.0095
MOUNTING	MOUNT_PIVOT_PIN_XMAX_YMIN	Al6063T6	3,534.292	0.0095
MOUNTING	MOUNT_PIVOT_PIN_XMIN_YMAX	Al6063T6	3,534.292	0.0095
MOUNTING	MOUNT_PIVOT_PIN_XMIN_YMIN	Al6063T6	3,534.292	0.0095
MOUNTING	MOUNT_TILT_RAIL_YMAX	Al6063T6	1,114,690.708	3.0097
MOUNTING	MOUNT_TILT_RAIL_YMIN	Al6063T6	1,114,690.708	3.0097
CFD_REFERENCE	CFD_ABSORBER_SOLID_MASTER	AA1050	2,607,284.779	7.0527
CFD_REFERENCE	CFD_FLUID_CIRCUIT_EVEN_CH	Water300K	219,815.369	0.2192
CFD_REFERENCE	CFD_FLUID_CIRCUIT_ODD_CH	Water300K	219,815.369	0.2192
CFD_STRIP_2CH_PERIODIC	STRIP2_FLUID_CH05	Steel	18,643.963	0.1464
CFD_STRIP_2CH_PERIODIC	STRIP2_FLUID_CH06	Steel	18,643.963	0.1464

Group	Body	Material	Volume	Mass
CFD_STRIP_2CH_PERIODIC	STRIP2_SOLID_2CH	Steel	187,817.19	1.4744
CFD_STRIP_3CH_MASTERR	STRIP_FLUID_CH05	Steel	18,643.963	0.1464
CFD_STRIP_3CH_MASTERR	STRIP_FLUID_CH06	Steel	18,643.963	0.1464
CFD_STRIP_3CH_MASTERR	STRIP_FLUID_CH07	Steel	18,643.963	0.1464
CFD_STRIP_3CH_MASTERR	STRIP_SOLID_3CH	Steel	281,693.638	2.2113

Table D1. Every body in the assembly

Appendix E. Earlier (Rev 2) figures kept for the record

Read before using these figures

These figures were made at Rev 2 with the superseded Nusselt closure 2.92 and other early settings. Their thermal numbers are NOT the project results. They are kept only to show the history of the work. Use Sections 4–14 for all results.

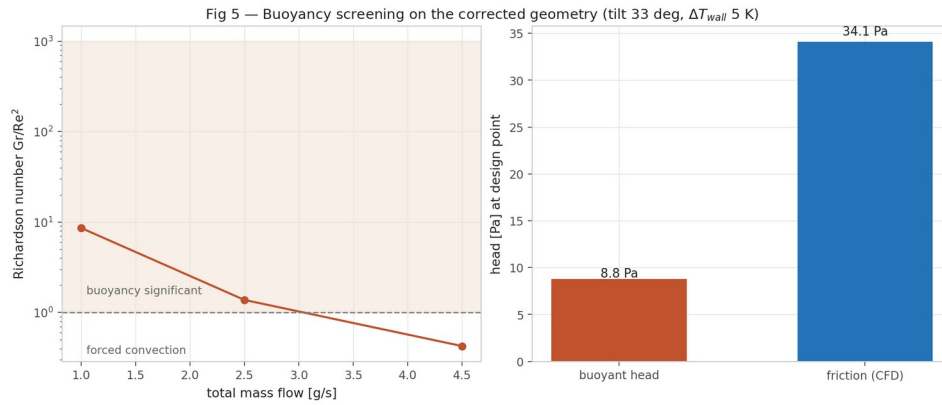


Figure E.1. Buoyancy check (Rev 2) — superseded numbers

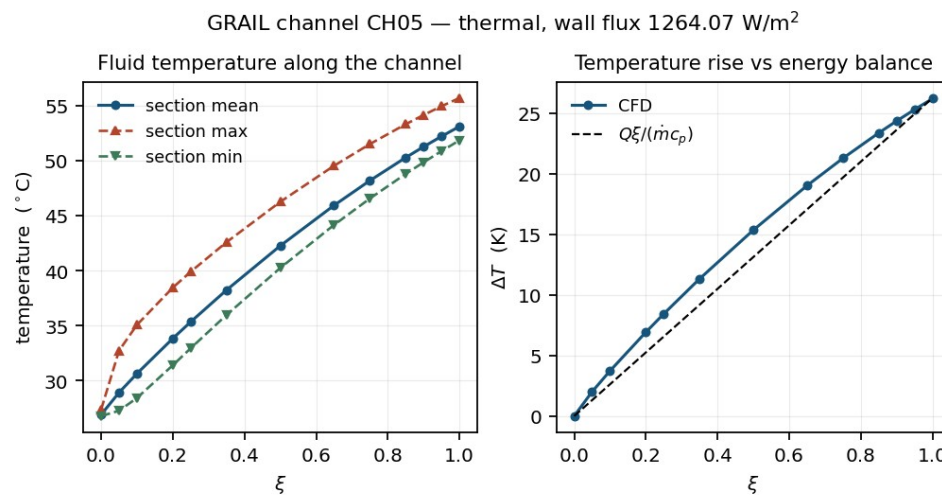


Figure E.2. 3-D thermal result (Rev 2) — superseded numbers

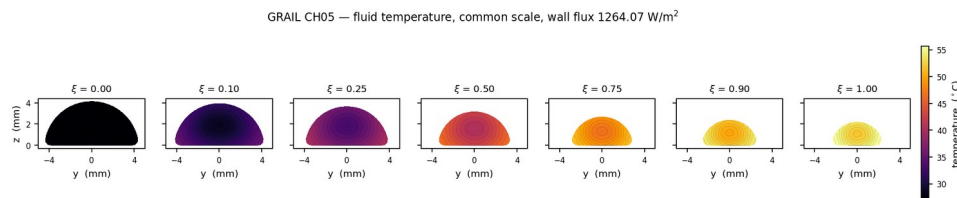


Figure E.3. 3-D temperature (Rev 2) — superseded numbers

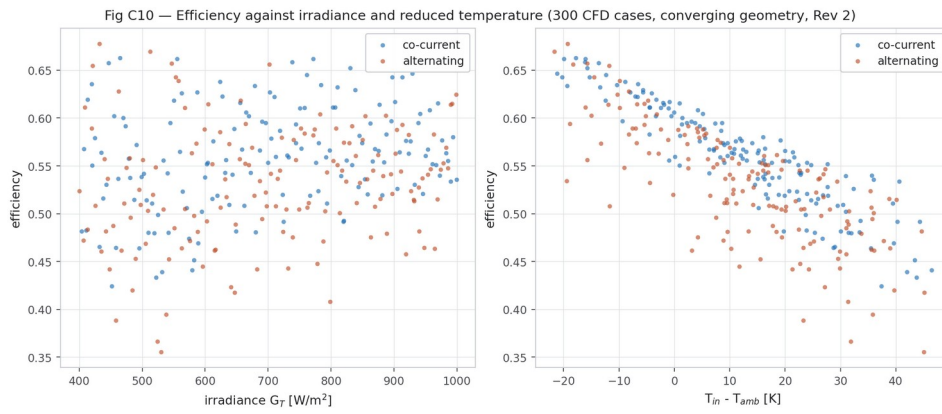


Figure E.4. Efficiency (Rev 2) — superseded numbers

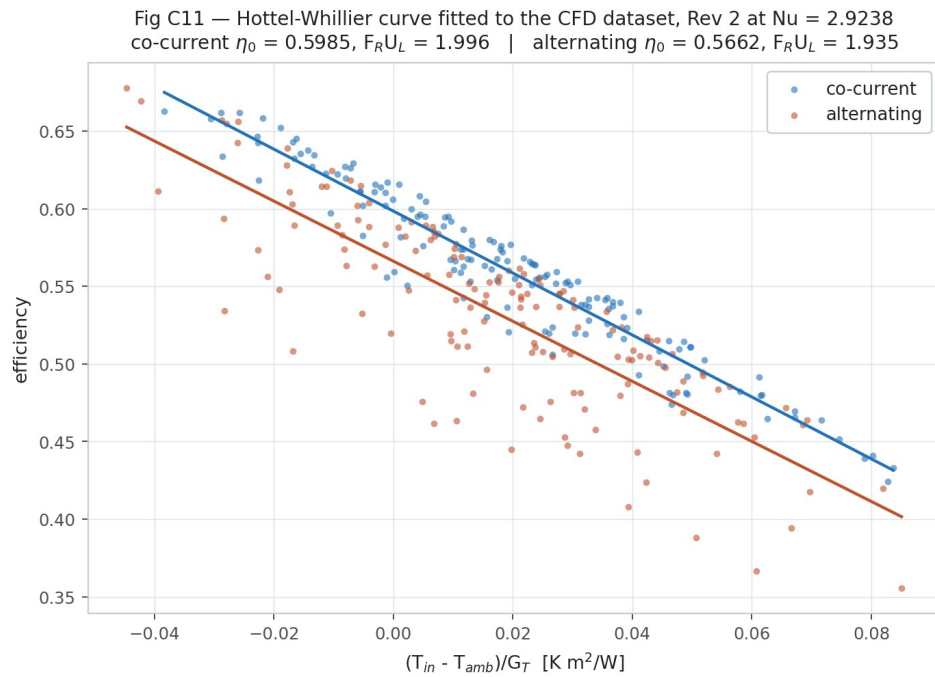


Figure E.5. Hottel-Whillier comparison (Rev 2) — superseded numbers

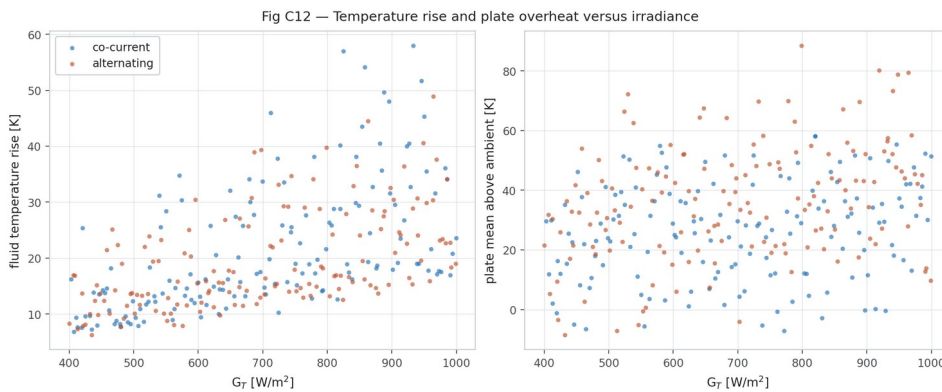


Figure E.6. Temperature rise (Rev 2) — superseded numbers

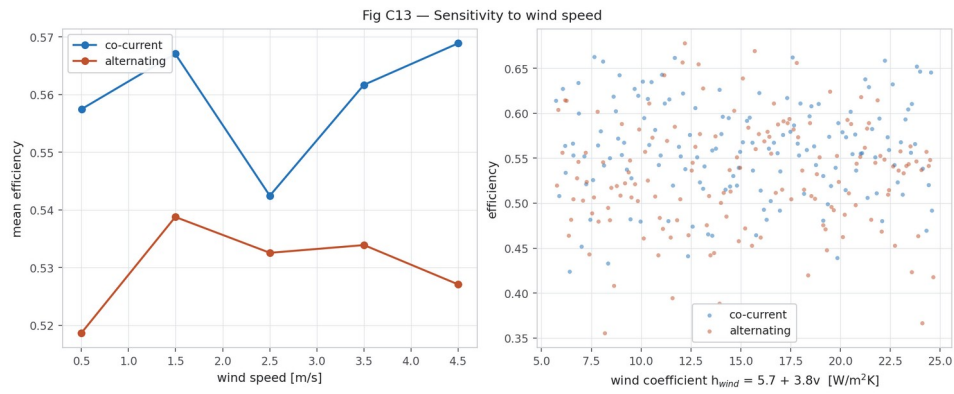


Figure E.7. Wind effect (Rev 2) — superseded numbers

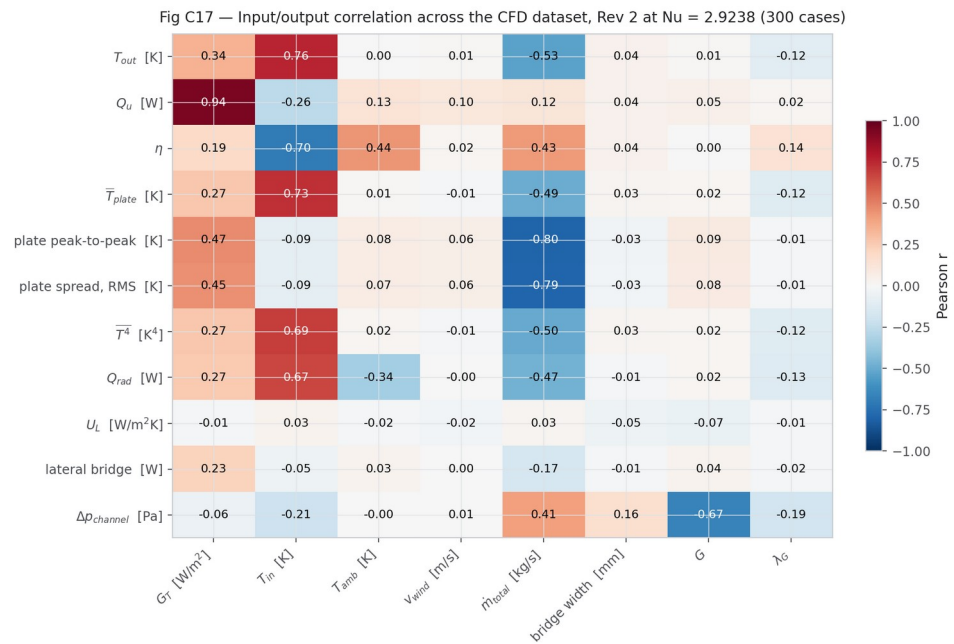


Figure E.8. Correlation (Rev 2) — superseded numbers

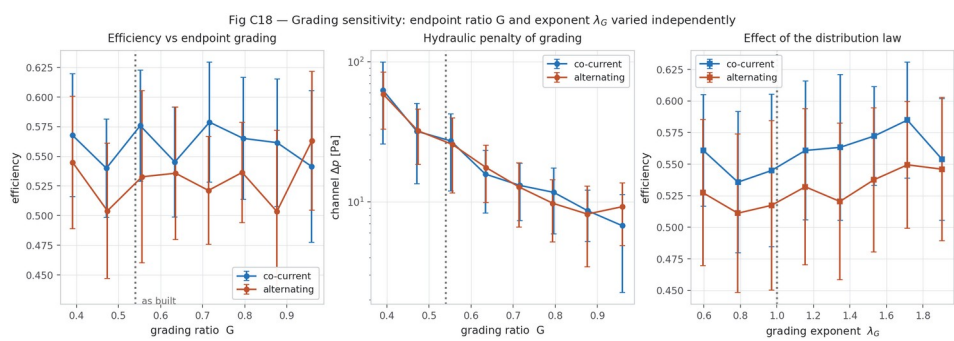


Figure E.9. Grading effect (Rev 2) — superseded numbers

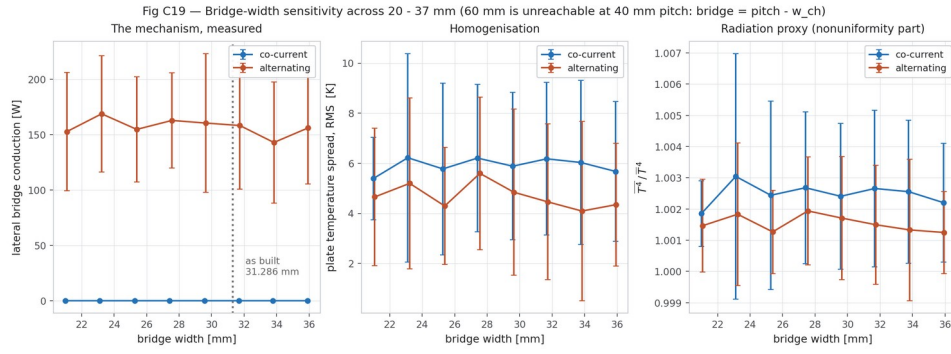


Figure E.10. Bridge effect (Rev 2) — superseded numbers

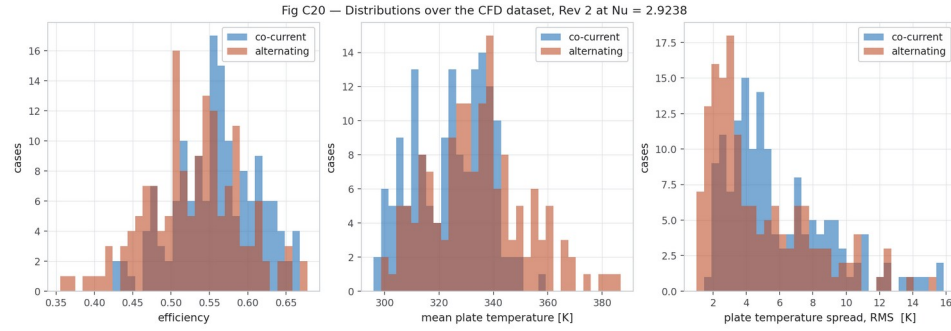


Figure E.11. Distributions (Rev 2) — superseded numbers

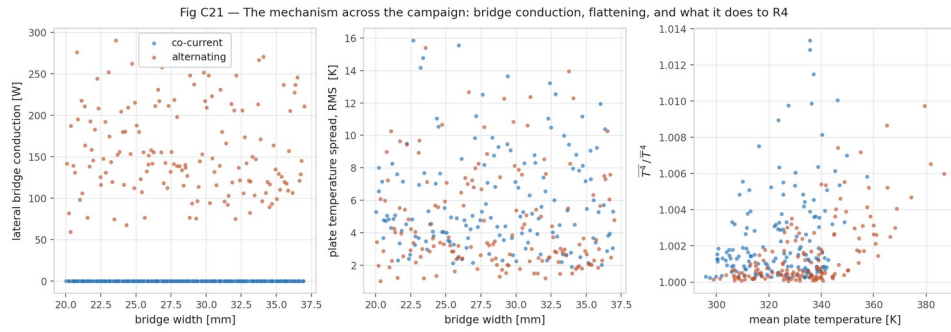


Figure E.12. Mechanism in campaign (Rev 2) — superseded numbers

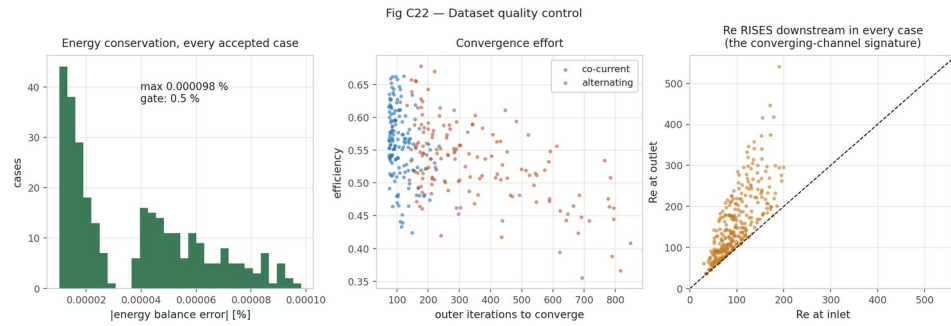


Figure E.13. Campaign quality (Rev 2) — superseded numbers

Fig D1 — Domain equivalence (CR-03). Under alternating flow only the 2-channel PERIODIC domain reproduces the full plate; an adiabatic symmetry plane inflates bridge conduction by 67 %.

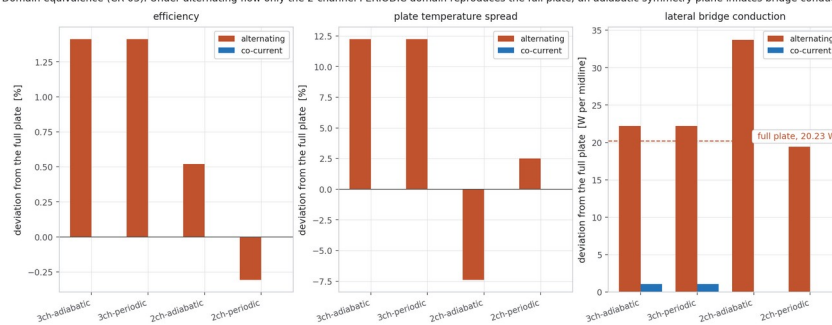


Figure E.14. Domain equivalence, 3-channel vs 2-channel (Rev 2) — superseded numbers

Fig D2 — The same alternating solution on three domains, as deviation from each field's own axial mean. Adiabatic symmetry planes force 67 % more flux through the one interior bridge

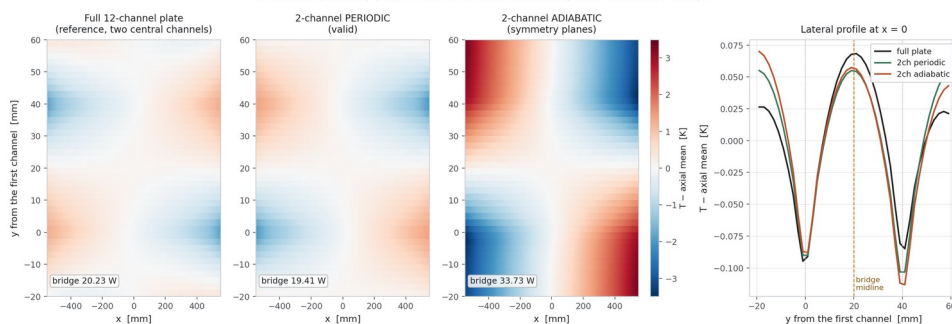


Figure E.15. Domain fields (Rev 2) — superseded numbers

GRAIL mechanism study — matched mass flow, irradiance, geometry and ambient

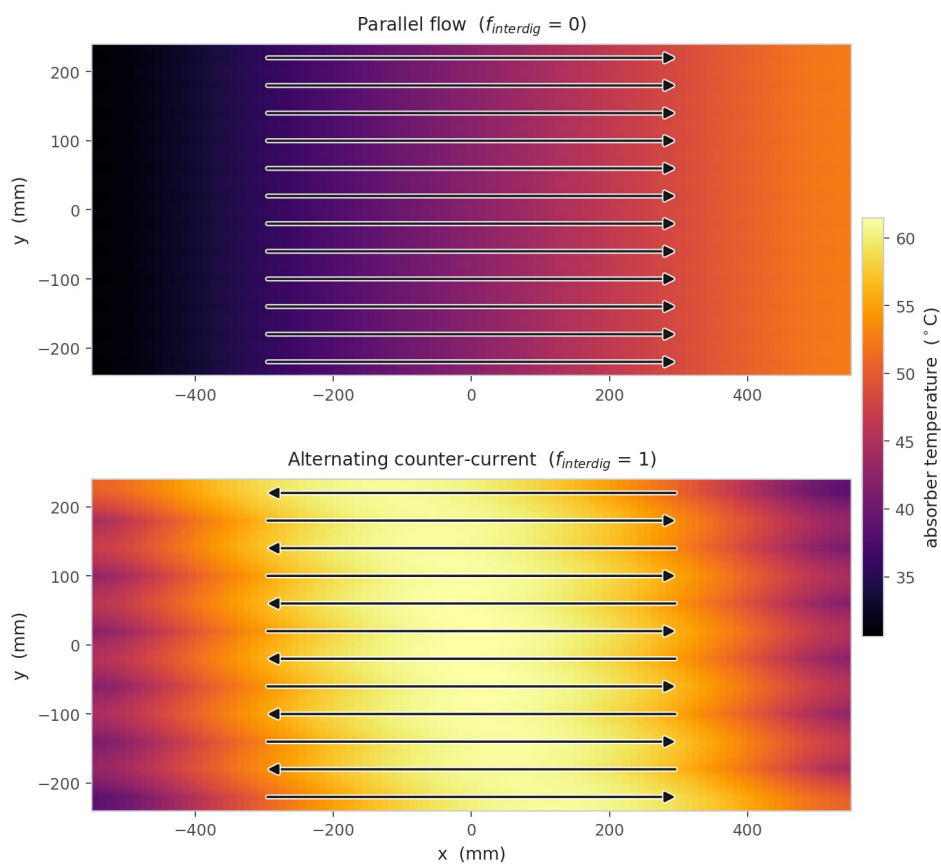


Figure E.16. Mechanism fields (Rev 2) — superseded numbers

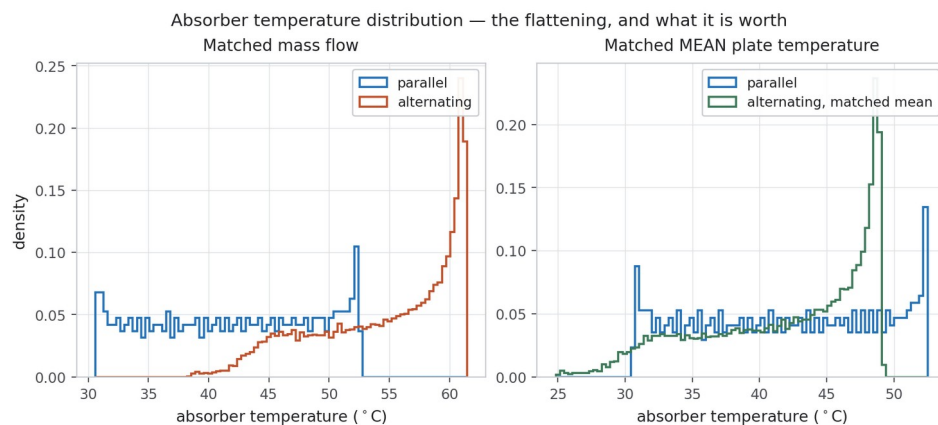


Figure E.17. Temperature distribution (Rev 2) — superseded numbers

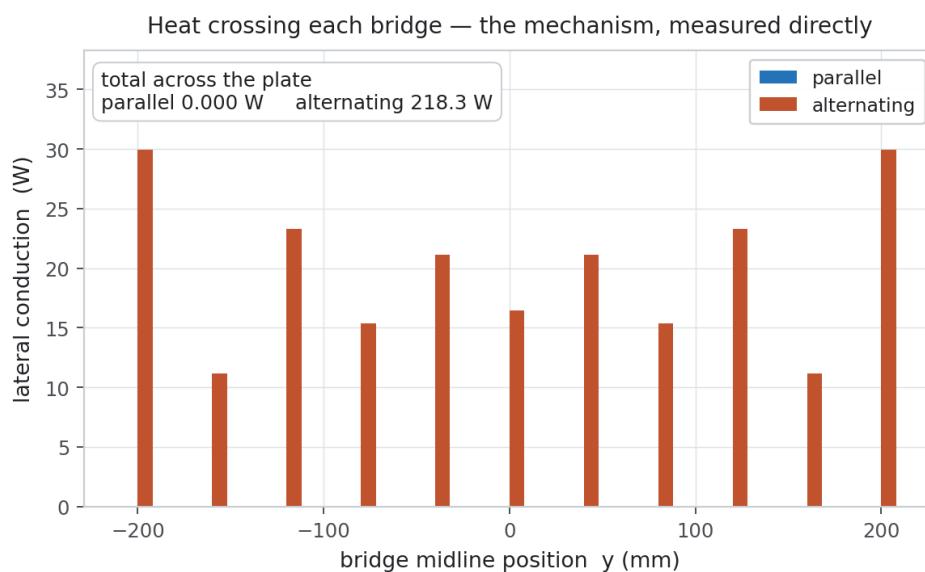


Figure E.18. Bridge conduction (Rev 2) — superseded numbers

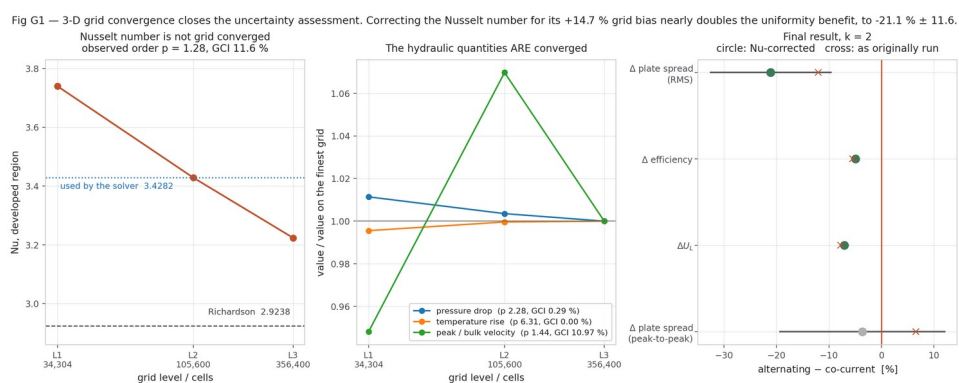


Figure E.19. 3-D grid convergence (Rev 2) — superseded numbers

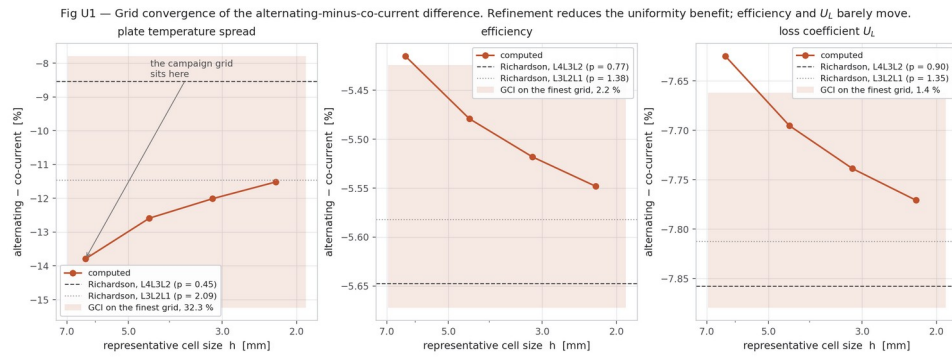


Figure E.20. Grid convergence (Rev 2) — superseded numbers

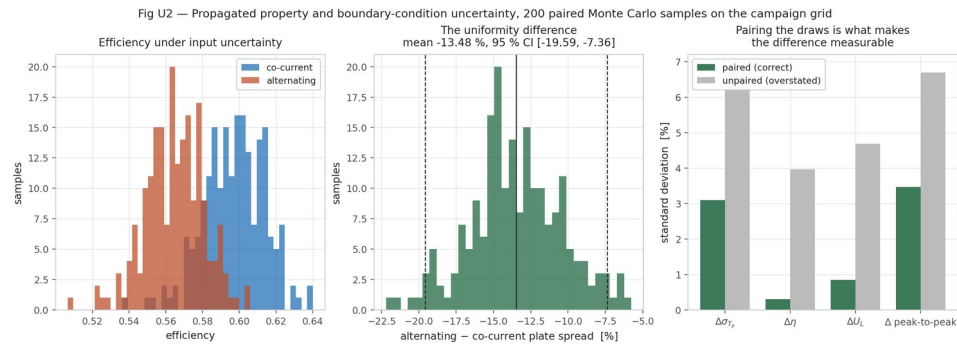


Figure E.21. Input uncertainty (Rev 2) — superseded numbers

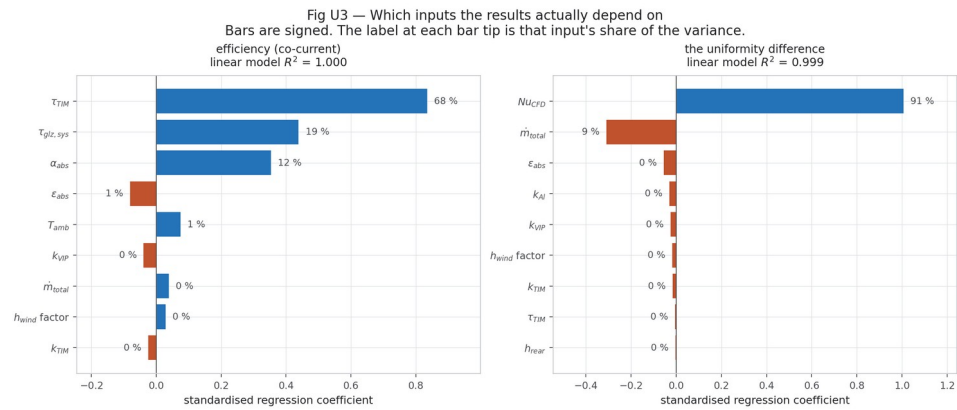


Figure E.22. Sensitivity (Rev 2) — superseded numbers

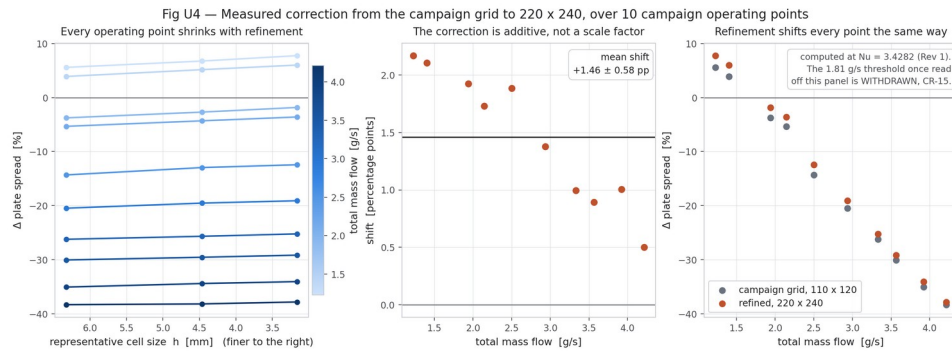


Figure E.23. Campaign correction (Rev 2) — superseded numbers

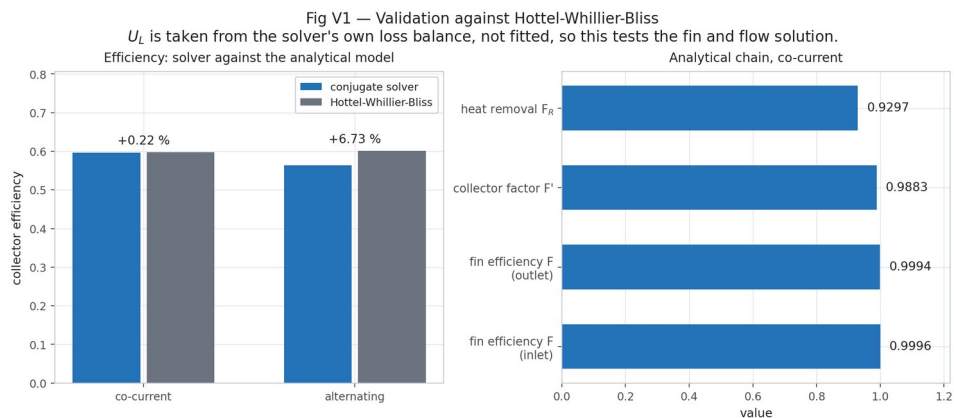


Figure E.24. HWB check (Rev 2) — superseded numbers

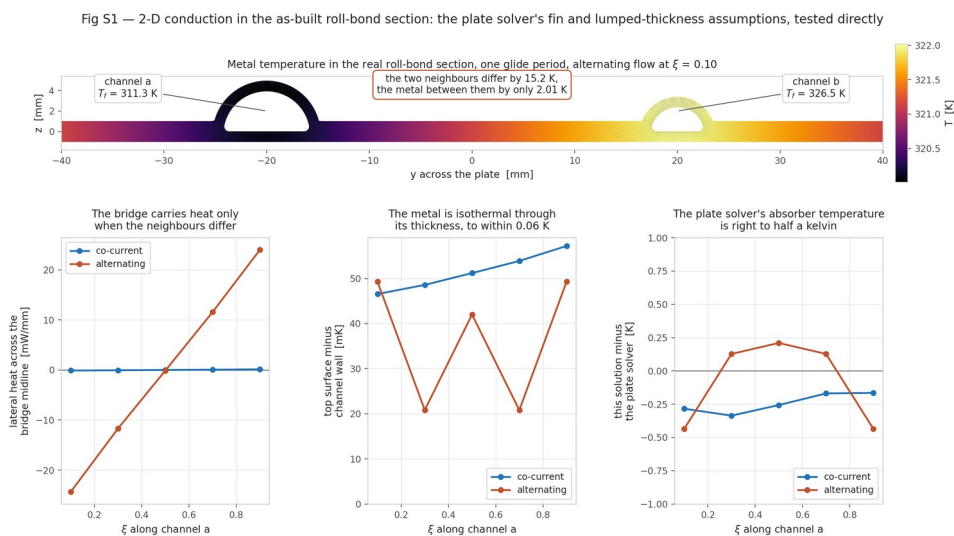


Figure E.25. Section conduction check (Rev 2) — superseded numbers

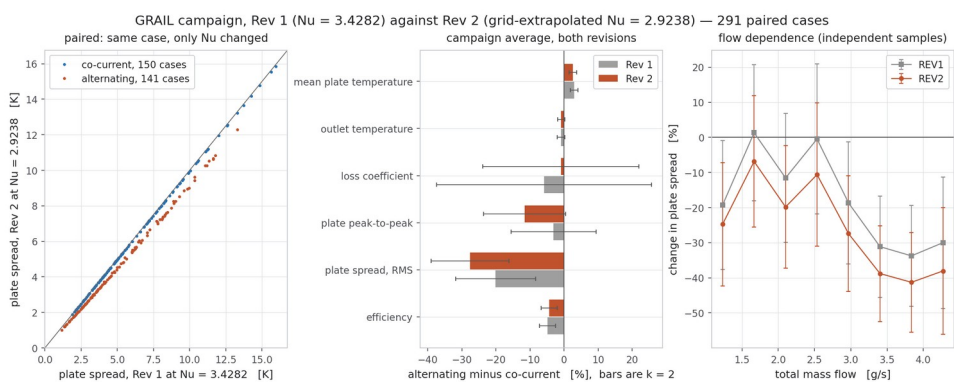


Figure E.26. Rev 1 vs Rev 2 — superseded numbers

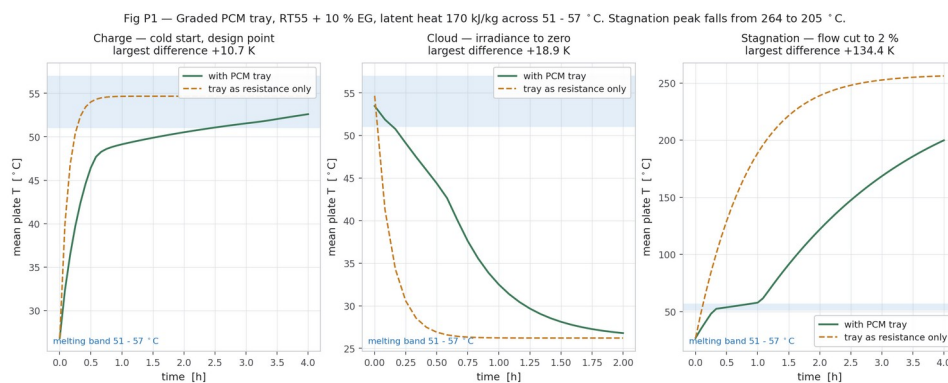


Figure E.27. PCM transient side study (Rev 2) — superseded numbers

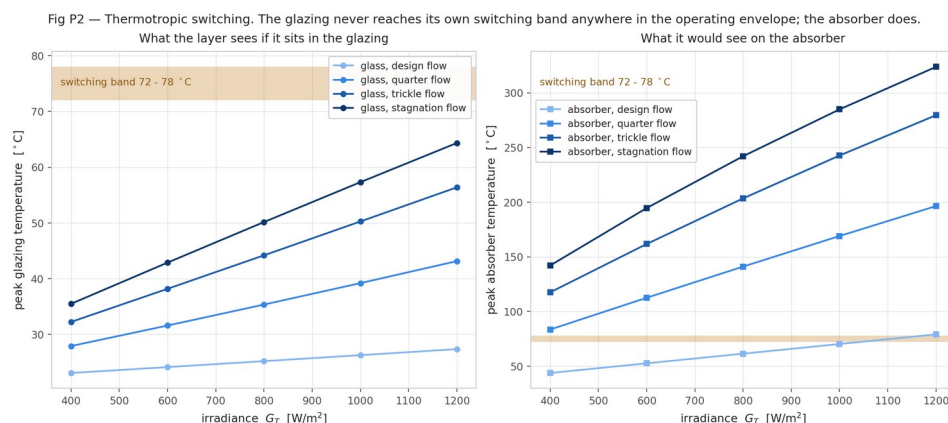


Figure E.28. Thermotropic layer side study (Rev 2) — superseded numbers

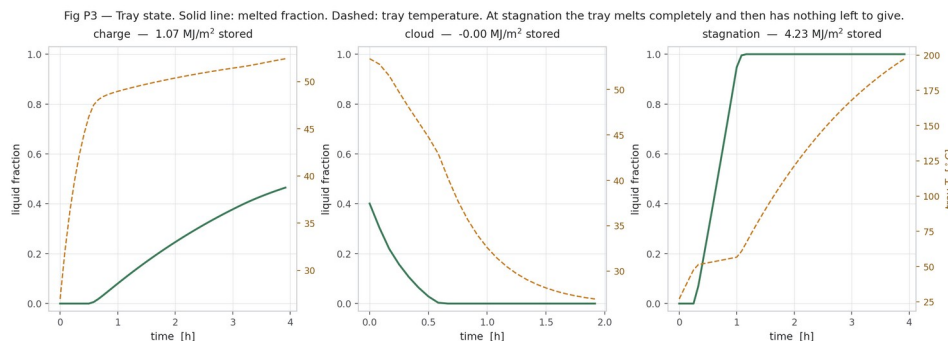


Figure E.29. PCM state (Rev 2) — superseded numbers

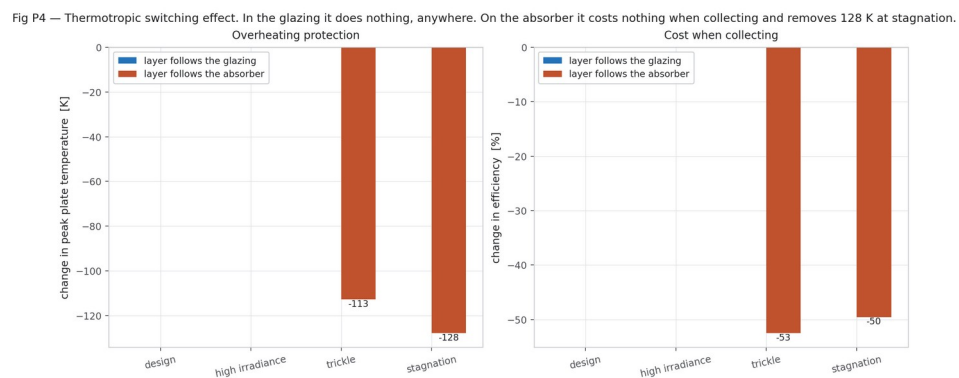


Figure E.30. Thermotropic effect (Rev 2) — superseded numbers

Appendix F. References

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- [11] Other screened papers: see Table 14 (Gunjo 2017; Bharathiraja 2024; Vahidinia 2024; Zareie 2024; Parthiban 2025; Ding 2025; Li 2024; Alawi 2024; Mansour 2013; Zheng 2024; Sakib 2026; Kessentini; Robles 2014; Oyinlola & Shire 2018).